DEPARTMENT OF DEFENSE
SMALL BUSINESS TECHNOLOGY TRANSFER (STTR) PROGRAM

STTR 25.A Annual Program Broad Agency Announcement (BAA)

IMPORTANT DATES

December 4, 2024: Topics Pre-release
January 8, 2025: Topics Open; DoD begins accepting proposals in DSIP
January 22, 2025: DSIP Topic Q&A closes to new questions at **12:00 p.m. ET**
February 5, 2025: Topics Close; Deadline for receipt of proposals is **12:00 p.m. ET**

Table of Contents

1.0 PROGRAM DESCRIPTION 1

1.1 Objectives and Context 1
1.2 A Three Phased Program 1
1.3 OUSD(R&E) Critical Technology Areas 2
1.4 Eligibility and Performance Requirements 2
1.5 Majority Ownership in Part by Multiple Venture Capital, Hedge Fund, and Private Equity Firms 3
1.6 Performance Benchmark Requirements/Increased Minimum Performance Standards for Experienced Firms 3
1.7 Program on Innovation Open Topics 4
1.8 Discretionary Technical and Business Assistance (TABA) 4
1.9 Phase II Enhancement Policy 4
1.10 Commercialization Readiness Program (CRP) 5
1.11 State and Other Available Assistance 5
1.12 Fraud and Fraud Reporting 5

2.0 CERTIFICATIONS AND REGISTRATIONS 5

2.1 System for Award Management (SAM) Registration 5
2.2 SBA Company Registry 6
2.3 Defense SBIR/STTR Innovation Portal (DSIP) Registration 6
2.3.1 DSIP Assistance and Support 6
2.4 Required Certifications 7
2.5 Due Diligence Program to Assess Security Risks 7
2.5.1 Training for Understanding FOCI 7
2.6 Joint Ventures 8
2.7 Conflicts of Interest 8
2.8 Organizational Conflicts of Interest (OCI) 8
2.9 Research Involving Human Subjects/Human Subject Research (RIHS/HSR) 9
2.10 Research Involving Animal Subjects 10
2.11 Research Involving Recombinant DNA Molecules 10
2.12 Federal Acquisition Supply Chain Security Act Orders 10

3.0 PROPOSAL PREPARATION INSTRUCTIONS AND REQUIREMENTS 10

3.1 Introduction 10
3.2 Export-Controlled Topic Requirements 11
3.3 Classified Proposals 11
3.4 Promotional Materials 11
3.5 Prior, Current, or Pending Support of Similar Proposals or Awards 11
3.6 Marking Proprietary Proposal Information 12
3.7 Phase I Proposal Instructions 12
3.8 Phase II Proposal Information 17
3.8.1 Phase II Commercialization Strategy 17
3.8.2 Phase II Adequate Accounting System 18

4.0 METHOD OF SELECTION AND EVALUATION CRITERIA 18

4.1 Evaluation Process 18
4.2 Evaluation Criteria 19
4.3 Proposal Status & Feedback 19

4.4 Award Denials 19
4.5 Pre-Award and Post Award BAA Protests 19

5.0 ADDITIONAL CONSIDERATIONS 20

5.1 Award Information 20
5.2 Contract Requirements 20
5.3 Agency Recovery Authority and Ongoing Reporting 21
5.4 Copyrights 21
5.5 Patents 21
5.6 Invention Reporting 21
5.7 Technical Data Rights 22
5.8 Final Technical Reports - Phase I through Phase III 22

6.0 PROPOSAL SUBMISSION 23

6.1 Submission Details 23
6.2 Technical Questions 23

7.0 PARTICIPATING COMPONENT INSTRUCTIONS & RESEARCH TOPICS 25

Appendix A: TECHNICAL PROPOSAL TEMPLATE (VOLUME 2) ..... A-1 – A-6
Appendix B: DEFINITIONS ..... B-1 – B-8
Appendix C: POTENTIAL APPLICABLE FEDERAL ACQUISITION REGULATION, DEFENSE FEDERAL ACQUISITION REGULATION SUPPLEMENT CLAUSES ..... C-1 – C-5

1.0 PROGRAM DESCRIPTION

1.1 Objectives and Context

The Defense Small Business Technology Transfer (STTR) Program's objectives include stimulating technological innovation, strengthening the role of small business to meet DoD research and development (R&D) needs, fostering and encouraging minority and disadvantaged persons' participation in technological innovation, and increasing the commercial application of DoD-supported research or R&D results. DoD invites proposing SBCs with the capability to conduct R&D and commercialize the results in any of the defense-related topic areas described in this STTR Program BAA to submit proposals.

The STTR Program, although modeled substantially after the Small Business Innovation Research (SBIR) Program, is a separate program and is separately financed. Subject to availability of funds, DoD Components will support high quality cooperative research and development proposals of innovative concepts to solve the listed defense-related scientific or engineering problems, especially those concepts that also have high potential for commercialization in the private sector. Partnerships between small businesses and Historically Black Colleges and Universities (HBCUs) or Minority Institutions (MIs) are encouraged, although no special preference will be given to STTR proposals from such proposers.

The Small Business Administration (SBA), through its SBIR/STTR Policy Directive, purposely departs from normal government solicitation formats and requirements, which simplifies the SBIR/STTR award process and minimizes the regulatory burden on small business. Consistent with the SBA SBIR/STTR Policy Directive, DoD is soliciting proposals as a broad agency announcement (BAA). The guidelines in this BAA incorporate and make use of the SBA SBIR/STTR Policy Directive's flexibility to encourage scientific and technical approaches proposals most likely to yield significant results for DoD and the private sector.

This BAA is for research topics accepting Phase I proposals. A separate BAA will not be issued requesting Phase II proposals, and unsolicited proposals will not be accepted. All proposing SBCs that receive a Phase I award from this BAA will be eligible to participate in Phase II competitions and potential Phase III awards. DoD Services/Components will notify Phase I awardees of the Phase II proposal submission requirements.

DoD is not obligated to make any awards under Phase I, Phase II, or Phase III, and all awards are subject to both a risk-based due diligence security review and funds availability. DoD is not responsible for any monies the proposing small business concern (SBC) spends before any award issuance. Proposals must conform to this announcement's terms. DoD is under no obligation to fund any proposal or any specific number of proposals in each topic. It also may elect to fund several or none of the proposed approaches to the same topic.

1.2 A Three Phased Program

The STTR Program has three phases, Phases I, II, and III. Phase I determines, to the extent possible, an idea's scientific, technical, and commercial merit and feasibility within the STTR program. Phase I and II awards are made adhering to current SBA Policy Directive guidelines.

The Phase I period of performance is generally between six to twelve months. Proposals should focus on Research or Research & Development (R/R&D)to prove the proposed effort's scientific and technical feasibility, and commercialization potential, the successful completion of which is a prerequisite for further DoD support in Phase II. Proposing SBCs are encouraged to consider whether the research or R&D being proposed to DoD Services/Components also has private sector potential, either for the proposed application or as a base for other applications.

Phase II awards will be made to proposing SBCs based on results of Phase I awards and the Phase II proposal's scientific merit, technical merit, and commercialization potential. The period of performance is generally 24 months. The objective of Phase II is to continue and further develop the R/R&D effort from the completed Phase I award.

Phase III refers to work that derives from, extends, or completes an effort made under prior STTR funding agreements, but is funded by sources other than the STTR Program. Under Phase III, the SBC should focus on commercializing previously STTR-funded technology and is required to obtain funding from either the private sector, a non-STTR federal source, or both, to develop the prototype into a viable product or non-R&D service for sale in military or private sector markets.

1.3 OUSD(R&E) Critical Technology Areas

Although each DoD Service/Component develops SBIR and STTR topics tailored to their mission needs, topics generally align with the Office of the Under Secretary of Defense, Research & Engineering (OUSD(R&E)) critical technology areas. While many technologies may cross between these categories, these areas represent the broad and different approaches required to advance technologies crucial to the Department, with a focus on accelerating key capabilities' transitions to the Military Services and Combatant Commands.

OUSD(R&E) critical technology areas include:

- FutureG
- Trusted AI and Autonomy
- Biotechnology
- Advanced Computing and Software
- Integrated Sensing and Cyber
- Directed Energy (DE)
- Hypersonics
- Microelectronics
- Integrated Network Systems-of-Systems
- Quantum Science
- Space Technology
- Renewable Energy Generation and Storage
- Advanced Materials
- Human-Machine Interfaces

Below are additional technology areas supporting DoD Services/Component-specific mission-critical areas:

- Advanced Infrastructure & Advanced Manufacturing
- Combat Casualty Care
- Emerging Threat Reduction
- Military Infectious Diseases
- Military Operational Medicine
- Mission Readiness & Disaster Preparedness
- Nuclear
- Sustainment & Logistics

Full descriptions of the above technology areas can be reviewed here.

1.4 Eligibility and Performance Requirements

Each proposing SBC must qualify as an SBC as defined in 13 C.F.R §§ 701-705 at time of award and certify to this in the proposal's cover sheet. The eligibility requirements for the SBIR/STTR programs are unique and do not correspond to those of other small business programs.

a. Proposing SBC must meet eligibility requirements for Small Business Ownership and Control (see 13 CFR § 121.702).
b. A minimum of 40% of each STTR project must be conducted by the small business concern and a minimum of 30% of the effort performed by the single research institution, as defined in Appendix B. The percentage of work is measured by both direct and indirect costs. Deviations

from these STTR requirements are not allowed, as the performance of work requirements are specified in statute at 15 USC 638(e). For more information on the percentage of work calculation during proposal submission, refer to section 3.7.

c. For both Phase I and II, the principal investigator's primary employment must be with the SBC or the research institution at the time of award and during the conduct of the proposed effort. Primary employment means that more than one-half of the principal investigator's time is spent with the small business (based on a 40-hour work week). At the time of award of a Phase I or Phase II contract, the SBC must have at least one employee in a management position whose primary employment is with the small business and who is not also employed by the research institution. Primary employment with an SBC precludes full-time employment at another organization. Deviations from this requirement or changes to the principal investigator are subject to the Funding Agreement officer approval.
d. For both Phase I and Phase II, the SBC and its subcontractors must perform all research or R&D work in the United States.
e. Joint ventures and limited partnerships are permitted, provided that the entity qualifies as small business in accordance with the ownership requirements in 13 CFR 121.702(a)(1)(iii) and the size requirements in 13 CFR 121.702(c)(6). Proposing SBC must disclose joint ventures with existing (or planned) relationships/partnerships with any foreign entity or any foreign government-controlled companies. See sections 2.6 and 3.7 for more detail.
f. A small business concern must negotiate a written agreement between the small business and the research institution allocating intellectual property rights and rights to carry out follow-on research, development, or commercialization (see Model Agreement for the Allocation of Rights).

1.5 Majority Ownership in Part by Multiple Venture Capital, Hedge Fund, and Private Equity Firms

Unless otherwise noted in the participating Service/Component instructions, proposing SBCs that are multiple venture capital operating companies (VCOCs), hedge funds, or private equity funds majority owned are **ineligible** to submit applications to or receive awards under this BAA. If a Service/Component authorizes such participation, any proposing VCOC, hedge fund, and/or private equity fund-owned SBC, whether in part or in whole, must identify each foreign national, foreign entity, or foreign government holding or controlling greater than a 5 percent, either directly or indirectly held, equity stake in the proposing SBC. The proposing SBC must also identify any ultimate parent owner(s) and other entities and/or individuals owning more than a 5 percent equity stake in its ownership chain.

In accordance with the requirements of 13 CFR 121.702(a)(2), no single venture capital operating company, hedge fund, or private equity firm may own more than 50 percent of the concern unless that single venture capital operating company, hedge fund, or private equity firm qualifies as a small business concern that is more than 50 percent directly owned and controlled by individuals who are citizens or permanent resident aliens of the United States.

1.6 Performance Benchmark Requirements/Increased Minimum Performance Standards for Experienced Firms

Proposing SBCs with multiple prior SBIR/STTR awards must meet minimum performance requirements to be eligible to apply for a new Phase I or Direct-to-Phase II award. The Phase I to Phase II Transition Rate addresses the extent to which an awardee progresses a project from Phase I to Phase II. The Commercialization Benchmark addresses the extent to which an awardee has moved past Phase II work towards commercialization.

The SBIR and STTR Extension Act of 2022 (Pub. L. 117-183) amended the benchmarks' application for more experienced firms. Find detailed information on benchmark calculations, increased performance standards for more experienced firms and consequence of failure to meet benchmarks here. SBA will notify companies failing either benchmark and the relevant officials at the participating agencies.

The SBIR/STTR Policy Directive defines the Departments of the Army, Navy, and Air Force each as its own federal agency, and the remaining DoD Components as an executive agency of the Department of Defense. Therefore, companies that fail to meet either of the benchmarks under the Increased Performance Standards for More Experienced Firms may not receive more than an overall total of 80 awards from DoD, as detailed in the breakdown below:

Army – 20 total Phase I and Direct to Phase II awards
Navy – 20 total Phase I and Direct to Phase II awards
Air Force – 20 total Phase I and Direct to Phase II awards
All other DoD Components combined – 20 total Phase I and Direct to Phase II awards

1.7 Program on Innovation Open Topics

15 U.S.C. §638 (ww) requires DoD establish innovation open topic activities to:

a. increase the transition of commercial technology to the DoD;
b. expand the small business nontraditional industrial base;
c. increase commercialization derived from DoD investments; and
d. expand the ability for qualifying SBCs to propose technology solutions to meet DoD needs.

Unlike conventional topics, which specify the desired technical objective and output, open topics use generalized mission requirements or specific technology areas to adapt commercial products or solutions to close capability gaps, improve performance, or provide technological advancements in existing capabilities.

Open topics released under this BAA will be clearly identified as such in the title and topic objective. Proposal preparation instructions for open topics may vary significantly across DoD Services/Components. Proposing SBCs must carefully read and follow all instructions from the DoD Service/Component for the open topic of interest. Unless specifically noted in the Service/Component instructions, all requirements outlined in this BAA remain in effect for open topics.

An SBC may only submit one proposal to each open topic. If an SBC submits more than one proposal for a single open topic, only the most recent certified proposal submitted prior to the submission deadline will receive an evaluation. All previously submitted proposals for the same open topic will be marked nonresponsive and will not receive an evaluation.

1.8 Discretionary Technical and Business Assistance (TABA)

DoD has not mandated the use of discretionary technical and business assistance (TABA). The proposing SBCs should review individual Service/Component-specific instructions to determine if TABA is offered by the Service/Component and follow instructions for requesting TABA funding.

1.9 Phase II Enhancement Policy

To further encourage the transition of SBIR/STTR research into both DoD acquisition programs and the private sector, certain DoD Services/Components developed their own Phase II Enhancement policies. Under this policy, the Service/Component will provide a Phase II awardee with additional Phase II

SBIR/STTR funding if the proposing SBC can match the additional SBIR/STTR funds with non-SBIR/STTR funds from DoD acquisition programs or the private sector.

See Service/Component instructions for more details on Phase II Enhancement opportunities.

1.10 Commercialization Readiness Program (CRP)

The SBIR/STTR Reauthorization Act of 2011 established the Commercialization Pilot Program (CPP) as a long-term program called the Commercialization Readiness Program (CRP). Each Military Department (Army, Navy, and Air Force) has a CRP; please check the Service/Component instructions for further information.

The Defense SBIR/STTR Program also established the OSD Transitions SBIR/STTR Technology (OTST) Pilot Program as an interim technology maturity phase (Phase II) inserted into the SBIR/STTR development. For more information contact osd.ncr.ousd-r-e.mbx.sbir-sttr-tech-transition@mail.mil.

1.11 State and Other Available Assistance

Many states have established programs to provide services to those proposing SBCs and individuals wishing to participate in the Federal SBIR/STTR Program. These services vary from state to state, but may include:

a. Information and technical assistance;
b. Matching funds to SBIR/STTR recipients;
c. Assistance in obtaining Phase III funding.

Contact your State SBIR/STTR Support office for further information. SBCs may seek general administrative guidance from small and disadvantaged business utilization specialists located in various defense contract management activities throughout the continental United States.

1.12 Fraud and Fraud Reporting

Knowingly and willfully making any false, fictitious, or fraudulent statements or representations may be a felony under the Federal Criminal False Statement Act (18 U.S.C. Sec 1001), punishable by a fine of up to $10,000, up to five years in prison, or both.

The DoD Office of Inspector General Hotline ("Defense Hotline") is an important avenue for reporting fraud, waste, abuse, and mismanagement within the DoD. The Office of Inspector General operates this hotline to receive and investigate complaints or information from contractor employees, DoD civilians, Service members, and the public. Individuals who wish to report fraud, waste or abuse may contact the Defense Hotline at (800) 424-9098 between 8:00 a.m. and 5:00 p.m. Eastern Time or visit their website to submit a complaint. Mailed correspondence should be addressed to the Defense Hotline, The Pentagon, Washington, DC 20301-1900, or email addressed to hotline@dodig.mil.

2.0 CERTIFICATIONS AND REGISTRATIONS

2.1 System for Award Management (SAM) Registration

The System for Award Management (SAM) allows proposing SBCs to provide basic information on business structure, capabilities, and financial and payment information with the Federal Government. Proposing SBCs must register in SAM here. Registration in SAM will generate the Unique Entity ID (UEI) number and the Commercial and Government Entry (CAGE) code. The UEI is required for registration in the U.S. Small Business Administration's (SBA) Company Registry. A proposing SBC

that is already registered in SAM should verify the registration is active, and its representations and certifications are current to avoid award delay.

2.2 SBA Company Registry

Proposing SBCs must be registered in the SBA Company Registry. SBCs will be required to verify registration by providing the SBC Control ID and Proof of Registration/Certification during proposal submission.

2.3 Defense SBIR/STTR Innovation Portal (DSIP) Registration

Individuals from proposing SBCs must be registered in the DSIP to prepare and submit proposals. Proposing SBCs submitting through this site for the first time will be asked to register. All users are required to have an individual user account to access DSIP. It is recommended proposing SBCs register as soon as possible upon identification of a proposal opportunity to avoid delays in the proposal submission process.

DSIP user accounts are authenticated by Login.gov. Users who do not already have a Login.gov account will be required to create one. Users who already have a Login.gov account can link their existing Login.gov account with their DSIP account. Job Aids and Help Videos to walk you through the process are in the Learning & Support section of DSIP.

Be advised the sharing of accounts and passwords is a violation of the Terms of Use for Login.gov and DoD policy.

Registered SBCs will have a designated DSIP Firm Admin responsible for creating the Firm PIN, controlling access for other users in the SBC and completing and maintaining the Firm-level forms, which must be completed before any proposals can be submitted.

Users should complete their account registrations as soon as possible to avoid any delays in proposal submissions.

NOTE: The DSIP application is only accessible from within the United States, which is defined as the fifty states, the territories and possessions of the Federal Government, the Commonwealth of Puerto Rico, the Republic of the Marshall Islands, the Federated States of Micronesia, the Republic of Palau, and the District of Columbia.

2.3.1 DSIP Assistance and Support

For assistance with the DSIP application, please visit the Learning & Support section of DSIP. Email DSIP Support at DoDSBIRSupport@reisystems.com only for further assistance with issues pertaining directly to the DSIP application. Questions submitted to DSIP Support will be addressed in the order received during normal operating hours (Monday through Friday, 9:00 a.m. to 5:00 p.m. ET). Please include information on your SBC, a proposal number (if applicable), and screenshots of any pertinent errors or issues encountered.

DSIP Support cannot provide updates to proposal status after submission, such as proposal selection/non-selection status or contract award status. Contact the DoD Service/Component that originated the topic following the Service/Component-specific instructions given at the beginning of that Service/Component-specific instructions.

2.4 Required Certifications

At the time of proposal submission, each SBC must certify via the Cover Sheet of the proposal that it meets the size, ownership, and other requirements of the STTR Program. In addition, the Policy Directive includes certification requirements set forth in the SBIR and STTR Extension Act of 2022 (Public Law 117-183). SBCs are required to certify that they are meeting the Program's requirements during the life cycle of the funding agreement: at the time of the Phase I and Phase II award, prior to final payment on the Phase I award, prior to receiving 50 percent of the total Phase II award amount, and prior to final payment on the Phase II award.

2.5 Due Diligence Program to Assess Security Risks

15 U.S.C. §638 (vv) requires the DoD, in coordination with the SBA, to establish and implement a due diligence program to assess and, when possible, mitigate national security risks from SBCs seeking a federally funded award. The Department will use the proposal and information in response to the "Disclosures of Foreign Affiliations or Relationships to Foreign Countries" (proposal submission Volume 7) to conduct a risk-based due diligence review of the following areas: cybersecurity practices; patent analysis; employee analysis and foreign ownership, including the SBC's financial ties and obligations (which shall include surety, equity, and debt obligations); and SBC employees' ties to a foreign country, foreign person, or foreign entity. The Department will also assess proposals utilizing open-source analysis and analytical tools, for the purpose of confirming the accuracy of the information provided as well as determining if the proposing SBC failed to disclose the information set forth in 15 U.S.C. 638(g)(13).

After reviewing the proposing SBC's responses to the Disclosures of Foreign Affiliations or Relationships to Foreign Countries, if DoD determines it appropriate the Department may ask the SBC to provide true copies of any contractual or financial obligation or other agreement specific to a business arrangement or joint-venture like arrangement with an enterprise owned by a foreign state or any foreign entity in effect during the five-year period preceding the SBC's proposal submission.

The DoD may not make awards that pose an unacceptable risk to national security. If the risk-based due diligence review concludes that an SBC should not be eligible for the specific SBIR or STTR award due to a national security risk that cannot be adequately mitigated, the proposal will not receive consideration for possible award regardless of the results of the technical review of the proposal. Attachment 2: Defense SBIR and STTR Due Diligence Program Common Risk Matrix of the memo titled "Defense Small Business Innovation Research and Small Business Technology Transfer Due Diligence Program", dated 13 May 2024, provides details on the factors for assessing SBC risk during the due diligence review.

2.5.1 Training for Understanding FOCI

DoD has partnered with Project Spectrum to provide an online course on Understanding Foreign Ownership, Control, or Influence (FOCI). This course defines FOCI, explains what it means to be under FOCI, and details FOCI's effect on a company seeking initial or continued eligibility for access to a federally funded award. Small business concerns can register and access this course by following the instructions below:

1. Go to projectspectrum.io
2. Click "Profile/Dashboard" in the top right and then click "Sign Up" from the dropdown menu.
3. Follow the instructions to sign up for an account. Descriptions of the account types are provided below each option.

4. Verify your email by entering the code sent to the email address you provided when signing up.
5. Log in to Project Spectrum by clicking "Profile/Dashboard > Login" in the top right.
6. Hover over "Courses" in the Navigation Bar, and then select "FOCI" from the dropdown listing.
7. Copy the provided password.
8. Click on the "Understanding Foreign Ownership, Control, or Influence (FOCI)" course, which will open a new browser tab.
9. From the new tab, log in to Encite.io using your email address and the copied password.
10. Enroll in the course and click "Enter" to begin.

For Project Spectrum registration or access assistance, please email support@projectspectrum.io.

2.6 Joint Ventures

A small business joint venture entity must submit, with its proposal, the representation required in paragraph (c) of FAR solicitation provision 52.212-3, Offeror Representations and Certifications-Commercial Products and Commercial Services, and paragraph (c) of FAR provision 52.219-1, Small Business Program Representations, in accordance with 52.204-8(d) and 52.212-3(b) for the following categories:

a. Small business;
b. Service-disabled veteran-owned small business;
c. Women-owned small business (WOSB) under the WOSB Program;
d. Economically disadvantaged women-owned small business under the WOSB Program; or
e. Historically underutilized business zone small business.

These representations can be found here and must be uploaded to Volume 5, Supporting Documents of the proposal submission in DSIP, if applicable.

2.7 Conflicts of Interest

Contract awards to an SBC owned by or employing current, or previous, Federal Government employees could create conflicts of interest for those employees, which may be a violation of federal law.

2.8 Organizational Conflicts of Interest (OCI)

FAR 9.5 Requirements

In accordance with FAR 9.5, proposing SBCs are required to identify and disclose all facts relevant to potential organizational conflicts of interest (OCIs) involving the proposing SBC's organization and any proposed team member (sub-awardee, consultant). The proposing SBC is responsible for providing this disclosure with each submitted proposal. The disclosure must include the proposing SBC's, and as applicable, proposed team member's OCI mitigation plan. The OCI mitigation plan must include a description of the actions the proposing SBC has taken, or intends to take, to prevent the existence of conflicting roles that might bias the proposing SBC's judgment, and to prevent the proposing SBC from having unfair competitive advantage. The OCI mitigation plan will specifically discuss the disclosed OCI in the context of each of the OCI limitations outlined in FAR 9.505-1 through FAR 9.505-4.

Agency Supplemental OCI Policy

DoD Services/Components also may have a supplemental OCI policy prohibiting contractors/performers from concurrently providing scientific engineering technical assistance (SETA), advisory and assistance services (A&AS), or similar support services, and being a technical performer. As part of the FAR 9.5 disclosure requirement, a proposing SBC must affirm whether the proposing SBC or any proposed team

member (sub-awardee, consultant) is providing SETA, A&AS, or similar support to any DoD Service/Component office(s) under: (a) a current award or sub-award; or (b) a past award or sub-award that ended within one calendar year prior to the proposal's submission date. If SETA, A&AS, or similar support is or was provided to any DoD Service/Component office(s), the proposal must include:

a. The name of the DoD Service/Component office receiving the support;
b. The prime contract number;
c. Identification of proposed team member (sub-awardee, consultant) providing the support; and
d. An OCI mitigation plan in accordance with FAR 9.5.

Government Procedures

In accordance with FAR 9.503, 9.504 and 9.506, the Government will evaluate OCI mitigation plans to avoid, neutralize, or mitigate potential OCI issues before award and determine whether it is in the government's interest to grant a waiver. The U.S. Government will only evaluate OCI mitigation plans for proposals determined selectable under the BAA evaluation criteria and funding availability.

The government may require proposing SBCs provide additional information to support evaluation of the proposing SBC's OCI mitigation plan.

If the government determines a proposer failed to fully disclose an OCI; or failed to provide a government waiver as described above; or failed to reasonably provide additional information the government requested when evaluating the proposer's OCI mitigation plan, the government may reject the proposal and withdraw it from consideration for award.

2.9 Research Involving Human Subjects/Human Subject Research (RIHS/HSR)

All research involving human subjects, to include use of human biological specimens and human data, shall comply with the applicable federal and state laws, and agency policy/guidelines for human subject protection (see Section 5.2 and Appendix B).

Institutions receiving funding for research involving human subjects must provide documentation of a current federal assurance of compliance with federal regulations for human subject protection; for example a Department of Health and Human Services, Office for Human Research Protections federal-wide assurance (http://www.hhs.gov/ohrp). The awarding DoD Service/Component may also request additional federal assurance documentation. All institutions engaged in human subject research, to include subcontractors, must also have a valid assurance. In addition, personnel involved in human subjects research must provide documentation of completed appropriate training for the protection of human subjects. Institutions proposing to conduct human subject research that meets one of the exemption criteria in 32 CFR 219.101 are not required to have a federal assurance of compliance. Proposing SBCs should clearly segregate research activities involving human subjects from other R&D activities in their proposal.

If selected, institutions must also provide documentation of institutional review board (IRB) approval, or a determination from an appropriate official in the institution, that the work meets one of the exemption criteria with 32 CFR 219. As part of the IRB review process, evidence that all investigators are appropriately trained should accompany the protocol. The protocol, separate from the proposal, must include a detailed description of the research plan, study population, risks and benefits of study participation, recruitment and consent process, and data collection and analysis.

The amount of time required for the IRB to review and approve the protocol will vary based on the IRB's procedures, the complexity of the research, the level of risk to study participants and the responsiveness of the investigator. The average IRB approval process can last between one and three months. Once the

IRB has approved the research, the awarding DoD Service/Component will review the protocol and the IRB's determination to ensure that the research will be conducted in compliance with both DoD and Service/Component policies. The DoD review process can last between three to six months. Ample time should be allotted to complete both the IRB and DoD approval processes prior to recruiting subjects. **No funding can be used towards human subject research until ALL approvals are granted. Submitters proposing research involving human and/or animal use are encouraged to separate these tasks in the technical proposal and cost proposal to avoid potential contract award delays.**

2.10 Research Involving Animal Subjects

All research, development, testing, experimentation, education, or training involving the use of animals shall comply with the applicable federal and agency rules on animal acquisition, transport, care, handling, and use (see Section 5.2 and Appendix B).

For submissions containing animal use, proposals should briefly describe plans for their institutional animal care and use committee (IACUC) review and approval.

All recipients must receive their IACUC's approval, as well as secondary or headquarters-level approval from a DoD veterinarian trained or experienced in laboratory animal medicine and science. **No animal research may be conducted using DoD funding until all appropriate DoD office(s) grant approval. Submitters proposing research involving human and/or animal use are encouraged to separate these tasks in the technical proposal and cost proposal to avoid potential contract award delays.**

2.11 Research Involving Recombinant DNA Molecules

All research involving recombinant DNA molecules shall comply with the applicable federal and state law, regulation, and additional agency guidance. An institutional biosafety committee must approve the research.

In addition to the standard federal and DoD procurement certifications, the SBA SBIR/STTR Policy Directive requires the proposing business concerns provide certain information at time of award and during the award life cycle. Each proposing SBC must provide this additional information at the time of the Phase I and Phase II award, prior to final payment on the Phase I award, prior to receiving 50 percent of the Phase II total award amount, and prior to final payment on the Phase II award.

2.12 Federal Acquisition Supply Chain Security Act Orders

FAR 52.204-29 Federal Acquisition Supply Chain Security Act (FASCA) Orders—Representation and Disclosures and FAR 52.204-30 FASCA Orders—Prohibition are included in this solicitation. In accordance with FAR 52.204-29 and FAR 52.204-30, proposing SBCs must review FASCSA orders here for covered articles, or any products or services produced or provided by a source, that an applicable FASCSA order prohibits.

During contract performance, the contractor shall review SAM.gov at least once every three months, or as the contracting officer advises, to check for covered articles, or products or services produced subject as part of any new FASCSA order(s) that could impact their supply chain, and report to the contracting officer any covered article, or product or service produced or provided by a source provided to the government or used during the contract performance.

The proposing SBC represents that, via proposal submission under this BAA, it conducted a reasonable inquiry, and it does not propose to provide or use any covered article, or any products or services produced or provided by a source, if an applicable FASCSA prohibited the covered article or the source effective this BAA's issue date.

3.0 PROPOSAL PREPARATION INSTRUCTIONS AND REQUIREMENTS

3.1 Introduction

The proposal must provide sufficient information to demonstrate to the evaluator(s) that the proposed work represents an innovative approach to an important scientific or engineering problem and is worthy of support under the stated criteria. The proposed research or R&D must be responsive to the chosen

topic, although it does not need to use the exact approach specified in the topic. SBCs should consider the following:

a. Does the technical approach have a reasonable chance of meeting the topic objective?
b. Is this approach innovative, not routine, with potential for commercialization?
c. Does the proposing SBC have the capability to implement the technical approach or can it obtain the appropriate people and equipment for the task?

DSIP provides a structure for providing the following proposal volumes:

a. Volume 1: Proposal Cover Sheet
b. Volume 2: Technical Volume
c. Volume 3: Cost Volume
d. Volume 4: Company Commercialization Report
e. Volume 5: Supporting Documents
f. Volume 6: Fraud, Waste and Abuse Training
g. Volume 7: Disclosures of Foreign Affiliations or Relationships to Foreign Countries

Each Service/Component guidance on allowable proposal content may vary. A completed proposal submission in DSIP does NOT indicate that each proposal volume has been completed in accordance with the Service/Component-specific instructions. Accordingly, it is the proposing SBC's responsibility to consult the Service/Component-specific instructions for detailed guidance, including required proposal documentation and structure, cost and duration limitations, budget structure, TABA allowance and proposal page limits.

3.2 Export-Controlled Topic Requirements

For proposals submitted under export-controlled topics, either International Traffic in Arms or Export Administration Regulations (ITAR/EAR), a copy of the certified DD Form 2345, Militarily Critical Technical Data Agreement, or evidence of application submission must be included. The form, instructions and FAQs may be found at the United States/Canada Joint Certification Program website, https://www.dla.mil/Logistics-Operations/Services/JCP/DD23%2045Instructions/.

DD Form 2345 approval will be required if a proposal submitted to an ITAR/EAR-marked topic receives an STTR award.

3.3 Classified Proposals

Classified proposals will not be accepted under the DoD STTR Program. If topics require classified work during Phase II, the proposing SBC must have a facility clearance to perform the work. For more information on facility and personnel clearance procedures and requirements, please visit the Defense Counterintelligence and Security Agency (DCSA) website at: https://www.dcsa.mil/mc/ctp/fc/.

3.4 Promotional Materials

Promotional and non-project related discussion is discouraged, and additional information provided via website links or on computer disks, CDs, DVDs, video tapes or any other medium will not be accepted or considered in the proposal evaluation.

3.5 Prior, Current, or Pending Support of Similar Proposals or Awards

While it is permissible, with proposal notification, to submit identical proposals or proposals containing a significant amount of essentially equivalent work for consideration under numerous federal program BAAs or solicitations, it is unlawful to enter negotiation for contracts requiring essentially equivalent effort. If there is any question concerning prior, current, or pending support of similar proposals or awards, it must be disclosed to the soliciting agency or agencies as early as possible and declared on the proposal cover sheet.

3.6 Marking Proprietary Proposal Information

Proposing SBCs that include data in their proposals they do not want disclosed to the public for any purpose, or only used for government evaluation purposes, shall:

a. Mark the first page of each volume of the proposal submission with the following legend:

   "This proposal includes data that shall not be disclosed outside the Government and shall not be duplicated, used, or disclosed-in whole or in part-for any purpose other than to evaluate this proposal. If, however, a contract is awarded to this proposing SBC as a result of-or in connection with-the submission of this data, the Government shall have the right to duplicate, use, or disclose the data to the extent provided in the resulting contract. This restriction does not limit the Government's right to use information contained in this data if it is obtained from another source without restriction. The data subject to this restriction are contained in pages [insert numbers or other identification of sheets]"; and

b. Mark each sheet of data it wishes to restrict with the following legend:

   "Use or disclosure of data contained on this page is subject to the restriction on the first page of this volume."

The DoD assumes no liability for disclosure or use of unmarked data and may use or disclose such data for any purpose.

Restrictive notices notwithstanding, support contractors may handle proposals and final reports submitted through DSIP for administrative purposes only; they are required to adhere to appropriate non-disclosure agreements.

3.7 Phase I Proposal Instructions

a. **Proposal Cover Sheet (Volume 1)**

   The proposal cover sheet is prepared on DSIP. The cover sheet must include a brief technical abstract that describes the proposed R&D project and an anticipated benefits and potential commercial applications discussion. Each section should be no more than 3,000 characters. **Do not include proprietary or classified information in the proposal cover sheet**. If your proposal is selected for negotiation and possible award, the technical abstract and anticipated benefits discussion may be publicly released online. DSIP will assign a proposal number once the cover sheet is saved. You may modify the cover sheet as needed until the BAA closes.

   NOTE: the amounts listed in the percentage of work (POW) certification question on the proposal cover sheet are derived from SBC-entered information in the cost volume (Volume 3). Details on the calculation can be viewed in DSIP during proposal submission.

   If the POW calculations fall below eligibility requirements, the small business concern will not be able to proceed with proposal submission. Deviations from the POW minimum requirements for STTR proposals are not allowed.

b. **Technical Volume Format (Volume 2)**

   1. **File Type.** The Technical Volume must be a single PDF file, including graphics. Perform a virus check before uploading the technical volume file. If a virus is detected, the proposal may be rejected. **Do not lock, password protect or encrypt the uploaded file.**

Do not include or embed active graphics, such as videos, moving pictures, or other similar media, in the document.

2. **Length.** It is the proposing SBC's responsibility to verify that the technical volume does not exceed the page limit after upload to DSIP. Please refer to Service/Component-specific instructions for how a technical volume is handled if the stated page count is exceeded. Some Services/Components will reject the entire technical proposal if the proposal exceeds the stated page count.

3. **Layout.** Number all proposal pages consecutively. Submit a direct, concise, and informative research or R&D proposal (no type smaller than 10-point on standard 8-1/2" x 11" paper with one-inch margins, including the header). Each header on each page in the technical volume should contain the proposing SBC's name, topic number, and the DSIP-assigned proposal number from the cover sheet.

c. **Technical Volume Content (Volume 2)**

The Technical Volume should cover the following items in the order given below:

1. Identification and Significance of the Problem or Opportunity
2. Phase I Technical Objectives
3. Phase I Statement of Work
4. Related Work
5. Relationship with Future Research or Research and Development
6. Commercialization Strategy
7. Key Personnel
8. Foreign Citizens
9. Facilities/Equipment
10. Subcontractors/Consultants
11. Prior, Current, or Pending Support of Similar Proposals or Awards
12. Identification and Assertion of Restrictions on the Government's Use, Release, or Disclosure of Technical Data or Computer Software

A Phase I technical volume template is available in Appendix A to provide details and helpful guidelines for completing each section of your Phase I technical proposal.

d. **Cost Volume Content (Volume 3)**

Complete the cost volume using the DSIP cost volume form. Some items in the cost breakdown may not apply to the proposed project. There is no need to provide information on each individual item; make sure to provide enough information for evaluators to understand the requested funds' planned use if a contract is awarded.

1. List all key personnels' names and include their individual hours dedicated to the project as direct labor.
2. While special tooling and test equipment and material cost may be included under Phase I, equipment and material inclusion will be carefully reviewed relative to need and appropriateness for the work proposed. Special tooling and test equipment purchases must, in the Service/Component contracting officer's opinion, be advantageous to the U.S. Government and should relate directly to the specific topic. These may include such items as innovative instrumentation or automatic test equipment. Title to property the U.S. Government furnished or acquired with government funds will be vested with the DoD

Service/Component, unless it is determined that title transfer to the contractor would be more cost effective than the DoD Service/Component equipment recovery.

3. Cost for travel funds must be justified and related to the project needs.
4. Cost sharing is permitted for proposals under this BAA; cost sharing is not required, nor will it be an evaluation factor in the Phase I proposal consideration.
5. A Phase I option (if applicable) should be fully costed separately from the Phase I (base) approach.
6. All subcontractor costs and consultant costs, such as labor, travel, equipment, materials, must be detailed at the same level as prime contractor costs. Provide detailed subcontractor costs substantiation in your cost proposal. Supporting Documents (Volume 5) may be used if additional space is needed.

If a proposal is selected for negotiation and possible award, you must be prepared to submit further documentation to the Service/Component contracting officer to substantiate costs (e.g., a cost estimates explanation for equipment, materials, and consultants or subcontractors). For more information about cost proposals and accounting standards, see visit DCAA's website.

e. **Company Commercialization Report (Volume 4)**

The company commercialization report (CCR) allows companies to report funding outcomes resulting from prior SBIR and STTR awards. SBA requires SBIR and STTR awardees to update and maintain their organization's CCR on SBIR.gov. Commercialization information is required upon the last deliverable's completion under the funding agreement. Thereafter, SBIR and STTR awardees are requested to voluntarily update the information in the database annually for a minimum 5-year period.

If the proposing SBC has prior DoD and/or non-DoD Phase I and/or Phase II SBIR/STTR awards, regardless of whether the project has any commercialization to date, the firm admin must download the CCR's PDF copy from SBIR.gov and upload it to DSIP's "Firm Forms" section. The DSIP firm admin completes the firm forms are completed and are applies them to all proposals the proposing SBC submits. To fulfill the DSIP CCR requirement complete the following:

1. Log into the firm account at https://www.sbir.gov/.
2. Navigate to My Dashboard > My Documents to view or print the information currently contained in the "Company Registry Commercialization Report."
3. Create or update the commercialization record, from the company dashboard, scroll to the "My Commercialization" section and click the "Create/Update Commercialization" tab under "Current Report Version." Please refer to the "Instructions" and "Guide" documents contained on dashboard for more detail on completing and updating the CCR. **Ensure the report is certified and submitted.**
4. Click the "Company Commercialization Report" PDF under the dashboard's "My Documents" section to download the CCR PDF.
5. Upload the CCR PDF (downloaded from SBIR.gov in previous step) to the "Company Commercialization Report" in DSIP's "Firm Forms" section. The firm admin must complete this upload action.

In Volume 4 of the DSIP proposal submission, the proposing SBC will be prompted to answer: "Do you have a new or revised Company Commercialization Report to upload?" There are three possible courses of action:

a. If the proposing SBC has prior DoD and/or non-DoD Phase I and/or Phase II SBIR/STTR awards and **has a new or revised CCR from SBIR.gov to upload to DSIP**, select YES.
   1. If the user is the firm admin, they can upload the CCR PDF from SBIR.gov directly on this page. It will also be updated in the "Firm Forms" and be associated with all new or in-progress proposals the proposing SBC submitted. If the user is not the firm admin, they will receive a message that they do not have access and must contact the firm admin to complete this action.
   2. **WARNING:** Uploading a new CCR under the DSIP "Firm Forms" section or clicking "Save" or "Submit" in one proposal's Volume 4 is considered a change for ALL proposals under any open BAAs or CSOs. If a proposing SBC has previously certified and submitted any Phase I or Direct to Phase II proposals under ***any*** BAA or CSO ***still open***, those proposals will be automatically reopened. Proposing SBCs will have to recertify and resubmit affected proposals. If a proposing SBC does not recertify or resubmit affected proposals, they will not be considered fully submitted and will not be evaluated.

b. If the proposing SBC has prior DoD and/or non-DoD Phase I and/or Phase II SBIR/STTR awards, and **no new or revised CCR from SBIR.gov to upload to DSIP**, select NO.
   1. If a prior CCR was uploaded to the "Firm Forms", the proposing SBC will see a file dialog box at the bottom of the page and can view the previously uploaded CCR. This read-only access allows the proposing SBC to confirm the firm admin uploaded the CCR.
   2. If no file dialog box appears at the bottom of the page **there is no previously uploaded CCR in the DSIP "Firm Forms**." To fulfill the DSIP CCR requirement the firm admin must follow steps 1-5 listed above to download a CCR PDF from SBIR.gov and upload it to the DSIP "Firm Forms" to be included with all proposal submissions.

c. If the proposing SBC has **NO** prior DoD and/or non-DoD Phase I and/or Phase II SBIR/STTR awards, the CCR upload from SBIR.gov is not required and SBC will select NO. The proposal's CCR section will be marked complete.

Please refer to the Service/Component-specific instructions for details on how the CCR information will be considered during proposal evaluations.

f. **Supporting Documents (Volume 5)**

Volume 5 is provided for proposing SBCs to submit additional documentation to support the Coversheet (Volume 1), Technical Volume (Volume 2), and the Cost Volume (Volume 3).

The following documents may be included in Volume 5, if applicable to the proposal. Refer to Service/Component-specific instructions for additional Volume 5 requirements. **Reminder: A completed proposal submission in DSIP does NOT indicate the mandatory supporting documents have been uploaded in accordance with the Service/Component-specific instructions.**

1. Letters of support
2. Additional cost information
3. Funding agreement certification
4. Technical data rights (assertions)

5. Lifecycle certification
6. Allocation of rights
7. Verification of Eligibility of Small Business Joint Ventures, if applicable
8. DD Form 2345, Militarily Critical Technical Data Agreement, if applicable (see section 3.2)

g. **Fraud, Waste, and Abuse Training (Volume 6)**

The fraud, waste, and abuse (FWA) training is **required** for DoD SBIR/STTR proposals. FWA training provides information on what represents FWA in the SBIR/STTR program, the most common mistakes that lead to FWA, as well as the penalties and ways to prevent FWA. The training currently consists of a 3-page PDF, consistent with the tutorial provided by the SBA. This training material must be thoroughly reviewed once per year and can be found here and in the DSIP proposal submission module for Volume 6. Plan time to review the tutorial during completion of Volume 6, prior to the proposal submission deadline. One DSIP firm user for the proposing SBC with read/write access (Proposal Owner, Corporate Official or Firm Admin) must complete this training.

h. **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)**

In accordance with 15 U.S.C. §638 (vv) and the SBA SBIR/STTR Policy Directive, the DoD will review all proposals submitted in response to this BAA to assess security risks of SBCs seeking a federally funded award. SBCs must complete the DSIP Volume 7 webform "Disclosures of Foreign Affiliations or Relationships to Foreign Countries" (NOTE: PDF uploads are no longer accepted). The corporate official cannot certify and submit the full proposal until the Volume 7 webform is fully completed and submitted.

Please be aware that the "Disclosures of Foreign Affiliations or Relationships to Foreign Countries" form WILL NOT be accepted as a supporting document in DSIP's Volume 5 proposal submission. Do not upload any previous versions of this form to Volume 5.

For additional details, please refer to Section 2.5. The disclosure questions are below:

1. Is any owner or covered individual of the applicant or awardee party to any malign foreign talent recruitment program? If yes, disclose the first and last name of each owner or covered individual, identify their role (i.e., owner or covered individual), and the malign foreign talent recruitment program.
2. Is there a parent company, joint venture, or subsidiary, of the applicant or awardee that is based in or receives funding from, any foreign country of concern? If yes, disclose the name, full address, applicant or awardee relationships (i.e., parent company, joint venture, or subsidiary) of each entity based in, or funded by, any foreign country of concern.
3. Does the applicant or awardee have any current or pending contractual or financial obligation or other agreement specific to a business arrangement, or joint venture-like arrangement with an enterprise owned by a foreign state or any foreign entity? If yes, disclose the name of each enterprise or foreign entity, type of obligation, agreement, or arrangement (*i.e.,* contractual, financial, or other), description of obligation, agreement, or arrangement, and the foreign state(s) and/or the country of the foreign entity (or entities).
4. Is the applicant or awardee wholly owned in a foreign country? If yes, disclose the foreign country.

5. Does the applicant or awardee have any venture capital or institutional investment? If yes, proceed to question 5a. If no, proceed to question 6.

    5a. Does the investing entity have a general partner or any other individual holding a leadership role who has a foreign affiliation with any foreign country of concern? If yes or unable to determine, disclose the venture capital or institutional investing entity's name, the percentage of ownership obtained by the investing entity, and the type of investment (i.e., equity, debt, or combination of equity and debt).

6. During the previous 5-year period, did the applicant or awardee have any technology licensing or intellectual property sales or transfers, to a foreign country of concern? If yes, disclose the name, address, and country, of the institution or entity that licensed, purchased, or received the technology or intellectual property.
7. Is there any foreign business entity, offshore entity, or entity outside the United States related to the applicant or awardee? If yes, disclose the entity name, relationship type (i.e., foreign business entity, offshore entity, entity outside the United States), description of the relationship to the applicant or awardee, and entity address and country.
8. Does the applicant or awardee have an owner, officer, or covered individual that has a foreign affiliation with a research institution located in a foreign country of concern? If yes, disclose the first and last name of each owner, officer, or covered individual that has a foreign affiliation with a foreign country of concern, identify their role (i.e., owner, officer, or covered individual), and the name of the foreign research institution and the foreign country of concern where it is located.

3.8 Phase II Proposal Information

Only Phase I awardees may submit Phase II proposals. Phase II proposals submission must follow individual Service/Component instructions. Awarding DoD Services/Component, either in the Phase I award or via subsequent notification, will provide details on Phase II proposal due date, content, and submission requirements. If a proposing SBC submits their Phase II proposal prior to the individual Service/Component's dates, it may be rejected without evaluation.

Due to specific limitations on the amount of funding and number of awards awarded to a particular proposing SBC per topic using SBIR/STTR program funds, head of agency determinations are now required before a different agency may make an award using another agency's topic. This limitation does not apply to Phase III funding. Please contact your original sponsoring agency before submitting a Phase II proposal to an agency other than the one who sponsored the original topic.

SBIR/STTR Policy Directive Section 4(b)(1)(i) allows that, at the agency's discretion, projects awarded a Phase I under a solicitation for SBIR may transition in Phase II to STTR and vice versa. A proposing SBC wishing to transfer from one program to another must contact their designated technical monitor to discuss the reasons for the request and the agency's ability to it. The transition may be proposed prior to award or during the Phase II effort performance. Agency disapproval of a request to change programs shall not be grounds for granting relief from any contractual performance requirement. All approved transitions between programs must be noted in the contracting officer-signed Phase II award or award modification that indicates the removal or addition of the research institution and the revised percentage of work requirements.

3.8.1 Phase II Commercialization Strategy

At a minimum, the commercialization strategy must address the following five questions:

a. What will be the first product to use this technology?
b. Who will be the customers, and what is the estimated market size?
c. How much money is needed to bring the technology to market, and how will that money be raised?
d. Does the proposing SBC contain marketing expertise and, if not, how will the SBC acquire that expertise?
e. Who are the proposing SBC's competitors, and what is the price and/or quality advantage over those competitors?

The commercialization strategy must also include a schedule showing the anticipated quantitative commercialization results at 1) one year after the Phase II project starts, 2) at Phase II completion, and 3) after Phase II completion (i.e., additional investment amount, sales revenue, etc.). After Phase II award, the proposing SBC is required to report actual sales and investment data in its SBA company commercialization report via SBIR.gov's "My Dashboard" on a minimum annual basis. Please refer to the Service/Component-specific instructions for guidance on formatting, page count and other details.

3.8.2 Phase II Adequate Accounting System

To reduce the small business's risk and avoid potential contracting delays, companies interested in pursuing Phase II SBIR/STTR contracts and other contracts of similar size with the DoD, must have an adequate accounting system in place per General Accepted Accounting Principles, Generally Accepted Government Auditing Standards, Federal Acquisition Regulation (FAR) and Cost Accounting Standards. The Defense Contract Audit Agency (DCAA) will audit the accounting system. See DCAA's website for requirements and standards, the audit process overview, and a pre-award system adequacy checklist.

4.0 METHOD OF SELECTION AND EVALUATION CRITERIA

4.1 Evaluation Process

All proposals will be evaluated and judged on a competitive basis in terms of technical capability and technical value. Proposals will be initially screened to determine responsiveness to the topic objective. Proposals passing this initial screening will be technically evaluated by engineers, scientists, or subject matter experts to determine the most promising technical and scientific approaches. As a common statement of work does not exist, each proposal will be assessed on the merit of the approach in achieving the technical objectives established in the topic.

4.2 Evaluation Criteria

Proposals will be evaluated based on the criteria outlined below, unless otherwise specified in the Service/Component-specific instructions. Selections will be based on a determination of the overall technical value of each proposal and an evaluation of the cost volume, with the appropriate method of analysis given the contract type to be awarded, for selection of the proposal(s) most advantageous to the Government, considering the following factors which are listed in descending order of importance:

a. The soundness, technical merit, and innovation of the proposed approach and its incremental progress toward topic or subtopic solution.
b. The qualifications of the proposed principal/key investigators, supporting staff, and consultants. Qualifications include not only the ability to perform the R&D but also the ability to commercialize the results.

c. The potential for commercial (government or private sector) application and the benefits expected to accrue from this commercialization.

Cost or budget data submitted with the proposals will be considered during evaluation.

Technical reviewers will base their conclusions only on information contained in the proposal. Do not assume reviewers are acquainted with the proposing SBC, key individuals, or any referenced experiments. Relevant supporting data such as journal articles, literature, including Government publications, etc., should be included based on requirements listed in Service/Component-specific instructions.

4.3 Proposal Status & Feedback

Proposing SBCs will be notified of selection or non-selection status for an award by the DoD Service/Component that originated the topic no later than 90 days of the closing date for this BAA. Please refer to the Service/Component-specific instructions for details.

After final selection decisions have been announced, the SBC may be provided proposal feedback in the form of a written debrief. This debriefing process varies across the DoD Services/Components. Please refer to the Service/Component-specific instructions for details on the debriefing processes.

4.4 Award Denials

The DoD will not make an award under the SBIR/STTR program if it determines:

a. The SBC submitting the proposal
   i. has an owner or covered individual that is party to a malign foreign talent recruitment program;
   ii. has a business entity, parent company, or subsidiary located in the People's Republic of China or another foreign country of concern; or
   iii. has an owner or covered individual that has a foreign affiliation with a foreign entity located in the People's Republic of China or another foreign country of concern; and
b. The relationships and commitments described in clauses (i) through (iii) of subparagraph (A)
   i. interfere with the capacity for DoD-supported activities to be carried out;
   ii. create duplication with DoD-supported activities;
   iii. present concerns about conflicts of interest;
   iv. were not appropriately disclosed to the DoD;
   v. violate Federal law or terms and conditions of DoD-awarded contracts or other agreements; or
   vi. pose a risk to national security.

4.5 Pre-Award and Post Award BAA Protests

Interested parties have the right to protest via procedures in FAR Subpart 33.1; protests exclusively related to this BAA's terms must be served to: osd.ncr.ousd-r-e.mbx.SBIR-STTR-Protest@mail.mil

For protests filed with the Government Accountability Office (GAO), a copy of the protest shall be submitted to the email address listed above (pre-award ONLY) or DoD Service/Component POC (post-selection/award decision ONLY) within one day of filing with the GAO. Protests of small business status of a selected proposing SBC may also be made to the SBA via the procedures in FAR § 19.302.

For the purposes of a protest related to a particular topic selection, non-selection or award decision, protests should be served to the point-of-contact (POC) listed in the instructions of the DoD Component that authored the topic.

5.0 ADDITIONAL CONSIDERATIONS

5.1 Award Information

The number of awards will be consistent with the Component's RDT&E budget. No contracts will be awarded until evaluation of all qualified proposals for a specific topic is completed.

Each proposal selected for negotiation and possible award will be funded under negotiated contracts, purchase orders, or Other Transactions and will include a reasonable fee or profit consistent with normal profit margins provided to profit-making proposing SBCs for R/R&D work. Firm-Fixed-Price, Firm-Fixed-Price Level of Effort, Labor Hour, Time & Material, or Cost-Plus-Fixed-Fee type contracts can be negotiated and are at the discretion of the Component Contracting Officer.

Contract value varies among the DoD Services/Components; it is important for proposing SBCs to review Service/Component-specific instructions regarding award size.

The SBA SBIR/STTR Policy Directive, Section 7(c)(1)(ii), states agencies should issue the award no more than 180 days after the closing date of the BAA.

5.2 Contract Requirements

Upon contract award, the contractor will be required to make certain legal commitments through acceptance of U.S. Government contract clauses in the Phase I contract. The examples below are illustrative of the types of provisions the Federal Acquisition Regulation requires in the Phase I contract. This is not an exhaustive provisions list that could be included in Phase I contracts, nor does it contain specific clause wording. Appendix C of this BAA contains additional potential required Federal Acquisition Regulation (FAR) and Defense Federal Acquisition Regulation Supplement (DFARS) clauses. Copies of complete general provisions will be made available prior to award.

Examples of general provisions:

a. **Standards of Work**. Work performed under the Funding Agreement must conform to high professional standards.
b. **Inspection**. Work performed under the Funding Agreement is subject to Government inspection and evaluation at all times.
c. **Examination of Records**. The Comptroller General (or a duly authorized representative) must have the right to examine any pertinent records of the Awardee involving transactions related to this Funding Agreement.
d. **Default**. The Federal Government may terminate the Funding Agreement if the contractor fails to perform the work contracted.
e. **Termination for Convenience**. The Funding Agreement may be terminated at any time by the Federal Government if it deems termination to be in its best interest, in which case the Awardee will be compensated for work performed and for reasonable termination costs.
f. **Disputes.** Any dispute concerning the Funding Agreement that cannot be resolved by agreement must be decided by the contracting officer with right of appeal.
g. **Contract Work Hours**. The Awardee may not require an employee to work more than 8 hours a day or 40 hours a week unless the employee is compensated accordingly (for example, overtime pay).
h. **Equal Opportunity**. The Awardee will not discriminate against any employee or applicant for employment because of race, color, religion, sex, or national origin.
i. **Equal Opportunity for Veterans**. The Awardee will not discriminate against any employee or application for employment because he or she is a disabled veteran or veteran of the Vietnam era.

j. **Equal Opportunity for People with Disabilities**. The Awardee will not discriminate against any employee or applicant for employment because he or she is physically or intellectually disabled.
k. **Officials Not to Benefit**. No Federal Government official may benefit personally from the SBIR/STTR Funding Agreement.
l. **Covenant Against Contingent Fees**. No person or agency has been employed to solicit or secure the Funding Agreement upon an understanding for compensation except bona fide employees or commercial agencies maintained by the Awardee for the purpose of securing business.
m. **Gratuities**. The Funding Agreement may be terminated by the Federal Government if any gratuities have been offered to any representative of the Government to secure the award.
n. **Patent Infringement**. The Awardee must report each notice or claim of patent infringement based on the performance of the Funding Agreement.
o. **American Made Equipment and Products**. When purchasing equipment or a product under the SBIR/STTR Funding Agreement, purchase only American-made items whenever possible.

5.3 Agency Recovery Authority and Ongoing Reporting

In accordance with Section 5 of the SBIR and STTR Extension Act of 2022, the DoD will:

a. require an SBC receiving an award under its STTR program to repay all amounts received from the federal agency under the award if,
   1. the SBC makes a material misstatement that the federal agency determines poses a risk to national security; or
   2. there is a change in the SBC's ownership, entity structure, or other substantial change in circumstances that the federal agency determines poses a risk to national security; and

b. require an SBC receiving an award under its STTR program to regularly report to the federal agency and the administration throughout the duration of the award on
   1. any change to a disclosure required under the Disclosures of Foreign Affiliations or Relationships to Foreign Countries form;
   2. any material misstatement made under paragraph (A) above; and
   3. any change described in paragraph (B) above.

5.4 Copyrights

With prior written permission of the contracting officer, the awardee may copyright (consistent with any appropriate national security considerations) material developed with DoD support. DoD receives a royalty-free license for the Federal Government and requires that each publication contain an appropriate acknowledgment and disclaimer statement.

5.5 Patents

SBCs normally may retain the principal worldwide patent rights to any invention developed with U.S. Government support. The government receives a royalty-free license for its use, reserves the right to require the patent holder to license others in certain limited circumstances, and requires that anyone exclusively licensed to sell the invention in the United States must normally manufacture it domestically. 35 U.S.C. § 205 authorizes that the government will not make public any information disclosing a government-supported invention for a period of five years to allow the awardee to pursue a patent. See also Section 6.8, Invention Reporting.

5.6 Invention Reporting

STTR awardees must report inventions to the Service/Component within two months of the inventor's report to the awardee, via either paper documentation submission, including fax, or through the Edison Invention Reporting System at www.iedison.gov for participating agencies.

5.7 Technical Data Rights

Rights in technical data, including software, developed under the terms of any contract resulting from proposals submitted in response to this BAA generally remain with the contractor, except that the U.S. Government obtains a royalty-free license to use such technical data only for government purposes during the period commencing with contract award and ending not less than twenty years after that date. This data should be marked with the restrictive legend specified in DFARS 252.227-7018 Class Deviation 2020-O0007. Upon expiration of the twenty-year restrictive license, the government has government purpose rights in the STTR data. During the license period, the U.S. Government may not release or disclose STTR data to any person other than its support services contractors except: (1) for evaluation purposes; (2) as expressly permitted by the contractor; or (3) a use, release, or disclosure that is necessary for emergency repair or overhaul government-operated items. See DFARS clause 252.227-7018 Class Deviation 2020-O0007 "Rights in Noncommercial Technical Data and Computer Software – Small Business Innovation Research (SBIR) Program."

If a proposing SBC plans to submit assertions in accordance with DFARS 252.227-7017 Class Deviation 2020-O0007, those assertions must be identified and assertion of use, release, or disclosure restriction must be included with your proposal submission, at the end of the technical volume. The contract cannot be awarded until assertions are approved.

5.8 Final Technical Reports - Phase I through Phase III

a. **Content**: A final report is required for each project phase. The reports must contain in detail the project objectives, work performed, results obtained, and estimates of technical feasibility. A completed SF 298, "Report Documentation Page," will be used as the first page of the report. The DoD Service/Component also may require monthly status and progress reports.

b. **SF 298 Form "Report Documentation Page" Preparation**:

   a. If desirable, the proposing SBC may also use language from its Phase II proposal to cover Phase I progress in the final report.

   b. For each unclassified report, the proposing SBC submitting the report should fill in Block 12 (Distribution/Availability Statement) of the SF 298, "Report Documentation Page," with the following statement: "Distribution authorized to U.S. Government only; Proprietary Information, (Date of Determination). Other requests for this document shall be referred to the Service/Component SBIR/STTR Program Office."

      Note: Data developed under an STTR contract is subject to SBIR/STTR Data Rights, under which DFARS 252.227-7018 Class Deviation 2020-O0007 (see Section 5.7, Technical Data Rights) provides protection. The sponsoring DoD activity, after reviewing the proposing SBC's entry in Block 12, has final responsibility for assigning a distribution statement.

      For additional information on distribution statements see the following Defense Technical Information Center (DTIC) website.

   c. Block 14 (Abstract) of the SF 298, "Report Documentation Page" must include as the first sentence, "Report developed under SBIR contract for topic [insert BAA topic number. [Follow with the topic title, if possible.]" The abstract must identify the work's purpose and briefly describe the work conducted, the findings or results, and the effort's potential applications. **Since DoD will publish the abstract, it must not contain any proprietary or classified data, and type "UU" in Block 17.**

d. Block 15 (Subject Terms) of the SF 298 must include the term "STTR Report".

c. **Submission**: In accordance with DFARS 252.235-7011, submit an electronic copy of the approved final scientific or technical report, not a summary, delivered under the contract to the Defense Technical Information Center (DTIC) through the web-based input system at https://discover.dtic.mil/submit-documents/ as required by DoD Instruction 3200.12, DoD Scientific and Technical Information Program (STIP). Include a completed Standard Form (SF) 298, Report Documentation Page, in the document, or complete the web-based SF 298. Additional submission resources are available here.

Delivery will normally be within 30 days after completion of the Phase I technical effort.

Other requirements regarding reports and/or other deliverables submission will be defined in each contract's contract data requirements list (CDRL). Special instructions for submitting CLASSIFIED reports will be defined in the contract's delivery schedule.

DO NOT email classified or controlled unclassified reports, or reports containing STTR Data Rights protected under DFARS 252.227-7018 Class Deviation 2020-O0007.

6.0 PROPOSAL SUBMISSION

6.1 Submission Details

DSIP is the official portal for DoD SBIR/STTR proposal submission. Proposers are required to submit proposals via DSIP; proposals submitted by any other means will be disregarded. Detailed guidance on DSIP proposal submission is found here.

Deadline for Receipt: Complete proposals must be certified and submitted in DSIP no later than the close date listed on the cover page of this BAA. Proposals cannot be submitted in DSIP after the deadline is reached and will not be accepted or evaluated.

The final proposal submission includes successful completion of all firm level forms, all required proposal volumes, and electronic corporate official certification. Although signatures are not required on the electronic forms at the time of submission, the proposal must be certified electronically by the corporate official for it to be considered submitted. If the proposal is selected for negotiation and possible award, the DoD Component program will contact the proposing SBC for signatures prior to award.

Please plan to submit proposals as early as possible to allow time for troubleshooting any possible issues before the BAA close. DSIP Support is unable to assist with submission issues once a deadline has passed and cannot provide submission extensions. DoD is not responsible for missed proposal submission due to system latency.

If the proposal status is "In Progress" or "Ready to Certify" it will NOT be considered submitted, even if all volumes are added prior to the BAA close date. The proposing SBC may modify all proposal volumes prior to the BAA close date.

6.2 Technical Questions

a. **Direct Contact with Topic Authors**. During the pre-release period, the names of the topic authors, their phone numbers and/or email addresses are published with the topic on the DSIP Topics and Topic Q&A page. During this time, proposing SBCs can contact topic authors via

telephone or email to ask technical questions about specific BAA topics. Questions must be limited to specific information related to understanding a particular topic's requirements. Proposing SBCs may not ask for advice or guidance on solution approach and may not submit additional material to the topic author.

If information provided during an exchange with the topic author is deemed necessary for proposal preparation, that information will be made available to all parties through DSIP Topic Q&A.

After the pre-release period, questions must be asked through DSIP Topic Q&A as described below. No further direct contact is allowed between proposing SBCs and topic authors, unless the topic author is responding to a question submitted during the pre-release period.

b. **DSIP Topic Q&A.** Proposing SBCs may submit written questions through DSIP Topic Q&A here, where all questions and answers are posted on a non-attribution basis for public viewing. DSIP Topic Q&A opens on the pre-release date and closes two weeks prior to the topic close date.

Proposing SBCs may use the topic search feature on DSIP to locate a topic of interest. Use the form at the bottom of the topic description, enter and submit the question. Answers are generally posted within seven business days of question submission and also e-mailed directly to the inquirer.

Questions submitted through the DSIP Topic Q&A are limited to technical information focused on understanding a topic's requirements. Any other questions, such as asking for advice or guidance on solution approach, or administrative questions, such as SBIR or STTR program eligibility, technical proposal/cost proposal structure and page count, budget and duration limitations, or proposal due date are not appropriate and will not receive a response; for administrative questions, refer to a topic's Service/Component-specific instructions.

Once the BAA proposal submission closes, no communication of any kind is allowed either with the topic author or through topic Q&A regarding submitted proposals.

Throughout the BAA period, proposing SBCs should frequently monitor DSIP for updates and amendments to the topics and DSIP Topic Q&A for questions and answers.

7.0 Participating Component Instructions & Research Topics

The following section contains all Component-specific proposal preparation instructions and research topics this BAA.

DoD SBIR 25.1 BAA

December 4, 2024: Topics Pre-release
January 8, 2025: Topics Open; DoD begins accepting proposals in DSIP
January 22, 2025: DSIP Topic Q&A closes to new questions at **12:00 p.m. ET**
February 5, 2025: Topics Close; Deadline for receipt of proposals is **12:00 p.m. ET**

Participating Services/Components:

- Department of the Navy (Navy)
- Department of the Air Force (Air Force)
- Defense Health Agency (DHA)

Version 2

DEPARTMENT OF THE NAVY (DoN)
25.A Small Business Technology Transfer (STTR)
Proposal Submission Instructions

IMPORTANT

- **The following instructions apply to STTR topics only:**
  - **N25A-T001 through N25A-T028**
- Information on the 25.1 SBIR and 25.A STTR Topics Workshop can be found at https://navysbir.com/nw25_1.htm.
- Submitting small business concerns are encouraged to thoroughly review the DoD SBIR/STTR Program Broad Agency Announcement (BAA) and register for the DSIP Listserv to remain apprised of important programmatic changes.
  - The DoD Program BAA is located at: https://www.defensesbirsttr.mil/SBIR-STTR/Opportunities/#announcements. Select the tab for the appropriate BAA cycle.
  - Register for the DSIP Listserv at: https://www.dodsbirsttr.mil/submissions/login.
- The information provided in the DoN Proposal Submission Instructions takes precedence over the DoD Instructions posted for this BAA.
- **DoN Phase I Technical Volume (Volume 2) page limit is not to exceed 10 pages.**
- Phase I Technical Volume (Volume 2) and Supporting Documents (Volume 5) templates, specific to DoN topics, are available at https://www.navysbir.com/links_forms.htm.
- The DoN provides notice that Basic Ordering Agreements (BOAs) may be used for Phase I awards, and BOAs or Other Transaction Agreements (OTAs) may be used for Phase II awards.
- This BAA is issued under regulations set forth in Federal Acquisition Regulation (FAR) 35.016 and awards will be made under "other competitive procedures". The policies and procedures of FAR Subpart 15.3 shall not apply to this BAA, except as specifically referenced in it. All procedures are at the sole discretion of the Government as set forth in this BAA. Submission of a proposal in response to this BAA constitutes the express acknowledgement to that effect by the proposing small business concern.

INTRODUCTION

The DoN SBIR/STTR Programs are mission-oriented programs that integrate the needs and requirements of the DoN's Fleet through research and development (R&D) topics that have dual-use potential, but primarily address the needs of the DoN. More information on the programs can be found on the DoN SBIR/STTR website at www.navysbir.com. Additional information on DoN's mission can be found on the DoN website at www.navy.mil.

For questions regarding this BAA, use the information in Table 1 to determine who to contact for what types of questions.

TABLE 1: POINTS OF CONTACT FOR QUESTIONS REGARDING THIS BAA

Type of Question	When	Contact Information
Program and administrative	Always	Navy SBIR/STTR Program Management Office usn.pentagon.cnr-arlington-va.mbx.navy-sbir-sttr@us.navy.mil or appropriate Program Manager listed in Table 2 (below)
Topic-specific technical questions	BAA Pre-release	Technical Point of Contact (TPOC) listed in each topic on the DoD SBIR/STTR Innovation Portal (DSIP). Refer to the Proposal Submission section of the DoD SBIR/STTR Program BAA for details.
	BAA Open	DoD SBIR/STTR Topic Q&A platform (https://www.dodsbirsttr.mil/submissions) Refer to the Proposal Submission section of the DoD SBIR/STTR Program BAA for details.
Electronic submission to the DoD SBIR/STTR Innovation Portal (DSIP)	Always	DSIP Support via email at dodsbirsupport@reisystems.com
Navy-specific BAA instructions and forms	Always	DoN SBIR/STTR Program Management Office usn.pentagon.cnr-arlington-va.mbx.navy-sbir-sttr@us.navy.mil

TABLE 2: DoN SYSTEMS COMMANDS (SYSCOM) SBIR PROGRAM MANAGERS

Topic Numbers	Point of Contact	SYSCOM	Email
N25A-T001 to N25A-T006	Ms. Kristi DePriest	Naval Air Systems Command (NAVAIR)	navair-sbir@us.navy.mil
N25A-T007 to N25A-T012	Mr. Jason Schroepfer	Naval Sea Systems Command (NAVSEA)	NSSC_SBIR.fct@navy.mil
N25A-T013 to N25A-T028	Mr. Steve Sullivan	Office of Naval Research (ONR)	usn.pentagon.cnr-arlington-va.mbx.onr-sbir-sttr@us.navy.mil

PHASE I SUBMISSION INSTRUCTIONS

The following section details requirements for submitting a compliant Phase I Proposal to the DoD SBIR/STTR Programs.

(NOTE: Proposing small business concerns are advised that support contract personnel will be used to carry out administrative functions and may have access to proposals, contract award documents, contract deliverables, and reports. All support contract personnel are bound by appropriate non-disclosure agreements.)

DoD SBIR/STTR Innovation Portal (DSIP). Proposing small business concerns are required to submit proposals via the DoD SBIR/STTR Innovation Portal (DSIP); and follow proposal submission

instructions in the DoD SBIR/STTR Program BAA on the DSIP at https://www.dodsbirsttr.mil/submissions. Proposals submitted by any other means will be disregarded. Proposing small business concerns submitting through DSIP for the first time will be asked to register. It is recommended that small business concerns register as soon as possible upon identification of a proposal opportunity to avoid delays in the proposal submission process. Proposals that are not successfully certified electronically in DSIP by the Corporate Official prior to BAA Close will NOT be considered submitted and will not be evaluated by DoN. Proposals that are encrypted, password protected, or otherwise locked in any portion of the submission will be REJECTED unless specifically directed within the text of the topic to which you are submitting. Please refer to the DoD SBIR/STTR Program BAA for further information.

Proposal Volumes. The following seven volumes are required.

- **Proposal Cover Sheet (Volume 1).** As specified in DoD SBIR/STTR Program BAA.

- **Technical Proposal (Volume 2)**
  - Technical Proposal (Volume 2) must meet the following requirements or the proposal will be REJECTED:
    - Not to exceed 10 pages, regardless of page content
    - Single column format, single-spaced typed lines
    - Standard 8 ½" x 11" paper
    - Page margins one inch on all sides. A header and footer may be included in the one-inch margin.
    - No font size smaller than 10-point
    - Include, within the 10-page limit of Volume 2, an Option that furthers the effort in preparation for Phase II and will bridge the funding gap between the end of Phase I and the start of Phase II. Tasks for both the Phase I Base and the Phase I Option must be clearly identified. Phase I Options are exercised upon selection for Phase II.
    - Work proposed for the Phase I Base must be exactly six (6) months.
    - Work proposed for the Phase I Option must be exactly six (6) months.

  - Additional information:
    - A Phase I proposal template specific to DoN to meet Phase I requirements is available at https://navysbir.com/links_forms.htm.
    - A font size smaller than 10-point is allowable for headers, footers, imbedded tables, figures, images, or graphics that include text. However, proposing small business concerns are cautioned that if the text is too small to be legible it will not be evaluated.

- **Cost Volume (Volume 3).**
  - Cost Volume (Volume 3) must meet the following requirements or the proposal will be REJECTED:
    - The Phase I Base amount must not exceed $140,000.
    - Phase I Option amount must not exceed $100,000.
    - Costs for the Base and Option must be separated and clearly identified on the Proposal Cover Sheet (Volume 1) and in Volume 3.
    - For Phase I a minimum of 40% of the work is performed by the proposing small business concern, and a minimum of 30% of the work is performed by the single research institution. The percentage of work requirement must be met in the Base costs as well as in the Option costs. The percentage of work is measured by both direct and indirect costs. To calculate the minimum percentage of effort for the proposing small business concern the sum of all direct

and indirect costs attributable to the proposing small business concern represent the numerator and the total cost of the proposal (i.e., Total Cost before Profit Rate is applied) is the denominator. The single research institution percentage is calculated by taking the sum of all costs attributable to the single research institution (identified as Total Subcontractor Costs (TSC) 1 in DSIP Cost Volume) as the numerator and the total cost of the proposal (i.e., Total Cost before Profit Rate is applied) as the denominator.

- Proposing Small Business Concern Costs (included in numerator for calculation of the small business concern):
  - Total Direct Labor (TDL)
  - Total Direct Material Costs (TDM)
  - Total Direct Supplies Costs (TDS)
  - Total Direct Equipment Costs (TDE)
  - Total Direct Travel Costs (TDT)
  - Total Other Direct Costs (TODC)
  - General & Administrative Cost (G&A)

  NOTE: G&A, if proposed, will only be attributed to the proposing small business concern.
- Research Institution (numerator for Research Institution calculation):
  - Total Subcontractor Costs (TSC) 1
- Total Cost (i.e., Total Cost before Profit Rate is applied, denominator for either calculation)

- **Cost Sharing: Cost sharing is not accepted on DoN Phase I proposals. If a value above or below $0.00 is entered in the Cost Sharing field the proposal will be deemed non-compliant and will be REJECTED by DoN.**

- Additional information:
  - Provide sufficient detail for subcontractor, material, and travel costs. Subcontractor costs must be detailed to the same level as the prime contractor. Material costs must include a listing of items and cost per item. Travel costs must include the purpose of the trip, number of trips, location, length of trip, and number of personnel.
  - Inclusion of cost estimates for travel to the sponsoring SYSCOM's facility for one day of meetings is recommended for all proposals.
  - The "Additional Cost Information" of Supporting Documents (Volume 5) may be used to provide supporting cost details for Volume 3. When a proposal is selected for award, be prepared to submit further documentation to the SYSCOM Contracting Officer to substantiate costs (e.g., an explanation of cost estimates for equipment, materials, and consultants or subcontractors).

- **Company Commercialization Report (Volume 4)**. DoD collects and uses Volume 4 and DSIP requires Volume 4 for proposal submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoD SBIR/STTR Program BAA for details to ensure compliance with DSIP Volume 4 requirements.

- **Supporting Documents (Volume 5).** Volume 5 is for the submission of administrative material that DoN may or will require to process a proposal, if selected, for contract award.

  All proposing small business concerns must review and submit the following items, as applicable:

  - Proposing small business concerns may include the following administrative materials in Supporting Documents (Volume 5); a template is available at

https://navysbir.com/links_forms.htm to provide guidance on optional material the proposing small business concern may want to include in Volume 5:
    - Additional Cost Information to support the Cost Volume (Volume 3)
    - SBIR/STTR Funding Agreement Certification
    - Data Rights Assertion
    - Allocation of Rights between Prime and Subcontractor
    - Disclosure of Information (DFARS 252.204-7000)
    - Prior, Current, or Pending Support of Similar Proposals or Awards
    - Foreign Citizens
  - Details of Request for Discretionary Technical and Business Assistance (TABA), if proposed, is to be included under the Additional Cost Information section if using the DoN Supporting Documents template.
  - Do not include documents or information to substantiate the Technical Volume (Volume 2) (e.g., resumes, test data, technical reports, or publications). Such documents or information will not be considered.
  - A font size smaller than 10-point is allowable for documents in Volume 5; however, proposing small business concerns are cautioned that the text may be unreadable.

- **Fraud, Waste and Abuse Training Certification (Volume 6)**. DoD requires Volume 6 for submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoD SBIR/STTR Program BAA for details.

- **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)**. In accordance with Section 4 of the SBIR and STTR Extension Act of 2022 and the SBA SBIR/STTR Policy Directive, the DoD will review all proposals submitted in response to this BAA to assess security risks presented by small business concerns seeking a Federally funded award. Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoD SBIR/STTR Program BAA for details.

PHASE I EVALUATION AND SELECTION

The following section details how the DoN SBIR/STTR Programs will evaluate Phase I proposals.

Proposals meeting DSIP submission requirements will be forwarded to the DoN SBIR/STTR Programs. Prior to evaluation, all proposals will undergo a compliance review to verify compliance with DoD and DoN SBIR/STTR proposal eligibility requirements. Proposals not meeting submission requirements will be REJECTED and not evaluated.

- **Proposal Cover Sheet (Volume 1).** The Proposal Cover Sheet (Volume 1) will undergo a compliance review to verify the proposing small business concern has met eligibility requirements and followed the instructions for the Proposal Cover Sheet as specified in the DoD SBIR/STTR Program BAA.

- **Technical Volume (Volume 2).** The DoN will evaluate and select Phase I proposals using the evaluation criteria specified in the Method of Selection and Evaluation Criteria section of the DoD SBIR/STTR Program BAA, with technical merit being most important, followed by qualifications of key personnel and commercialization potential of equal importance. The information considered for this decision will come from Volume 2. This is not a FAR Part 15 evaluation and proposals will not be compared to one another. Cost is not an evaluation criterion and will not be considered during the

evaluation process; the DoN will only do a compliance review of Volume 3. Due to limited funding, the DoN reserves the right to limit the number of awards under any topic.

The Technical Volume (Volume 2) will undergo a compliance review (prior to evaluation) to verify the proposing small business concern has met the following requirements or the proposal will be REJECTED:

- Not to exceed 10 pages, regardless of page content
- Single column format, single-spaced typed lines
- Standard 8 ½" x 11" paper
- Page margins one inch on all sides. A header and footer may be included in the one-inch margin.
- No font size smaller than 10-point, except as permitted in the instructions above.
- Include, within the 10-page limit of Volume 2, an Option that furthers the effort in preparation for Phase II and will bridge the funding gap between the end of Phase I and the start of Phase II. Tasks for both the Phase I Base and the Phase I Option must be clearly identified.
- Work proposed for the Phase I Base must be exactly six (6) months.
- Work proposed for the Phase I Option must be exactly six (6) months.

- **Cost Volume (Volume 3).** The Cost Volume (Volume 3) will not be considered in the selection process and will only undergo a compliance review to verify the proposing small business concern has met the following requirements or the proposal will be REJECTED:
  - Must not exceed values for the Base ($140,000) and Option ($100,000).
  - Must meet minimum percentage of work; 40% of the work is performed by the proposing small business concern, and a minimum of 30% of the work is performed by the single research institution. The percentage of work requirement must be met in the Base costs as well as in the Option costs.
  - **Cost Sharing: Cost sharing is not accepted on DoN Phase I proposals. If a value above or below $0.00 is entered in the Cost Sharing field the proposal will be deemed non-compliant and will be REJECTED by DoN.**

- **Company Commercialization Report (Volume 4).** The CCR (Volume 4) will not be evaluated by the DoN nor will it be considered in the award decision. However, all proposing small business concerns must refer to the DoD SBIR/STTR Program BAA to ensure compliance with DSIP Volume 4 requirements.

- **Supporting Documents (Volume 5).** Supporting Documents (Volume 5) will not be considered in the selection process and will only undergo a compliance review to ensure the proposing small business concern has included items in accordance with the PHASE I SUBMISSION INSTRUCTIONS section above.

- **Fraud, Waste, and Abuse Training Certificate (Volume 6).** Not evaluated.

- **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)**. Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7) will be assessed as part of the Due Diligence Program to Assess Security Risks. Refer to the DoD SBIR/STTR Program BAA to ensure compliance with Volume 7 requirements.

ADDITIONAL SUBMISSION CONSIDERATIONS

This section details additional items for proposing small business concerns to consider during proposal preparation and submission process.

Due Diligence Program to Assess Security Risks. The SBIR and STTR Extension Act of 2022 (Pub. L. 117-183) requires the Department of Defense, in coordination with the Small Business Administration, to establish and implement a due diligence program to assess security risks presented by small business concerns seeking a Federally funded award. Please review the Certifications and Registrations section of the DoD SBIR/STTR Program BAA for details on how DoD will assess security risks presented by small business concerns. The Due Diligence Program to Assess Security Risks will be implemented for all Phases.

Discretionary Technical and Business Assistance (TABA). The SBIR and STTR Policy Directive section 9(b) allows the DoN to provide TABA (formerly referred to as DTA) to its awardees. The purpose of TABA is to assist awardees in making better technical decisions on SBIR/STTR projects; solving technical problems that arise during SBIR/STTR projects; minimizing technical risks associated with SBIR/STTR projects; and commercializing the SBIR/STTR product or process, including intellectual property protections. Proposing small business concerns may request, in their Phase I Cost Volume (Volume 3) and Phase II Cost Volume, to contract these services themselves through one or more TABA providers in an amount not to exceed the values specified below. The Phase I TABA amount is up to $6,500 and is in addition to the award amount. The Phase II TABA amount is up to $25,000 per award, is to be included as part of the award amount, and is limited by the established award values for Phase II by the SYSCOM (i.e., within the $2,000,000 or lower limit specified by the SYSCOM). As with Phase I, the amount proposed for TABA cannot include any profit/fee by the proposing small business concern and must be inclusive of all applicable indirect costs. TABA cannot be used in the calculation of general and administrative expenses (G&A) for the STTR proposing small business concern. A Phase II project may receive up to an additional $25,000 for TABA as part of one additional (sequential) Phase II award under the project for a total TABA award of up to $50,000 per project. A small business concern receiving TABA will be required to submit a report detailing the results and benefits of the service received. This TABA report will be due at the time of submission of the final report.

Request for TABA funding will be reviewed by the DoN SBIR/STTR Program Management Office.

If the TABA request does not include the following items the TABA request will be denied.

- TABA provider(s) (firm name)
- TABA provider(s) point of contact, email address, and phone number
- An explanation of why the TABA provider(s) is uniquely qualified to provide the service
- Tasks the TABA provider(s) will perform (to include the purpose and objective of the assistance)
- Total TABA provider(s) cost, number of hours, and labor rates (average/blended rate is acceptable)

TABA must <u>NOT</u>:

- Be subject to any indirect costs, profit, or fee by the STTR proposing small business concern
- Propose a TABA provider that is the STTR proposing small business concern
- Propose a TABA provider that is an affiliate of the STTR proposing small business concern
- Propose a TABA provider that is an investor of the STTR proposing small business concern
- Propose a TABA provider that is a subcontractor or consultant of the requesting small business concern otherwise required as part of the paid portion of the research effort (e.g., research partner, consultant, tester, or administrative service provider)

TABA requests must be included in the proposal as follows:

- Phase I:

  - Online DoD Cost Volume (Volume 3) – the value of the TABA request.
  - Supporting Documents (Volume 5) – a detailed request for TABA (as specified above) specifically identified as "TABA" in the section titled Additional Cost Information when using the DoN Supporting Documents template.
- Phase II:
  - DoN Phase II Cost Volume (provided by the DoN SYSCOM) - the value of the TABA request.
  - Supporting Documents (Volume 5) – a detailed request for TABA (as specified above) specifically identified as "TABA" in the section titled Additional Cost Information when using the DoN Supporting Documents template.

Proposed values for TABA must NOT exceed:
- Phase I: A total of $6,500
- Phase II: A total of $25,000 per award, not to exceed $50,000 per Phase II project

If a proposing small business concern requests and is awarded TABA in a Phase II contract, the proposing small business concern will be eliminated from participating in the Navy SBIR Transition Program (STP), the DoN Forum for SBIR/STTR Transition (FST), and any other Phase II assistance the DoN provides directly to awardees.

All Phase II awardees not receiving funds for TABA in their awards must participate in the virtual Navy STP Kickoff during the first or second year of the Phase II contract. While there are no travel costs associated with this virtual event, Phase II awardees should budget time of up to a full day to participate. STP information can be obtained at: https://navystp.com. Phase II awardees will be contacted separately regarding this program.

Disclosure of Information (DFARS 252.204-7000). In order to eliminate the requirements for prior approval of public disclosure of information (in accordance with DFARS 252.204-7000) under this award, the proposing small business concern shall identify and describe all fundamental research to be performed under its proposal, including subcontracted work, with sufficient specificity to demonstrate that the work qualifies as fundamental research. Fundamental research means basic and applied research in science and engineering, the results of which ordinarily are published and shared broadly within the scientific community, as distinguished from proprietary research and from industrial development, design, production, and product utilization, the results of which ordinarily are restricted for proprietary or national security reasons (defined by National Security Decision Directive 189). A small business concern whose proposed work will include fundamental research and requests to eliminate the requirement for prior approval of public disclosure of information must complete the DoN Fundamental Research Disclosure and upload as a separate PDF file to the Supporting Documents (Volume 5) in DSIP as part of their proposal submission. The DoN Fundamental Research Disclosure is available on https://navysbir.com/links_forms.htm and includes instructions on how to complete and upload the completed Disclosure. Simply identifying fundamental research in the Disclosure does **NOT** constitute acceptance of the exclusion. All exclusions will be reviewed and, if approved by the Government Contracting Officer, noted in the contract.

Partnering Research Institutions. The Naval Academy, the Naval Postgraduate School, and other military academies are Government organizations but qualify as partnering research institutions. However, DoN laboratories DO NOT qualify as research partners. DoN laboratories may be proposed only IN ADDITION TO the partnering research institution.

System for Award Management (SAM). It is strongly encouraged that proposing small business concerns register in SAM, https://sam.gov, by the Close date of this BAA, or verify their registrations are

still active and will not expire within 60 days of BAA Close. Additionally, proposing small business concerns should confirm that they are registered to receive contracts (not just grants) and the address in SAM matches the address on the proposal. A small business concern selected for an award MUST have an active SAM registration at the time of award or they will be considered ineligible.

Notice of NIST SP 800-171 Assessment Database Requirement. The purpose of the National Institute of Standards and Technology (NIST) Special Publication (SP) 800-171 is to protect Controlled Unclassified Information (CUI) in Nonfederal Systems and Organizations. As prescribed by DFARS 252.204-7019, in order to be considered for award, a small business concern is required to implement NIST SP 800-171 and shall have a current assessment uploaded to the Supplier Performance Risk System (SPRS) which provides storage and retrieval capabilities for this assessment. The platform Procurement Integrated Enterprise Environment (PIEE) will be used for secure login and verification to access SPRS. For brief instructions on NIST SP 800-171 assessment, SPRS, and PIEE, please visit https://www.sprs.csd.disa.mil/nistsp.htm. For in-depth tutorials on these items please visit https://www.sprs.csd.disa.mil/webtrain.htm.

Human Subjects, Animal Testing, and Recombinant DNA. Due to the short timeframe associated with Phase I of the SBIR/STTR process, the DoN does **not** recommend the submission of Phase I proposals that require the use of Human Subjects, Animal Testing, or Recombinant DNA. For example, the ability to obtain Institutional Review Board (IRB) approval for proposals that involve human subjects can take 6-12 months, and that lengthy process can be at odds with the Phase I goal for time-to-award. Before the DoN makes any award that involves an IRB or similar approval requirement, the proposing small business concerns must demonstrate compliance with relevant regulatory approval requirements that pertain to proposals involving human, animal, or recombinant DNA protocols. It will not impact the DoN's evaluation, but requiring IRB approval may delay the start time of the Phase I award and if approvals are not obtained within two months of notification of selection, the decision to award may be terminated. If the use of human, animal, and recombinant DNA is included under a Phase I or Phase II proposal, please carefully review the requirements at: https://www.nre.navy.mil/work-with-us/how-to-apply/compliance-and-protections/research-protections. This webpage provides guidance and lists approvals that may be required before contract/work can begin.

Government Furnished Equipment (GFE). Due to the typical lengthy time for approval to obtain GFE, it is recommended that GFE is not proposed as part of the Phase I proposal. If GFE is proposed, and it is determined during the proposal evaluation process to be unavailable, proposed GFE may be considered a weakness in the technical merit of the proposal.

International Traffic in Arms Regulation (ITAR). For topics indicating ITAR restrictions or the potential for classified work, limitations are generally placed on disclosure of information involving topics of a classified nature or those involving export control restrictions, which may curtail or preclude the involvement of universities and certain non-profit institutions beyond the basic research level. Small businesses must structure their proposals to clearly identify the work that will be performed that is of a basic research nature and how it can be segregated from work that falls under the classification and export control restrictions. As a result, information must also be provided on how efforts can be performed in later phases if the university/research institution is the source of critical knowledge, effort, or infrastructure (facilities and equipment).

SELECTION, AWARD, AND POST-AWARD INFORMATION

Notifications. Email notifications for proposal receipt (approximately one week after the Phase I BAA Close) and selection are sent based on the information received on the proposal Cover Sheet (Volume 1). Consequently, the e-mail address on the proposal Cover Sheet must be correct.

Debriefs. Requests for a debrief must be made within 15 calendar days of select/non-select notification via email as specified in the select/non-select notification. Please note debriefs are typically provided in writing via email to the Corporate Official identified in the proposal of the proposing small business concern within 60 days of receipt of the request. Requests for oral debriefs may not be accommodated. If contact information for the Corporate Official has changed since proposal submission, a notice of the change on company letterhead signed by the Corporate Official must accompany the debrief request.

Protests. Interested parties have the right to protest in accordance with the procedures in FAR Subpart 33.1.

Pre-award agency protests related to the terms of the BAA must be served to: osd.ncr.ousd-r-e.mbx.SBIR-STTR-Protest@mail.mil. A copy of a pre-award Government Accountability Office (GAO) protest must also be filed with the aforementioned email address within one day of filing with the GAO.

Protests related to a selection or award decision should be filed with the appropriate Contracting Officer for an Agency Level Protest or with the GAO. Contracting Officer contact information for specific DoN Topics may be obtained from the DoN SYSCOM Program Managers listed in Table 2 above. For protests filed with the GAO, a copy of the protest must be submitted to the appropriate DoN SYSCOM Program Manager and the appropriate Contracting Officer within one day of filing with the GAO.

Awards. Due to limited funding, the DoN reserves the right to limit the number of awards under any topic. Any notification received from the DoN that indicates the proposal has been selected does not ultimately guarantee an award will be made. This notification indicates that the proposal has been selected in accordance with the evaluation criteria and has been sent to the Contracting Officer to conduct compliance review of Volume 3, to confirm eligibility of the proposing small business concern, and to take other relevant steps necessary prior to making an award.

Contract Types. The DoN typically awards a Firm Fixed Price (FFP) contract or a small purchase agreement for Phase I. In addition to the negotiated contract award types listed in the section of the DoD SBIR/STTR Program BAA titled Additional Considerations, for Phase II awards the DoN may (under appropriate circumstances) propose the use of an Other Transaction Agreement (OTA) as specified in 10 U.S.C. 4021/10 U.S.C. 4022 and related implementing policies and regulations. The DoN may choose to use a Basic Ordering Agreement (BOA) for Phase I and Phase II awards.

Funding Limitations. In accordance with the SBIR and STTR Policy Directive section 4(b)(5), there is a limit of one sequential Phase II award per small business concern per topic. Additionally, to adjust for inflation DoN has raised Phase I and Phase II award amounts. The maximum Phase I proposal/award amount including all options (less TABA) is $240,000. The Phase I Base amount must not exceed $140,000 and the Phase I Option amount must not exceed $100,000. The maximum Phase II proposal/award amount including all options (including TABA) is $2,000,000 (unless non-SBIR/STTR funding is being added). Individual SYSCOMs may award amounts, including Base and all Options, of less than $2,000,000 based on available funding. The structure of the Phase II proposal/award, including maximum amounts as well as breakdown between Base and Option amounts will be provided to all Phase I awardees either in their Phase I award or a minimum of 30 days prior to the due date for submission of their Initial Phase II proposal.

Contract Deliverables. Contract deliverables for Phase I are typically a kick-off brief, progress reports, and a final report. Required contract deliverables (as stated in the contract) must be uploaded to https://www.navysbirprogram.com/navydeliverables/.

Payments. The DoN makes three payments from the start of the Phase I Base period, and from the start of the Phase I Option period, if exercised. Payment amounts represent a set percentage of the Base or Option value as follows:

Days from Start of Base Award or Option	Payment Amount
15 Days	50% of Total Base or Option
90 Days	35% of Total Base or Option
180 Days	15% of Total Base or Option

Transfer Between SBIR and STTR Programs. Section 4(b)(1)(i) of the SBIR and STTR Policy Directive provides that, at the agency's discretion, projects awarded a Phase I under a BAA for SBIR may transition in Phase II to STTR and vice versa.

PHASE II GUIDELINES

Evaluation and Selection. All Phase I awardees may submit an **Initial** Phase II proposal for evaluation and selection. The evaluation criteria for Phase II is the same as Phase I (as stated in the BAA). The Phase I Final Report and Initial Phase II Proposal will be used to evaluate the small business concern's potential to progress to a workable prototype in Phase II and transition the technology to Phase III. Details on the due date, content, and submission requirements of the Initial Phase II Proposal will be provided by the awarding SYSCOM either in the Phase I contract or by subsequent notification.

Awards. The DoN typically awards a Cost Plus Fixed Fee contract for Phase II; but, may consider other types of agreement vehicles. Phase II awards can be structured in a way that allows for increased funding levels based on the project's transition potential. To accelerate the transition of SBIR/STTR-funded technologies to Phase III, especially those that lead to Programs of Record and fielded systems, the Commercialization Readiness Program was authorized and created as part of section 5122 of the National Defense Authorization Act of Fiscal Year 2012. The statute set-aside is 1% of the available SBIR/STTR funding to be used for administrative support to accelerate transition of SBIR/STTR-developed technologies and provide non-financial resources for the small business concerns (e.g., the Navy STP).

PHASE III GUIDELINES

A Phase III SBIR/STTR award is any work that derives from, extends, or completes effort(s) performed under prior SBIR/STTR funding agreements, but is funded by sources other than the SBIR/STTR programs. This covers any contract, grant, or agreement issued as a follow-on Phase III award or any contract, grant, or agreement award issued as a result of a competitive process where the awardee was an SBIR/STTR firm that developed the technology as a result of a Phase I or Phase II award. The DoN will give Phase III status to any award that falls within the above-mentioned description. Consequently, DoN will assign SBIR/STTR Data Rights to any noncommercial technical data and noncommercial computer software delivered in Phase III that were developed under SBIR/STTR Phase I/II effort(s). Government prime contractors and their subcontractors must follow the same guidelines as above and ensure that companies operating on behalf of the DoN protect the rights of the SBIR/STTR firm.

Navy STTR 25.A Topic Index

N25A-T001	High-Fidelity Computational Modeling of Fluid-Thermal-Structural Interactions in Hypersonic Air-Breathing Systems
N25A-T002	Lattice-Boltzmann Real-Time Dynamic Interface (LBRDI)
N25A-T003	Metamaterial Lens Additive Manufacturing
N25A-T004	Material System Design Tool to Enable Functionally Graded Materials
N25A-T005	Enhancing Naval Aviation Training Through Advanced Psychological and Physiological Techniques
N25A-T006	Reducing the Life Cycle Cost of Automatic Target Recognition Application
N25A-T007	Automated Data Synchronization Across Multiple Databases
N25A-T008	Modeling of Magnetic Properties of High Strength Steels
N25A-T009	Medium Voltage Alternating Current System Grounding Circuit
N25A-T010	Miniaturized Low Frequency Electric Field Sensing for Underwater Applications
N25A-T011	Marine Biofouling Mitigation and Innovative Broad Band Hydrophobic Glass Development
N25A-T012	Medium Voltage Direct Current Protection Relays and Associated Sensors
N25A-T013	Vitrimer Patches to Repair Composite Parts with a Complex Shape in a Contested Environment
N25A-T014	Removable Smart Tooling for High Temperature Composite Fabrication
N25A-T015	Lateral Shear and Strain Sensor for the Ocean Environment
N25A-T016	Persistent Airborne Counter Unmanned Aerial System Surveillance
N25A-T017	Mattresses for Improved Sleep Quality (MISQ)
N25A-T018	High-Power Microwave Phase Shifter
N25A-T019	Passive Automatic Dependent Surveillance-Broadcast Validation to Support Collision Avoidance During Missions Requiring Emissions Control
N25A-T020	Generation of Infrared Synthetic Data for System Training and Testing
N25A-T021	Non-Wearable and Off-Engine Jet Noise Reduction
N25A-T022	Photonic Bump Bonds
N25A-T023	Robotic Manipulation Learning for Repair of Marine Diesel Engines
N25A-T024	In-situ Seabed Physical Properties Probe (ISPPP)
N25A-T025	Expendable Air-sea Profiling Observations in Hazardous Weather Conditions

Version 2

N25A-T026	Automatic Readiness and Training Systems Enhanced with Artificial Intelligence, Machine Learning, and Optimization
N25A-T027	Dynamic Thermal Management Suit
N25A-T028	Rapid Cryogenic Cooling of Superconducting Systems

Version 2

N25A-T001 TITLE: High-Fidelity Computational Modeling of Fluid-Thermal-Structural Interactions in Hypersonic Air-Breathing Systems

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Hypersonics

OBJECTIVE: Develop robust, accurate, and efficient computational methods for predicting the coupled fluid-thermal-structural response of an air-breathing hypersonic system at moderate hypersonic speeds.

DESCRIPTION: Air-breathing hypersonic systems are typically powered by ramjet or scramjet engines. At hypersonic speeds (i.e., Mach 5+), compression of the air stream is performed without any turbomachinery simply by slowing the oncoming flow through an isentropic (i.e., minimal loss) compression and/or a series of oblique shock waves. For scramjet engines, designing an inlet, which provides uniform flow to the combustor with minimal aerodynamic losses, is critical to obtaining high-engine performance (e.g., high-inlet efficiency, high-combustion efficiency, thrust). In addition to engine performance, the design of scramjet inlets significantly influences engine operability. At off-design conditions, the contraction ratio of the inlet may cause strong shock waves to develop that can prohibit mass flow through the engine and ultimately expel the flow out the front of the engine leading to engine "unstart". Unstart can be catastrophic to the performance of a hypersonic system. Once an engine has unstarted, it is very difficult, if not impossible, to restart the engine in flight. Furthermore, due to the dynamics of the unstart process, maintaining control of the vehicle can be challenging and often leads to system failure.

In the hypersonic environment, the aerodynamics through the engine become increasingly coupled with the structural and thermal response of the vehicle. Vehicle deformation, both structural deformation and thermal deformation, can have significant impacts on the performance and operability of the propulsion system. To date, conventional design and analysis methodologies utilize isolated assessments of each physics domain (e.g., aerodynamics, structures, and thermodynamics) and do not properly capture the coupled effects between physics domains (e.g., fluid-thermal-structural interaction). If multiphysics interactions are not properly captured and accounted for during the design phase, the resulting hypersonic system may have reduced mission performance or may experience an unstart event leading to system failure.

Improvements in high-fidelity modeling and simulation of the fluid-thermal-structural response of a hypersonic vehicle are desired. The computational methods that are developed should address the impacts of coupling strategies between physics domains on the resultant simulation time and modeling fidelity for air-breathing hypersonic vehicles at moderate hypersonic speeds. The methods and tools shall be validated against experimental data and be capable of accurately accessing engine, air vehicle, and material/structural performance during flight. Any developed methods should be capable of transition and integration into government toolsets.

PHASE I: Develop and implement robust multiphysics coupling methodologies to simulate the coupled fluid-thermal-structural response of a body at moderate hypersonic speeds. Review the robustness, computational cost, and accuracy of coupling strategies for high-fidelity multiphysics simulation. Coordinate with government POCs on interface design to ensure the developed methodology can transition into government toolsets (e.g., CREATE-AV Kestrel). Validate the coupled multiphysics simulation capability against existing experimental data where the coupling of either fluid-structural, fluid-thermal, or fluid-thermal-structural interaction is captured. Assess the accuracy and computational cost of various multiphysics coupling strategies relative to conventional, single-physics analyses. The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Demonstrate the coupled multiphysics simulation capability for a Navy-relevant, air-breathing, hypersonic vehicle. Assess the design impacts and improvements realized by the coupled multiphysics simulation capability relative to existing approaches. Further enhance the robustness, accuracy, and efficiency of the methodology. Continue coordination with government POCs to ensure an effective transition to government toolsets. Any developed methodologies should be able to execute on conventional CPU-based high-performance computing (HPC) hardware. Deliver prototype software tools on HPC, and document the mathematical theory, assumptions, and guidance for tool execution. Conduct a training session for potential government users on the new capability.

PHASE III DUAL USE APPLICATIONS: Verification and validation (V&V) of the new methods based on available test data. Methods should be updated based on the V&V effort. Additional analyses should be performed on a Navy relevant configuration. Developed algorithms and code are sufficiently validated to transition to HPCMP CREATE-AV for integration.

With the push for commercial aircraft operating at hypersonic speeds now part of the national discussion, the tools and methods developed under this STTR topic will have utility to the design and development of future commercial hypersonic and reusable space access platforms. Additionally, developments from this work can be applicable to computational analysis of supersonic and subsonic reacting flows.

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KEYWORDS: Hypersonic; Fluid-Thermal-Structural Interaction; Computational Aerothermodynamics; Multidisciplinary Physics-Based Modeling; Digital Engineering; Air-Breathing Hypersonic Vehicle

N25A-T002 TITLE: Lattice-Boltzmann Real-Time Dynamic Interface (LBRDI)

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software

OBJECTIVE: Develop an innovative hybrid hardware/software system, based on Lattice-Boltzmann aerodynamic modeling techniques, capable of accurately modeling interactional aerodynamic effects on rotary-wing vehicles in the shipboard environment that is suitable for real-time, blade element-based rotary-wing aircraft flight simulations of urban and shipboard launch and recovery operations.

DESCRIPTION: Rotorcraft operations near obstacles (e.g., buildings, revetments, and ships) are characterized by complex aerodynamic interactions between the flow field created by the passage of air over the obstacle (airwake) and the flow field created by the air vehicle (rotor wake). Typically, real-time piloted simulations use simplified aerodynamic models in order to meet real-time execution requirements. The effects of obstacle proximity and unsteady, non-uniform wind flow (airwake) are accounted for using separate, precomputed models that are superimposed on the rotor inflow model. While this approach works well in many situations, important interactional aerodynamic effects that occur when a rotorcraft is operating in close proximity to obstacles are neglected. This compromise in fidelity is a barrier to using modeling and simulation for launch and recovery envelope development or for pilot shipboard certification. Coupled rotor/obstacle aerodynamics can significantly affect handling qualities; therefore, modeling these effects accurately is essential for predictive ship launch and recovery, and urban environment flight simulations. Interactional aerodynamics can be modeled using high fidelity Navier-Stokes (N-S) computational fluid dynamics (CFD) methods; however, the computations are significantly slower than real time. Lattice-Boltzmann CFD methods (LBM CFD) offer the possibility of calculating the evolving flow field in the loop with the piloted simulation because the equations set is reduced (as compared to N-S), and the algorithms are highly scalable on GPU computer hardware.

This STTR topic seeks to leverage ongoing research related to modeling-coupled aircraft/ship environments with LBM CFD methods to develop and demonstrate an aerodynamic coupling “compute on the fly” flight simulator. A combined hardware/software LBM CFD capability that captures average and standard deviation of rotor thrust to within 90% of N-S CFD and executes at speeds sufficient to model the evolving physics in a real-time piloted simulation environment is sought. The final system must be demonstrated in the Manned Flight Simulator environment at NAVAIR, NAS Patuxent River, Maryland.

PHASE I: Demonstrate accuracy of helicopter rotor thrust prediction for in-ground-effect hover ladder and for stationary hover behind a ship or ship-like structure for no-wind and with-wind conditions by comparing to GFE data. If the model does not execute fast enough to meet real-time simulation requirements, determine technical feasibility and develop a plan to achieve execution speed requirements. Demonstrate execution of LBM CFD integrated into a flight simulation environment for an (auto) piloted rotary-wing vehicle attempting to hold a hover in the wake of a bluff body obstacle. Create a plan to develop and demonstrate a deployable module. The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Extend the Phase I approach to model shipboard launch and recovery operations for a single-main-rotor and a multiple-main-rotor aircraft as specified by the Navy Technical Point of Contact (TPOC). Demonstrate desktop (auto-piloted) simulations using contractor or university hardware. Develop and implement an approach to interface with the Controls Analysis and Simulation Test Loop Environment (CASTLE) operating environment. Implement the software and models in the NAS Patuxent River Manned Flight Simulator (MFS) CASTLE environment on a GFI GPU cluster at the MFS. Demonstrate integration of hardware/software system in real-time flight simulation for at least one Navy relevant ship/aircraft combination. Accuracy of the results will be judged through comparisons with GFI

experimental and/or high-fidelity, fully coupled, Navier-Stokes CFD simulations. The ship configuration, the aircraft model, and CASTLE will be provided GFI.

PHASE III DUAL USE APPLICATIONS: Enable application for future Navy ship and aircraft configurations by developing methods to expedite new aircraft and ship model development and implementation. Determine and document best practices required to assure accuracy requirements are achieved for new aircraft and ship models.

Developments from this STTR topic are marketable to both commercial and private sectors to improve fidelity of helicopter flight simulators for shipboard, urban environments, and mountainous terrain.

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KEYWORDS: Lattice-Boltzmann Methods; LBM; Rotorcraft; Shipboard; Computational Fluid Dynamics; CFD; Flight Simulation; Coupled Aerodynamics; Real-Time; Dynamic Interface; DI; Launch and Recovery Envelopes; LRE

N25A-T003 TITLE: Metamaterial Lens Additive Manufacturing

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials;Space Technology;Sustainment

OBJECTIVE: Develop and demonstrate additive manufacturing (AM) processes to produce metamaterial lenses process in Middle Wavelength Infrared (MWIR) (3–5 µm) or Long Wavelength Infrared (LWIR) (8–12 µm) bands derived from full-wave electrodynamic simulation.

DESCRIPTION: Meta-optics present several benefits to modern optical technologies. First, they allow for extreme miniaturization of optical components, reduced size, weight, and power (SWaP) for optical systems. Second, they enable smart functions that are not readily available with conventional optics, such as integrating multiple functions in a single optical element. Third, the nanoscale volume of the resonators offers fast and real-time reconfigurability of optical functions. Most importantly, meta-optic provides a means for the Navy to develop optical components in the MWIR and LWIR, which is not accessible by convectional glass.

With all of the benefits of meta-optics highlighted above, it is a challenge to produce meta-optical components. The current practice of producing meta-optics relies on simulations, creating a mask, using photo-lithography process, and then etching away the final product. This process is time consuming and has significant cost associated with it. Additionally, the small business awardee is relegated to materials that are complementary metal-oxide semiconductor (CMOS) friendly. If there is an error in the design or fabrication cycle the small business awardee must start the whole process from the beginning.
To mitigate the exhaustive development and manufacturing cycle of a meta-optical component, the Navy is seeking an AM solution that would take a meta-optic design and 3D print the final product. The Navy also seeks to work with any material of choice that supports the MWIR and LWIR spectrums. The Navy also seeks to work with multimaterial meta-optic solutions.

Create a meta-optical design using a n ($10 < n < 3$) sided polygon and demonstrate that the simulation produces a 10% diffraction limited lens of focal length 50 mm. Develop a method of preforming AM and simulate the production of meta-optical lens design using the AM process proposed by the awardee. Using the Phase I simulations produce a 25 mm meta-optical lens that is 10% diffraction limited with a focal length of 50 mm using AM process and equipment proposed by the awardee.

PHASE I: Propose a concept for broadband LWIR metalens AM materials that meets the objectives stated in the Description. The concept will include specific prototypes, by which the proposed AM process technology will be demonstrated. These prototypes will subsequently be produced and used (in Phase II) to verify, by testing and analysis, the efficacy of the proposed AM concept. Demonstrate feasibility by a combination of analysis, modelling, simulation, and evaluation of proposed process steps against established manufacturing methods. The Phase I Option, if exercised, will include the initial process specifications, AM equipment requirements, test specifications, and capabilities description to build a prototype AM facility in Phase II. Determine necessary equipment and manufacturing process to fabricate planar optical elements. Produce initial proof-of-concept lenses using the design and manufacturing processes to show viable path to meeting Navy program requirements. The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Develop and demonstrate a prototype facility for AM of LWIR metalens. In this context, “facility” refers to the combination of equipment, tooling, and process steps required to demonstrate the end-to-end AM capability provided by the proposer, not the actual physical facility. Demonstration of the AM process (or multiple processes) will be accomplished by fabrication and evaluation of the prototype components and materials identified during Phase I. Multiple prototype components and samples are

expected during execution of this Phase as the process development is assumed to be necessarily iterative in nature. However, at the conclusion of Phase II, at least one example of each proposed prototype component or material sample will be delivered to the U.S. Government with no fewer than five total prototype samples delivered overall. Test data will also be delivered with each prototype sample delivered.

PHASE III DUAL USE APPLICATIONS: Provide representative prototype samples using the awardee's AM process to a U.S. Government laboratory and to a U.S. Government depot. Evaluate, by conventional metrology, the innovative optical surface with the flatness as detailed in the Description to ensure the AM process is on par with an optical flatness produced by common practice. Transition the AM process to a U.S. Government laboratory and to a U.S. Government depot.

Perform testing and make improvements to the AM process based upon the U.S. Government's evaluations and results.

Begin producing optical MWIR and LWIR AM components for field testing and use in military systems. Laser manufacturers, camera manufacturers, and imaging technology manufacturers will benefit from this AM technology because they can now specify custom-size optical components with unique MWIR and LWIR transmission profiles that are not currently available with conventional optical processing.

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KEYWORDS: Additive Manufacturing; AM; lens; LWIR; 3D Printing; Material Science; Metallurgy

N25A-T004 TITLE: Material System Design Tool to Enable Functionally Graded Materials

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Advanced Materials;Sustainment

OBJECTIVE: Develop an Integrated Computational Material Engineering (ICME)-based material system design tool for multi-material functionally graded materials (FGMs).

DESCRIPTION: Additive manufacturing (AM) is a process of "printing" a 3D solid object from a digital model and enables the production of geometrically complex components that would be too costly, time-consuming, and/or impossible to produce by other manufacturing processes. However, most of the current AM technologies have not yet begun to achieve their full potential. One example of this is the ability to use multiple materials to fabricate components that are multifunctional. Functionally graded materials (FGMs) are materials whose composition (elements and/or phases) or microstructure (grain size, shape, and/or orientation) varies across a volume. The gradient of material properties results in unique mechanical and functional properties.

FGMs can enable multifunctional material systems with multiple functions. Multi-material FGMs can strategically allocate properties such as strength, density, and wear resistance over a single part. For example, wear and corrosive resistant alloys have been placed on tool steel die/mold for performance enhancements to elongate service life [Refs 1–3]. However, current multi-material AM processes produce discrete and relatively thick layers, which make the creation of a smooth gradient challenging. Additionally, while current multi-material AM processes are sufficient for some material combinations (e.g., deposition of stainless steel onto low-carbon steel), other material combinations have poor compatibility (e.g., Ti-6Al-4V and Inconel 718). A combination of material compatibility and residual stresses from processing can result in cracking and delamination of layers. However, a true FGM with gradual transition from one material to the next can mitigate this issue. The results can be further improved if a buffer material is incorporated into the FGM. For example, Thiriet et al. (2019) demonstrated the potential of fabricating an FGM using Ti-6Al-4V and Inconel 718 by using a molybdenum buffer layer [Ref 5]. Therefore, the buffer material selection will be critical for FGM design and fabrication for certain material combinations.

An analytical toolset to design functionally graded material systems is being sought to improve mechanical performance (strength, ductility, fatigue, fracture toughness) and environmental protection (corrosion, wear/fretting) for structural and subsystem components on U.S. Navy and Marine Corps aircraft. More specifically, FGMs are desired to provide transition zones from a substrate material to a desired coating material to reduce stiffness mismatches between the two materials to promote a longer service life via reducing stresses. The ICME-based analytical tool must address material selection by considering the compatibility of material properties, such as thermal expansion coefficients, as well as the need for buffer materials. Material selection must also consider undesirable high-temperature chemical reactions may result in intermetallics that may degrade the mechanical properties of the material system. The analytical tool must also predict the thicknesses of a transition zone for a FGM and any required buffer materials, and must consider different types of gradients (e.g., continuous and discrete gradients). The material system design must also ensure the fulfillment and optimization of application requirements, such as the overall strengths, fatigue, fracture toughness, weight, corrosion resistance, wear resistance, thermal and electrical conductivity, and so forth, of the fabricated FGM part.

PHASE I: Develop, design, and demonstrate feasibility of an ICME-based material system design tool to facilitate the fabrication of multi-material FGMs once the based material and target material(s) are identified. The focus of Phase I must be to explore the feasibility of a robust methodology for designing the proper FGM material system, including all buffer materials. Such an FGM material system design tool

must be demonstrated by fabricating at least two different FGM samples with buffer materials. FGM samples must be designed and fabricated to satisfy different and complementary/supplementary functionalities, such as strength, fatigue, fracture toughness, density, corrosion, or wear/fretting resistance, and so forth. The produced FGM samples must be evaluated based on microstructure and mechanical properties, such as hardness and tensile and fatigue strength. The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Fully develop and validate the prototype tool with various material systems and applications, including more performance requirements, such as fracture toughness, fatigue resistance, corrosion resistance, and so forth. The design tool must be fully demonstrated with at least five separate FGM examples that are applicable and useful for Navy applications. The developed methodology and tool must be demonstrated for its robustness in handling various applications, including the cases that selected base material and the target material(s) are not non-compatible, thus buffer material(s) is needed. FGM samples must be fabricated and validated through proper testing, including mechanical and environmental testing to validate the multi-functional requirements.

PHASE III DUAL USE APPLICATIONS: Fully develop the FGM design and fabrication tool to produce naval aircraft components that can be integrated into the fleet. Conduct final component level testing to fabricate the FGM parts with geometry and material properties of AM components meeting the Navy's specifications.

The process will be directly applicable to a wide range of AM process applications due to the high amount of usage of AM parts in the commercial/private aerospace industry. The proposed process will allow the industry to apply the benefits of AM technology to many critical aircraft components.

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KEYWORDS: Functionally Graded Materials; Additive Manufacturing; Multi-Material Design; High-Fracture Resistance; Integrated Computational Material Engineering; Corrosion

N25A-T005 TITLE: Enhancing Naval Aviation Training Through Advanced Psychological and Physiological Techniques

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Human-Machine Interfaces;Trusted AI and Autonomy

OBJECTIVE: Research, develop, and validate advanced training techniques and technologies that leverage cutting-edge psychological and physiological insights to improve human performance and psychological well-being within naval aviation training environments.

DESCRIPTION: The Navy seeks a multi-disciplinary approach to the design development of novel manned and unmanned pilot operation training solutions drawing upon insights from high-performance domains such as Drone Racing League's [Refs 1–5, and 7]. This civilian population demonstrates exceptional hand-eye coordination, reaction times, and resilience to simulation sickness. These are highly useful traits and skills in naval aviation. Thorough research and examination of the unique physiological and psychological characteristics of individuals excelling in such environments, this effort aims to identify key factors contributing to their success including, but not limited to binocular vision, psychological resilience, and physiological adaptations [Refs 6 and 8].

To address these objectives, the Navy seeks a detailed investigation into the physiological and psychological profiles that distinguish high performers in extreme stress and high-demand environments in the initial phase. This exploration will not only encompass the intrinsic traits and capabilities that facilitate exceptional performance, but also the training methodologies and environmental factors that contribute to knowledge and skill development. The Navy seeks isolation of core components to identify transferable techniques and principles that can be adapted and integrated into naval aviation training programs. The investigation should include advanced data collection methods, such as eye-tracking technology and physiological monitoring, to gain insights into the cognitive and physical processes underlying peak performance and resilience to motion/simulation sickness.

The second phase of the effort will involve the development of a comprehensive training and intervention program, tailored to the specific needs and challenges of naval aviation training. Efforts should consider leveraging digital technologies, including mobile apps and websites, to deliver customized training modules focused on enhancing cognitive flexibility, stress tolerance and resilience, increased human performance in extreme environments, and physiological readiness [Refs 6 and 8]. The interventions will be designed to improve overall human performance and well-being, with a particular focus on mitigating the effects of motion/simulation sickness, a common challenge in aviation training in a manner that addresses human factors considerations [Ref 9]. Efforts should incorporate evidence-based techniques from existing relevant fields such as positive psychology and sports psychology, adapted for the unique context of naval aviation, to foster a positive training environment that supports mental health and peak performance [Refs 10 and 11].

In parallel, efforts should explore methods to establish a predictive model for individual performance variation, integrating a wide range of data points from physiological metrics to lifestyle factors such as diet, sleep, and physical activity. This model will serve as a foundational tool for customizing training approaches to the individual, optimizing learning and performance outcomes. By identifying key predictors of success and areas of potential improvement, the model will enable trainers to tailor interventions more effectively, ensuring that each aviator receives the support and guidance needed to reach their full potential. Findings should be considered for recommendations for selection and training. Finally, the initiative will explore the commercialization potential of the developed training program, considering its applicability not only within the military aviation context, but also in commercial and civilian aerospace sectors. As space travel and commercial aviation continue to evolve, there is a growing

demand for training solutions that can prepare individuals for the challenges of these environments, from motion sickness to the psychological stresses of long-duration flights. The project's outcomes could thus have broad implications, offering valuable tools and insights for a wide range of high-stress professions and activities. This multi-faceted approach, combining rigorous research with practical application and commercial potential, positions the initiative to make a significant impact on the effectiveness and well-being of naval aviators and beyond.

PHASE I: Research, develop, and validate the taxonomy detailing the knowledge, skills, attitudes, and individual factors that contribute to high resilience against motion/simulation sickness and peak performance in high-stress environments. This Phase will also investigate the feasibility and alignment of existing interventions from sports psychology and positive psychology for adaptation or integration into naval aviation training solutions. Initial considerations should be made for any software/technology solutions to be in compliance with Risk Management Framework policy and requirements. The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Leverage insights from Phase I to develop a comprehensive training program, integrating mobile app or website-based interventions, tailored to enhance the psychological well-being and performance of naval aviators. Include specific training techniques aimed at improving resilience to motion/simulation sickness, enhancing cognitive and physical performance, and promoting psychological health. Considerations should be made for any software/technology solutions to be in compliance with Risk Management Framework policy and requirements.

PHASE III DUAL USE APPLICATIONS: Conduct architecture hardening and engineering documentation to facilitate an Authority to Operate (ATO) approval for software/technology solutions as a standalone package or coordinate with a program to become part of a larger system approved ATO. Transition focused training effectiveness evaluations will offer input for data driven decision making associated with future acquisitions. Continued enhancements to technology may include automated processes for scenario generation, content conversion, sustainment of the system infrastructure, as well as commercialization of the capability to other stakeholders and domains.

Performance in high-stress environments or dynamic individual/team jobs that require increased interaction with remote and/or automated technology continue to increase. As technology adoption in a variety of fields increase, the potential utility of these types of training solutions will increase. In the near-term, commercial aviation offers a highly related domain for consideration of commercialization. The increased feasibility of remote and automated technology integration that impacts the complexity of jobs in homeland security or transportation security may benefit from similar technology. As a variety of commercial applications and domains consider automated human-in-the loop unmanned system integration (e.g., logistics, commercial goods transportation, food delivery, local area transit), considerations will be needed for the selection and training of personnel involved thereby increasing the commercialization applicability of the technology developed under this effort.

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KEYWORDS: Training; Human Performance; Psychological Well-Being; High-Stress Domains; Resilience; Motion/Simulation Sickness

N25A-T006 TITLE: Reducing the Life Cycle Cost of Automatic Target Recognition Application

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Integrated Sensing and Cyber;Sustainment;Trusted AI and Autonomy

OBJECTIVE: Utilize advances in 3D deep learning to dramatically lower the cost of developing and maintaining automatic target recognition template databases.

DESCRIPTION: Generating and/or collecting sufficient imagery to train neural-network-based automated ship recognition applications can be extremely costly. For classification purposes, the ships of the world can be adequately represented with approximately 1500 vessel types. Collecting sufficient imagery in situ for that many vessel types over a very wide range of viewing angles or building 3D models of that many vessels is not feasible from a cost and time perspective. Advances in 3D deep learning are revolutionizing computer vision and machine learning ability to synthesize views of complex scenes using only a limited number of input views. Approaches such as neural radiance fields (NeRF) [Ref 1] and Generative Query Networks [Ref 2] are capable of generating high-quality, photorealistic volumetric scenes from multiview input images by incorporating prior knowledge of the world to significantly reduce the input data requirements. Here we seek utilize these classes of techniques with minimal input imagery, potentially diverse in nature, to generate the 3D geometry of targets of interest such as combatant ships from which 2D projections of the target models from arbitrary views can be produced. These 2D dimensionally accurate projections can then be utilized to train radar and electro-optical/infra-red (EO/IR)-based neural-network target recognition applications or populate databases of classification templates for expert-system and/or hybrid artificial intelligence/machine learning (AI/ML) system automatic target recognition applications.

Consideration should be given to the nature and quantity of acceptable inputs. Candidate inputs include, but are not limited to, multiple view optical images or plan and profile view silhouettes. Limited physical dimension information such as vessel overall length from which the dimensions and physical location of other features on the vessel can be computed. Real-time, full-spectrum rendering is required. Comparing real-time sensor data to a target classification database in a tactical situation, for example, demands this. Inputs may require segmentation or other operations to make them suitable for processing. Once the nature of acceptable inputs are determined, the processing should be automated and include accuracy assessments using suitable additional information not included as input. The approach should be directly extendable to other target types such as aircraft. The goal is to make the entire process from input sourcing through 2D projections as efficient as possible in order to dramatically reduce the life cycle costs of these automatic target recognition applications.

PHASE I: Research, evaluate, and develop the overall system approach and architecture. Identify acceptable input requirements to produce the 3D representations and 2D projections. Consideration should be given to options for sourcing suitable inputs. The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Develop the complete processing chain including steps required to prepare inputs. Automate the processing as much as is feasible. Work with the Navy to produce a limited template database and conduct a comprehensive evaluation it being used in existing maritime automatic target recognition applications.

PHASE III DUAL USE APPLICATIONS: Complete the automated processing capability and integrate as a means to train a target recognition application or to generate a feature database for a template-matching algorithm.

Version 2

The 3D deep learning ability to synthesize views of complex scenes using only a limited number of input views can be utilized for a very wide range of computer vision applications.

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KEYWORDS: 3D Learning; Neural Radiance Fields (NeRF); Automatic Target Recognition; Machine Learning; Template Matching; Maritime Situational Awareness

N25A-T007 TITLE: Automated Data Synchronization Across Multiple Databases

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software

OBJECTIVE: Develop a software algorithm that provides a capability to automate data synchronization across multiple databases of software defects and problem reports of surface naval platforms.

DESCRIPTION: Databases now use a set of disaggregated decision criteria that do not support collaboration and must be aggregated by the operator to be useful. Current decision support database tools are fixed in time and rely on operators to assess various sources of information to maximize performance over time. Databases to be utilized for tasking and actions are within the maritime, and air and land mission domains and should not be limited in their ability to visually represent multi-unit or multi-domain temporal coordination. The Navy seeks an automated course of action (COA) capability that improves situational understanding using merging, removing, or automating synchronization. New approaches are needed with enhanced methods to support database alignment, clarity, and usability.

The Navy has multiple databases located at various locations. It invests in several different land-based testing facilities for various capabilities. These various sites maintain their own hardware and database configurations for tracking purposes. Many of the databases contain identical or almost identical data with the same intent. The data needing to be automated and synchronized are the various software defects that have been reported and problem reports associated with the various surface navy combat systems baselines or platforms. The duplication of data residing on a host of different databases builds to the unnecessary data redundancies. The redundancies cause searching for information harder by having to eliminate data through comparison with all data that is retrieved. The current process requires manual human labor to catch duplicates through Change Review Board and potential audits. The redundancy in data prevents a common picture of ground truth in reporting defect status and the insufficient process slows down the decision-making actions of stakeholders costing valuable time in a decision-making process. The Navy desires an automated data synchronization capability in real time. It must retrieve information that is then compared with only unique occurrences of the same data which is brought forward as a single source. This will remove redundancies and reduce decision time of stakeholders that drive capability.

The AEGIS combat system databases are comprised of various databases that have various capabilities for controlling components specific to their baseline. The various baselines control specific radars and weapons per their primary and secondary mission areas. The databases in use currently store data for tactical employment and replay capability for reconstruction for mission accomplishment. The databases are engineered to operate independently as stand-alone sources and synchronize interoperate with other databases and combat systems defining the interoperability capability. The various surface naval databases can vary in different ways that may provide difficulty for the sailor to access in a timely manner when comparing different types of data.

The Navy is seeking a software application that can automate and synchronize identical or almost identical data contained on various databases. The solution will enhance capabilities by bringing a complete accessibility to a variety of data from non-similar sources for Surface Navy planning. There are currently no available commercial capabilities that will solve the need for the technology.
One of the key challenges of big data is taking the enormous amounts of data and turning it into useful cognitive information. Although there are no defined limitations to hardware and software for use aboard ships, Navy resource sponsors are seeking to reduce lifecycle costs to support Fleet capability by developing hardware agnostic software and by employing software standards that facilitate updates without significant cost. Currently, the software defects are reported on one database and not synchronized to extend to another database, failing to provide overarching awareness across all

capabilities. Equally, a Test Observation Report (TOR), which is sometimes handwritten during testing and turns into Trouble Reports (TR)/ Problem Reports (PRs), reported on one database may not be shared to another, thus not being universally accessible.

The speed and accuracy of the solution must exceed existing database access performance attributes by 10% or better. The software code should be able to demonstrate the ability to scan or screen the physical copy into editable fields within databases.

For Change Requests (CRs) that need to enter into a defect database, currently a team of individuals manually input the CR data on a weekly basis. Ideally, it would be more efficient to provide a file in a pre-determined format that could then be imparted into the defect database to create the PRs.

To reduce redundancy, it currently requires manual human labor to catch duplicates through the Change Review Board (CRB) and CR review and potentially audits. The solution software application needs to demonstrate the ability to screen through multiple databases and find duplicates based on parameters like change request number, description of the data observed, and PR type; and highlight the potential duplicate so users can take the next step. In addition, the application should be able to prioritize which defects have higher impact and rank them from red to green scale based on priority and safety impact parameters.

PHASE I: Develop a concept for an automated data synchronization algorithm that meets the objectives stated in the Description. Demonstrate the feasibility of the concept in meeting the Navy's need. Feasibility shall be demonstrated by a combination of analysis, modeling, and simulation as stated in the Description. The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Develop and deliver a prototype automated data synchronization algorithm based on the results of Phase I. Demonstrate the prototype's functionality to improve situational visualization and situational understanding within a varied database context.

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the prototype database software applications to Navy use. The final product will allow for further experimentation and refinement. The prototype database software application will be incorporated into the AEGIS baseline database modernization process. This will consist of integration into a database definition, incorporation of the databases existing and new database capabilities for software defects and problem reports accessibility to support certification.

Database algorithms could provide assistance to additional Department of Defense areas and air traffic controllers in maximizing database capability to prevent future potential collisions.

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KEYWORDS: Multiple Databases; Identical Data; Software Defects; Database Alignment; Automating Synchronization; Data Redundancies

N25A-T008 TITLE: Modeling of Magnetic Properties of High Strength Steels

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials

OBJECTIVE: Develop physics-based material models for the understanding and prediction of stress-induced hysteretic magnetic behavior of high strength steels.

DESCRIPTION: The problem of stress-induced hysteretic magnetic behavior of high strength steels is challenging and a pressing one for the Navy. This is because Navy vessels, in general, and submarines, in particular, are subjected to stresses and complex changes in magnetic states that in turn causes their magnetic signature to evolve in a nonlinear, hysteretic fashion. Lack of fundamental understanding of this phenomenon prevents the Navy from designing magnetic signature reduction systems using a rigorous physics-based approach. Consequently, the Navy and industry address this issue by resorting to empirical approaches lacking sufficient physics-based understanding.

Computationally tractable models have been developed for the hysteretic magnetic behavior of steels of interest [Refs 4,5]. The commercial sector addresses this problem, e.g., for the effect of stresses on steels used in motor stators [Ref 5]. However, these models typically do not investigate the impact on the magnetic behavior due to hysteresis induced by large stresses of the order of 100's of mega Pascals (MPa). This effect is of great interest to the Navy.

While investigators have studied the effect of large stresses on hysteretic behavior of steels [Refs 1,2], fundamental understanding of this behavior that can be translated into a robust computational tool is still lacking.

Developing a computationally tractable model that accurately captures the physics is challenging because it involves the interplay of structural, stress, and magnetic properties spanning extremely varied length scales [Ref 3]. Therefore, this STTR topic calls for proposals that emphasize the need for a rigorous understanding of the effect of stress on the hysteretic behavior of high strength steels. Research and development are needed for the following inter-dependent areas:

1. Measurement of magneto-elastic and magneto-thermal hysteresis.
2. Development of both fundamental and phenomenological physics-based models of magneto-elastic and magneto-thermal hysteretic behavior based on measurements.
3. Implementation of the developed models into an efficient computational tool for rapid and robust prediction of magnetic behavior.

Taken together, the measurements, models, and computational predictions will be assessed against measurements and predictions based on physical scale and simple computational models.

PHASE I: Provide a concept of an approach to model the magnetic properties of high strength steels. Ensure that the concept clearly demonstrates how the challenges inherent in the three R&D areas in the Description section above will be addressed. Measure the effectiveness of the feasibility of the concept, e.g., by applying the methodology on idealized and geometrically simple objects such as spheres or shells with magnetically simple properties as test cases to serve as subscale prototypes or surrogates.
The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Build on the results obtained in Phase I to make a more robust framework that will consider the measurements and modeling of simple shapes to develop and implement measurement and computational tools that can be applied to a wide variety of magnetic and stressed environments. These measurement and computational tools will also be applied to objects of increasing geometric complexity.

The results of the application of the above tools will be verified or validated against comparable results obtained on Navy-owned physical scale models. The results of the verification and validation as well the computational tools (prototype) will be delivered to the Navy.

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the mature computational tool prototype as well as measurement techniques that have been developed in Phase II for use by the Navy. Routinely validate, test, qualify, and certify the tools for Navy use by a rigorous verification and validation process involving comparisons against physical scale model and full-scale measurements. In addition, the computational and measurement tools will be used for the understanding and characterization of commercial steels that are likely to be used for high-stress and low-ambient magnetic field applications.

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KEYWORDS: Ferromagnetism; Stress; Hysteresis; Isotropic Model; High Strength Steel Measurement; Multi-scale Modeling; Magneto-elastic

Version 2

N25A-T009 TITLE: Medium Voltage Alternating Current System Grounding Circuit

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Directed Energy (DE)

OBJECTIVE: Develop a compact system grounding circuit for Medium Voltage Alternating Current (MVAC) applications between 1 and 15 kV that can operate continuously with a ground fault, is air cooled, works with a ground fault having a direct current (dc) component, dissipates less than 3 kW of heat during a ground fault, and prevents high transient voltage spikes due to intermittent ground faults.

DESCRIPTION: To enable continued operation in the case of a single ground fault, shipboard MVAC power systems typically are high resistance grounded using a grounding transformer and a High Resistance Grounding (HRG) resistor. The HRG circuit ensures line to ground transient voltages during a ground fault (particularly during intermittent ground faults) are not excessive and that under normal operation, the neutral to ground voltage is nearly zero. During a ground fault, the current through the HRG resistor is equal or marginally higher than the capacitive charging current of the system. This current can be on the order of 10 amps and the voltage would be the line to neutral voltage (The line-to-line voltage divided by the square root of three). In a 13.8 kV system, this can result in the HRG resistor dissipating roughly 80 kW. This amount of heat requires special accommodations for heat removal (water cooling for example) that complicate ship integration and drive costs. Alternately, the ground fault must be cleared within a few minutes before the space overheats, possibly impacting the safety of the ship. Details on HRG systems may be found in IEEE Std 45.1 (Recommended Practice for Electrical Installations or Shipboard—Design) and IEEE Std 3003.1 (Recommended Practice for System Grounding of Industrial and Commercial Power Systems).

Continuous operation with a ground fault is required to ensure vital electrical loads such as propulsion and combat systems do not lose power during critical time periods. The goal is to allow for locating and clearing the fault(s) in a non-critical time period. Since the length of time the ship will be in a critical time period is unknown, the HRG system should have the ability to operate continuously.
Another issue is that if a rectifier load is directly attached to the MVAC system, a ground fault on the Direct Current (DC) side of the rectifier can result in a DC current in the grounding transformer; this DC current can result in saturation of the grounding transformer leading to the grounding transformer and resistor overheating and potential catastrophic failure.

Research and Development along with Innovation are required to fulfill the functionality of an HRG circuit while dissipating less than 3 kW, enabling an air-cooled solution with a heat load that can be accommodated by the ship's ventilation system.

In addition to the requirement to limit heat dissipation to 3 kW, the proposed solution should not require more than 2 square meters deck space (threshold) or 1 square meters deck space (objective). The proposed solution should be capable of accommodating 10 Amps (A) of ground current (threshold) or 20 A of ground current (objective). The power quality of the MVAC system should be assumed to be in accordance with MIL-STD-1399-300.2. The shipboard equipment should be shock qualified to Grade A (MIL-DTL-901) and constructed to operate in the shipboard environment (MIL-DTL-917).

The solution may include power electronic devices with advanced controls to achieve the desired properties. Repair parts should be capable of passing through Navy standard hatches 26" x 66" and doors 30" x 60".

This MVAC System Grounding Circuit should be applicable to all shipboard MVAC systems up to 15 kV (both commercial and naval) and terrestrial industrial power systems where high reliability power is required (such as for process controls).

PHASE I: Provide a concept design of an MVAC System Grounding Circuit fulfilling the desired functionality listed in the Description. The feasibility of the MVAC System Grounding Circuit shall be demonstrated through Computer Aided Design (CAD) models and circuit simulation.

The Phase I Option, if exercised, should include the identification of key knowledge gaps and risks, initial design specifications, capabilities description, test plan, and test procedures for a prototype solution in Phase II. The prototype should address the key knowledge gaps and risks. A draft procurement specification for the production system should also be delivered.

PHASE II: Update the initial design specifications (if necessary) developed during the Phase I Option and produce a prototype system for testing in accordance with the test plan and test procedures. Include, if required, component testing to close knowledge gaps prior to updating the design specifications. Conduct the tests and update the system design and draft procurement specification for the production system based on lessons learned from testing. Once all testing has been completed, deliver the prototype system and all design and test report documentation to the Government.

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the MVAC System Grounding Circuit for Navy use. Any naval ship with a MVAC distribution system (including CVN-68 class, CVN-78 class, LHD-8, LHA-6 class, DDG-51 flight III, FFG-62, and DDG(X)) is a candidate for transitioning the technology. Any commercial ship with an MVAC distribution system (such as cruise ships, ice breakers, etc.) is also a candidate for technology transition. Any terrestrial process control industrial application (oil refining, plastic production, chemical manufacturing, etc.) also can benefit from this MVAC System Grounding Circuit.

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KEYWORDS: High Resistance Grounding; Medium Voltage Alternating Current; Ground Fault; Grounding Resistor; Grounding Transformer; Power Continuity

N25A-T010 TITLE: Miniaturized Low Frequency Electric Field Sensing for Underwater Applications

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Microelectronics;Quantum Science

OBJECTIVE: Develop a novel, low SWaP (space, weight, and power) electric field sensor for installation in maritime environments.

DESCRIPTION: The Navy utilizes electric field sensing to monitor emissions from corrosion protection systems. Electric field measurements in maritime environments are traditionally acquired using pairs of sensitive electrodes separated by a relatively large baseline. State of the art sensing systems that have a direct current (DC) response employ silver chloride electrodes, but these electrodes are susceptible to drift while deployed in maritime environments. An example of current specifications for silver chloride electrode drift and accuracy can be found in the impressed current cathodic protection system military specification: MIL-DTL-23919.

The Navy desires a sensitive, low SWaP electric field sensor with a DC response that is stable while deployed in the ocean. The desired sensor will be expected to:

- Fit into dimensions of: 10 centimeters by 7.5 centimeters by 5 centimeters
- Consume less than 10W of power continuously

The objective is to develop a compact electric field sensor package, fitting into the dimensions listed above, that meets the following sensing requirements:

- High dynamic range: 0.001 to 10000 millivolts per meter
- High accuracy: 0.001 millivolts per meter
- Low frequency: DC through 500 Hz
- Low noise: 0.001 mV/m/rtHz at 1 Hz (Threshold), 10 nV/m/rtHz at 1 Hz (Objective)
- Resist biofouling and operate in conductivities from 0.1 to 10 S/m
- Biaxial or triaxial design
- Orthogonality shall be established to within 1 degree
- Accuracy and dynamic range requirements shall be met for each independent electric field component axis

It is essential that the sensors maintain these accuracies under environmental stresses in the ocean. These conditions include: temperature 0-50 °C, hydrostatic pressure up to 10,000 kPa, total suspended solids of 0-120 mg/L, fouling and biofouling over extended deployment periods.

Advanced technologies for accurate measurement of local electric fields are sought and may include quantum sensing technologies. Technologies can be revolutionary and physical in nature, such as using phase transitions to sense electric fields. Recently, it was demonstrated that the insulator to metal (IMT) transition in perovskite nickelates could be used to sense marine electric fields [Ref 1]. Technologies can also be evolutionary in nature, such as using improved silver chloride chemistry to reduce the accuracy or drift errors resulting in more accurate small scale electric field measurements. Technologies based on existing silver chloride chemistry but using novel, alternate data processing means to achieve desired accuracy will be considered as well.

PHASE I: Develop a concept for compact electric field sensing meeting the accuracy requirements outlined in the Description. Demonstrate the feasibly of the concept to meet the parameters listed in the Description through modeling, simulation, and analysis. The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Develop and deliver two prototypes of the proposed electric field sensor sensors. Demonstrate the prototype's performance under the necessary environmental stresses. Deliver two such packaged sensors for Navy evaluation. The Navy will evaluate the prototypes for all requirements outlined in the Description section in a relevant seawater environment.

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the technology through system integration and qualification testing. Implement cost reduction measures and install sensor at various locations. Demonstrate operation of the sensor package for a minimum of two years through the collection of background electric field data to support long term environmental data collections. Show that the system is able to collect data for magneto telluric surveys.

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KEYWORDS: Underwater Electromagnetic; Compact Ocean Sensing; Electric Field Measurement; Underwater Electromagnetic Measurement Range; Low SWaP; Quantum Sensing

N25A-T011 TITLE: Marine Biofouling Mitigation and Innovative Broad Band Hydrophobic Glass Development

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Biotechnology;Biotechnology

OBJECTIVE: Design an innovative approach to reduce or mitigate biofouling on submerged broad band optical head windows.

DESCRIPTION: Biofouling has been a long-lasting problem in a variety of marine systems. Optics and sensors suffer reduced performance and, in some cases, catastrophic failure due to biofouling. No suitable optical surface coating is available to mitigate biofouling and salt mineral depositions. Legacy coating (Ameral) was effective on imaging systems, but Ameral has been discontinued due to environmental and health hazards. A replacement solution (Rain-X) has a short lifetime and requires frequent beam director window cleaning applications. Neither solution is effective for Photonics periscope and Laser applications, since both solutions absorb light and need periodic applications. No current market options combine the required hydrophobicity and broadband anti-reflection (AR) properties when applied to broadband optical surfaces exposed to a marine environment.

Nanostructured materials with super-hydrophobic self-cleaning surfaces, and hyper-branched polymer structures with hydrophilic-ended or activated foulant releasing groups are incorporated in most current coating techniques on the glass or sensors surface. However, advanced nano-structured coatings are usually associated with reduced mechanical properties, sophisticated fabrication processes, and extensive use of chemicals resulting in higher costs and more environmental issues.

The Navy seeks an innovative nanotechnology to mitigate broadband AR coating challenges. In this STTR topic, the Navy is looking for a cost effective and environmentally friendly innovative approach to mitigating biofouling on submerged glass and sensors. The exposed broadband optical surfaces employed are either sapphire, aluminum oxide, spinal, gallium germinate, or Germanium Oxide (GeO) . The proposed innovative technology should address biofouling and salt mineral deposition mitigation. Both an active and passive solution will be considered. Solutions must provide self-cleaning of the optical and sensor surface with ultra hydrophobicity and broadband AR coating.

Proposals will be evaluated on:

1. Nano structured deposition optical surface transmission wavelength from 0.45 to higher than 5 micrometer with greater than 90% transmission and higher than 99% optical transmission between 1 to 1.5 micrometer Wavelength.
2. Nano structure coating on head window will have wave-front error less than wavelength of light/50.
3. Optical Surface size of minimum 12-inch diameter shall not introduce any polarization or birefringence.
4. Surface durability more than 720 hours of continuous operation against any salt/mineral and any marine particle.
5. Optical surface should introduce self-cleaning technology.

PHASE I: Develop a concept to solve the Navy's biofouling problem on Optical Head Window. Demonstrate the feasibility of the innovative concept to solve or mitigate the Navy's problem of broad band glass with nano structure for anti-reflection (AR), biofouling and salt mineral deposition mitigation. Deliverable will include the initial modeling of the biofouling mitigation on broadband developed glass optical surface with salt mineral deposition and broad band AR coating. The Phase I Option, if exercised, will include the validation modeling and capabilities description to build a sample prototype optical glass for testing to mitigate biofouling of the proposed broadband glass surface at Phase II.

PHASE II: Develop the biofouling mitigated broadband glass with self-cleaning optical surface for an HEL beam director or Photonics periscope for data collection analysis. Deliver the prototype to the Navy for further evaluation. In Phase II Base the awardee shall demonstrate the broad band glass GeO and its biofouling properties and hydrophobicity. Test of the glass shall be performed inhouse.

The biofouling and salt mineral deposition mitigated broad band HEL window shall support multi kW laser power transmission with self-cleaning surface technology. The wave front error of the biofouling mitigated surface shall not be bigger than optical wavelength/50 of the optical surface. The Bio fouling mitigated surface shall also have higher than 160-degree contact angle with ultra-hydrophobic surface for water shedding and have broadband (visible to Mid wave IR) anti reflection properties.

PHASE III DUAL USE APPLICATIONS: Complete the final design of the biofouling mitigated broad band glass with self-cleaning optical surface. Support the Navy in transitioning the technology to Navy use.

This technology can have application to both DoD and commercial sectors such as ship's hulls, underwater pipes, oceanographic sensors, terrestrial optical sensors and laser systems, drilling equipment, oil platforms, fishing industry, power plants, and aquaculture systems.

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KEYWORDS: Bio Fouling; Optical head windows; Hydrophobicity; Laser; HEL; AR (Anti reflection coating); Salt-mineral deposition

N25A-T012 TITLE: Medium Voltage Direct Current Protection Relays and Associated Sensors

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Directed Energy (DE)

OBJECTIVE: Develop a Fault Protection relay and associated sensors for 1 kV, 6 kV, and 12 kV Medium Voltage Direct Current (MVDC) distribution systems to implement overcurrent and differential / directional protection in both breaker and breakerless architectures.

DESCRIPTION: MVDC systems are an evolution of the DDG 1000 1 kV DC Integrated-Fight-Through-Power (IFTP) system combined with shared and distributed energy storage as well as advanced controls with active state anticipation data linkage between machinery and combat systems. An Integrated Power and Energy System (IPES) offers the potential to provide revolutionary warfighting capability at an affordable cost. An IPES utilizes integrated energy storage and power along with advanced controls to provide a distribution bus suitable for servicing highly dynamic mission loads and propulsion demands while keeping the lights on. Additionally, such a system can enhance survivability, reliability, and flexibility while providing new capabilities such as the ability to quietly maneuver solely on energy storage. Once technically mature, MVDC IPESs are intended for Future Surface Combatants (FSCs) to affordably improve warfighting capability to meet evolving threats over the ship's service life in an agile manner. The Navy is anticipating relying more and more on high power, highly dynamic, and pulsed weapons and sensors on FSCs. Because the need for generator synchronism is eliminated, MVDC is anticipated to be able to support these systems at lower cost, lower weight, and lower space requirements. Details on IPES are provided in the Naval Power & Energy Systems (NPES) Technology Development Roadmap.

One of the key enablers of an MVDC IPES is reliable and affordable MVDC fault protection systems compatible with both breaker and breakerless fault protection strategies [Refs 1-2]. One of the primary functions of a fault protection system is to sense a fault (such as line to line short circuit) and act quickly to localize and initiate isolating the fault. This function may be performed in both circuit breakers and in protection relays. Circuit breakers typically must decide whether to open or close using only locally measured voltages and currents. Also, protection relays can use more remote current and voltage sensors at the edge of a protection zone to determine if a fault has occurred, and if so, if the fault is within the protection zone. The protection relay initiates action to isolate the fault by sending commands to active rectifiers, circuit breakers, and/or disconnect switches. Designs of future MVDC systems are anticipated to require the use of protection relays to achieve power reliability requirements.

In differential protection, the currents on all conductors of the same polarity entering or leaving the protection zone are summed by the protection relay; if they do not sum to zero, then a fault is within the zone. For directional protection, protection relays examine the direction of current flow to determine if a fault is within the zone. One challenge is minimizing false alarms by coordinating multiple current readings when fault currents rise with high di/dt, particularly in a bus topology supporting pulsed power loads. If the current readings do not occur near simultaneously when currents are quickly ramping up, the sum of the current readings may not sum to zero even if the fault is not within the zone. An additional challenge is proving robustness and reliability of MVDC current and voltage sensors.

The Navy seeks protection relay and associated voltage and current sensors to detect a fault, determine if the fault is within the protection zone, and initiate commands to isolate the fault (if the fault is in the protection zone) within 8 microseconds (threshold) and 1 microsecond (objective). These times are to prevent solid state circuit breaker overcurrent protection from activating without coordination with other circuit breakers. The threshold requirement may require insertion of system inductance to slow the current ramp rate; added inductance can result in stability issues that then must be addressed. The protection relay and associated voltage and current sensors should be designed for use in a shipboard environment (as

defined in MIL-DTL-917) for a design life of forty years (threshold) or fifty years (objective). If there is a fault within the zone, the protection relay and associated sensors should operate as desired at least 99% of the time (threshold) or 99.9% of the time (objective). The protective relay should experience a false positive (initiating fault protection when no fault is present in the protection zone) on average no more than once every 30,000 operating hours (threshold) or 300,000 operating hours (objective). The protection relay should be able to differentiate between a fault and a pulsed load. Currently this application does not commercially exist.

PHASE I: Provide a concept design of an MVDC Protection Relay and associated sensors fulfilling the desired functionality listed in the Description. The feasibility of the MVDC Protection Relay and associated sensors should be demonstrated through circuit simulation.
The Phase I Option, if exercised, should include the identification of key knowledge gaps and risks, knowledge gap closure plans/risk mitigation steps, initial design specifications, and capabilities description. Provide a Phase II test plan and test procedures for a prototype solution that addresses the key knowledge gaps and risks. Deliver a draft procurement specification for the production system including the initial design specifications and capabilities description to build a prototype solution in Phase II.

PHASE II: Provide updated initial design specifications (if necessary) developed during the Phase I Option and produce a prototype system for testing in accordance with the test plan and test procedures. Include, if necessary, component testing to close knowledge gaps prior to updating the design specifications. Conduct the tests and update the system design and draft procurement specification for the production system based on lessons learned from testing. Deliver a fully tested prototype system to the Government.

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the technology to Navy use. Any naval ship with elements of a MVDC distribution system (including CVN-78 class, DDG-1000 class, DDG-51 flight III, FFG-62, and DDG(X)) is a candidate for transitioning the technology.

In addition to naval warships, MVDC systems using the protection relays are anticipated to apply to commercial ships such as ferries, cruise ships, and ice breakers. Commercial applications are likely to increase in the upcoming years as international treaties enforce lower emissions from commercial ships; MVDC systems enable more efficient operation of diesel generator sets and easier integration of energy storage systems. Any terrestrial process control industrial application (oil refining, plastic production, chemical manufacturing, etc.) or renewal energy system employing MVDC could also benefit from the protection relay and associated sensors.

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KEYWORDS: Medium Voltage Direct Current; Fault Protection Relay; Current Sensor; Fault Detection and Localization; Differential Protection; Directional Protection

N25A-T013 TITLE: Vitrimer Patches to Repair Composite Parts with a Complex Shape in a Contested Environment

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials;Sustainment

OBJECTIVE: Determine the feasibility of using vitrimer repair patches for repairing composite damage on a contoured surface in a contested environment.

DESCRIPTION: Repairing composites damaged on a contour is always a challenge. This is especially true when the fleet is forward deployed and operates in a contested environment. Usual go-to repair is a wet-layup patch. However, using wet layup patches results in a strength knockdown. The Navy is seeking a robust vitrimer resin based solution that can be deployed with the fleet and will be used to perform repairs in the theater, that will restore the full strength of the composite part.

Vitrimers are a relatively new class of resins developed over the last decade which has dynamic chemistry in which covalent bonds are exchanged and rearranged by providing stimuli such as heat. Referred to as "reversible thermosets," these polymers have thermoset properties below the glass transition temperature (Tg) yet behave like a thermoplastic at elevated temperatures, above the Tg. Tg can currently be tailored between 80° C to 180° C. However, the chemistry and the resulting Tg are an area of active research. Kits can be developed as flat patches and contoured to shape as needed in the theater. These contoured patches can be cured on the part or secondarily bonded using traditional epoxy vitrimer/epoxy based resins. These late resins can provide bond strength at least as good or better than traditional epoxy adhesives but also can be cleanly removed as needed by applying heat.

PHASE I: Identify suitable vitrimer resin pre-pregs and demonstrate its feasibility to be used for repair. Through suitable experimental campaign, map its formability and mechanical properties. In Phase I any limitations in obtaining tight tolerances and any property degradation during reforming the vitrimer patches should be identified. Potential methods to join it to a carbon/epoxy part should also be identified and its feasibility established. The feasibility study should be completed at the end of the Phase I base as the down select will be made before the Phase I option is exercised.

PHASE II: Using the results from Phase I, develop repair kits. Identify a control surface and an access panel. With TPOC input design and create damage on the parts. Repair the damage using the Vitrimer patches. Examine the quality of the patches through non-destructive inspection (NDI). Determine the strength of the patches through testing.

At this point NAVAIR depots are expected to be involved to inform the team of the needs for repairs. Using this information as well as results from the program, design and produce prototype repair kits and demonstrate it at a Navy depot.

PHASE III DUAL USE APPLICATIONS: The primary transition goal is NAVAIR depots. While interest is from PEO(A) the solution applies to all aircraft with composites.

In addition to repairs, vitrimer resins have great potential in manufacturing complex production parts. In addition to the aviation industry there is also interest in vitrimers by the automotive industry, urban mobility industry, and high performance sport industry.

Vitrimer composites are environmentally friendly and recyclable. Thus it has great potential as environmentally benign replacements for thermoset composite parts.

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KEYWORDS: Vitrimer; Composite Repair; Wet Patch Layup; contested environment; depot repair; kits; recyclable

Version 2

N25A-T014 TITLE: Removable Smart Tooling for High Temperature Composite Fabrication

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials;Sustainment

OBJECTIVE: Develop removable tooling to produce complex contours by Automated Tape Placement (ATP) and Automated Fiber Placement (AFP) for thermoplastic composite parts.

DESCRIPTION: Recent developments in ATP and AFP have revolutionized the composites industry. They have enabled affordable automated production with high dimensional tolerances. ATP/AFP operations often use a tool over which the material is laid. A removable tool allows for manufacture of complex parts in one operation rather having to make multiple parts and joining them secondarily. Such tooling remains a challenge as it needs to maintain its dimension and strength throughout the layup and consolidation process.

Two promising technologies that can be leveraged in this STTR topic are washout tools and the melt tools. In the first washout material is manually poured into tooling and allowed to harden. Once the part is laid up on the tool and consolidated, usually water is used to washout the tool. There are now washout material available that can operate at up to 370 °C which is sufficient for thermoplastics. However, additional work is needed to improve dimensional tolerances and develop it to a point that it can be used as an ATP/AFP tool. The second candidate technology is the meltout tooling. In this process the tool is molded using Resin Transfer Molding (RTM). Once the part is consolidated it is removed by heating. The challenge for this technology is that it currently uses materials that cannot be used at temperatures needed for thermoplastic manufacturing. Thus material discovery and/or development is needed to meet the topic objective.

The program should produce the tooling such that it is highly repeatable (tolerances of +0.005” on thickness and +0.010” on everything else), is low cost considering both materials and forming process ($40-$50/part of size 1.5” x 12”x 0.5”), does not require an oven post cure, has thermal shock and impact resistance equal to ceramics, and requires minimal processing time. The material should be capable of being used in autoclave processing with typical processing conditions of 350°F temperature and 100 psi pressure. In addition, the material used for the tooling should be environmentally friendly and should not create a hazardous waste stream.

The washout tooling must also provide the appropriate characteristics required for the production of high performance composite structures. In addition to the above mentioned tolerances, it is important that the tooling produces the finished part with the appropriate dimensions. This requires that the washout tooling have the appropriate Coefficient of Thermal Expansion (CTE) such that a finished part meet the tolerances required for the processed component. It should be noted that carbon/Polyetherketoneketone (PEKK) tape will be one of the materials that the washout tooling will be required to be compatible with.

PHASE I: Develop and demonstrate material and handling properties to produce proof of concept specimens with properties suitable for use in removable smart tooling applications, as detailed above. Demonstrate that the tooling material can be easily removed after a component has been processed at 700° F. Finally, demonstrate that the tooling material can be stored under ambient conditions.

PHASE II: Develop and demonstrate a repeatable process whereby the awardee can produce a soluble rectangular mandrel with dimensions of approximately 1.5” x 12” x 0.5” with tolerances of +0.005” on thickness and +0.010” on everything else. These parts must also demonstrate that they are capable of handling the processing conditions typical of tape placement. In addition, they need to demonstrate that the process is scalable to 2” x 20” x 0.5”. Demonstrate the ability to develop and demonstrate the process for manufacturing the removable tooling with more complex shapes, as determined by the topic sponsor.

The shape can include rectangular sections that are out of plane, or circular sections with jogs and protrusions, or foil shapes with complex curvature. Scale up the processing to be able to produce mandrels with planar dimensions of a minimum 12” x 24“. Demonstrate that the manufactured component meets the dimensional tolerances established by the Technical Point of Contact (TPOC).

PHASE III DUAL USE APPLICATIONS: Perform a functional demonstration to DoD stakeholders arranged by the Navy team. Help transition the technology to a DoD original equipment manufacturer. While focused on thermoplastic the tooling will also be useful for thermoset AFP and ATP. The commercial aircraft industry would benefit significantly from low cost tooling material that allows for the fabrication of reproducible low cost composite parts by reducing the part count for complex shaped components. The tooling can also be used in naval platforms (ships, subs) which could benefit from reduced part count and complex composite components.

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KEYWORDS: Removable tooling; Washable tooling; melt tooling; composites manufacturing; thermoplastic composite; low cost manufacturing; complex shapes

Version 2

N25A-T015 TITLE: Lateral Shear and Strain Sensor for the Ocean Environment

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Sustainment

OBJECTIVE: Develop a sensor to advance our capability of measuring finescale (1-10 m) lateral gradients in ocean velocity for environmental sensing of anisotropic turbulence. The sensor should be capable of direct measurement of lateral gradient fields within this scale range. While deployment via a ship is appropriate for the initial design phase, the design should consider future deployments on autonomous or moored platforms.

DESCRIPTION: Direct measurement of ocean turbulence depends on sampling of quantities at length scales where universal scaling laws hold. These measurements are typically made in the vertical over regimes that are thought to be isotropic (invariant with rotation), homogeneous (invariant in space), and stationary (invariant in time). In the stratified ocean, the assumption of isotropy breaks down when the vertical scale of turbulent overturns is suppressed by the fluid's stabilizing stratification. In contrast, horizontal shear is unrestrained and turbulent eddies persist. This anisotropic turbulence regime resides in between spatial scales dominated by internal gravity waves in the open ocean and the inertial subrange of isotropic turbulence [Ref 1]. It is characterized by what have been termed 'pancake' eddies or vortical motion as turbulent vortices have a 'flattened' aspect ratio in the vertical.

Direct measurement of velocity fluctuations over this spatial range are exceedingly rare as most existing Commercial off-the-shelf (COTS) sensors offer either a fixed volume or vertical profile of velocity. The Navy seeks development of a sensor capable of direct measurement of lateral velocity gradients over spatial ranges of order 1-10 meters. The Navy is agnostic to the approach, which may be acoustic [Ref 2], optical [Ref 3], or some alternative platform [Ref 4]. Proposed designs should either sample the lateral velocity vector while resolving differences over the scale of 1 m or measure lateral shear/strain directly on scales of at least 1 meter. The sensor should be capable of 1) resolving gradients of 0.0001 1/s (implying a noise floor below this value) and 2) distinguishing lateral motions from platform tilt either through motion correction or direct sensing. While deployment via a ship is appropriate for the initial design phase, the sensor should ultimately be deployable on autonomous or moored platforms.

PHASE I: Identify a design concept for a sensor along with the hardware components that can meet the stated requirements. Develop a concept for onboard and offline software processing. Develop a detailed power budget for the sensor. A detailed analysis for strengths and weaknesses of the proposed design should be included in Phase I, considerations should include resolution and range tradeoffs, endurance determined by power and storage needs, and physical footprint of the sensor. After assessment of strengths and weaknesses a final design review should be completed.

PHASE II: Develop and test a prototype system. Complete analysis of the performance of the system. Report on results. Perform multi-stage testing, allowing for redesign between tests with initial tests in a surrogate ocean environment (e.g., lake or tank) and final testing in the ocean under controlled conditions (e.g., coastal bay). Both hardware and software systems should be developed and tested during Phase II. Field testing in Phase II will constrain the parameter space under which the system is operationally capable for the Navy.

PHASE III DUAL USE APPLICATIONS: Support the transition to Navy use. This technology has potential use in DoD and commercial applications that require current velocity information at small scales (1-10 m). Possible applications include monitoring of contaminants shed from wind/wave farms and other offshore structures, monitoring fluxes for marine carbon dioxide removal (mCDR) development, and navigation of UUVs.

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Optical application

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KEYWORDS: stratified turbulence, anisotropic turbulence, ocean sensing, vorticity, lateral shear, lateral strain

Version 2

N25A-T016 TITLE: Persistent Airborne Counter Unmanned Aerial System Surveillance

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Integrated Sensing and Cyber

OBJECTIVE: Develop and demonstrate an innovative airborne counter-unmanned aerial system (C-UAS) concept that balances sensing performance and host unmanned airborne platform capabilities supporting persistent surveillance with overall system size, weight, power, and cost (SWAP-C).

DESCRIPTION: The Navy seeks an airborne C-UAS system capable of providing persistent coverage around either Naval ships or forward operating bases (FOBs). Such an airborne system could provide extended coverage beyond that possible from fixed surface sensors and enable the host platform to pursue intruders in an effort to locate launch and recovery locations. Significant investments have been made by the Department of Defense (DoD) and the Department of Homeland Security (DHS) in the development of ground/surface-based sensing systems. Group 2 and 3 UAS technology has matured significantly with long endurance systems widely available. Some of these systems are capable of being launched and recovered from small deck ships. Leveraging knowledge of these existing technologies, the Navy is seeking airborne C-UAS concepts that best balances sensing performance which includes detection and tracking performance against Group 1 through 3 intruders, means to mitigate false alarms, volume coverage and revisit rates with SWAP-C to provide persistence long-endurance coverage and the ability to pursue intruding UAS. Consideration should be given to common sensing systems including radar, radio frequency analyzers, acoustic and optical systems either alone or in combination on the host platform. The maximum sensor package SWaP allocation for group 3 UAS should not exceed 50 lbs, 750 watts and if pod mounted should be less than 12" in diameter and 60 inches long.

PHASE I: Complete a comprehensive analysis and trade-off study of sensor modality and performance with UAS host platform capabilities and SWAP-C. The analysis and study should address the nature of the C-UAS threat for both FOBs and Naval ships (both in-port and underway), sensing system considerations and performance assessment, and integration approaches on candidate host UASs. The study should recommend a sensor and platform solution to be develop and demonstrated in Phase II.

PHASE II: Develop and demonstrate a prototype of the recommended solution to achieve the capabilities analyzed in Phase I. The demonstration may be conducted on a surrogate platform with commercially available sensors if it can be shown that a feasible (in terms of cost, time and risk) path exists leading to a mature deployable system.

PHASE III DUAL USE APPLICATIONS: Work with UAS platform and sensor equipment manufacturers to fully integrate and mature the airborne C-UAS. The technology is suitable for a wide range of facility protection and airspace monitoring for civil applications.

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KEYWORDS: Counter-Unmanned Aerial Systems; Radar, Acoustic, Optical, Group 2 and 3 UAS; radio frequency analyzer

N25A-T017 TITLE: Mattresses for Improved Sleep Quality (MISQ)

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Sustainment

OBJECTIVE: Improve shipboard mattresses while meeting required Navy standards (MIL-PRF-32568) to improve sleep quality and enable increases in morale, reductions in fatigue/incident risk, and enhancement of resiliency. No human testing to be completed in Phase I.

DESCRIPTION: Insufficient sleep quality and quantity degrade health and performance outcomes. This is particularly problematic in high-risk occupations where long hours of continuous operations along with circadian rhythm disruption negatively affect sleep quality and quantity and in turn increase the probability of mishaps and mistakes to occur; for example, fatigue was a contributory factor in the two deadly mishaps in 2017 with the USS Fitzgerald and USS McCain. Research focused on exploring the operational factors driving Sailor fatigue found that uncomfortable mattresses are a key contributor negatively impacting sleep quality and quantity [Refs 1,2]. In addition, the Government Accountability Office also identified poor mattress quality as a key factor for addressing fatigue in the Navy [Ref 3]. The objective of this STTR topic is to develop an enhanced mattress for Sailors and Marines aboard Naval vessels to increase comfort and sleep, and ultimately lead to improvements in sleep quality/quantity and reductions in fatigue and operational risks. Innovative mattress technologies must comply with current berthing area dimensional requirements and standards (i.e., fire-proofing, floatation, etc.) outlined in MIL-PRF-32568. Mattress testing (e.g., ASTM E1590) needs be conducted in compliance with MIL-PRF-32568. The enhanced mattresses must demonstrate improved comfort, as subjectively rated by Sailors; and improved sleep quantity and quantity, as objectively defined by gold standard sleep assessment techniques in lab and shipboard-based settings (e.g., polysomnography/ electroencephalography, wearable devices, and subjective assessments) when compared to current shipboard mattresses.

To develop these mattresses and ensure that they provide a higher level of comfort and ultimately sleep quality/quantity, the proposer should partner with an academic and/or clinical institution that has experience in performing evaluations and human sleep research studies. The proposer should work with the academic/clinical institution to ensure there is a valid and reliable improvement in comfort and sleep quality/quantity using the newly developed prototype mattresses over the currently deployed standard shipboard mattresses. It is expected that mattresses will be examined initially in a lab-based setting and then in a shipboard environment once all necessary standards and regulations have been addressed; and demonstrate improvements in comfort and sleep quality.

PHASE I: Define and develop an enhanced shipboard mattress prototype model/concept which meets the dimensional and safety requirements for placement aboard Naval vessels (MIL-PRF-32568); and provide the theoretical justifications for why new mattresses will improve comfort and sleep quality/quantity over the current standard shipboard legacy mattresses. Determine feasibility of scalability of incorporating the new enhanced mattress across the Fleet to replace the legacy mattresses. Required Phase I deliverables to include, but are not limited to, enhanced shipboard mattress concept including planned materials, costs, manufacturing logistics, planned testing (e.g., ASTM E1590), and planned partnership with academic/clinical institution.

The academic/clinical institutional partner should work with the small business to develop a plan for human subjects testing that would take place in subsequent award phases. The deliverables from the small business and academic institution partnership should describe the means and methods to ensure valid and reliable assessment of the mattress performance compared to the legacy mattresses; these are to include

but are not limited to: targeted sample population for testing, lab space for testing, IRB and research protocol preparation, and analytical methods.

PHASE II: Develop and limited production of prototype enhanced mattresses that meet the dimensional, safety, and improved comfortability aspects defined in Phase I and the Description. The enhanced mattress should be developed and validated for improving sleep quality and quantity via standard sleep assessments in partnership with the academic/clinical institution (e.g., polysomnography, other wearable devices, subjective ratings). The prototype should be capable of being deployed aboard Naval vessels at the end of Phase II. Prototypes need be tested for compliance with Navy standards (e.g., ASTM E1590) in Phase II prior to deployment aboard ships. The deliverables from the Phase II are to include, but are not limited to: 12 prototype mattresses (6 of which are to be provided to the Navy for testing) and 6 of which can be used by the small business and academic/clinical partner to conduct internal testing outlined in the Phase I deliverables. Three (3) initial prototype mattresses should be delivered to the Navy within 3 months of Phase II award; with the additional 9 to be completed for the Navy within 6 months of award. Human testing should be completed and reported on within the second 6 months of the award (12 months after award).

PHASE III DUAL USE APPLICATIONS: Focus on refining the product for transition and integration into the U.S. Navy surface fleet. This is to include preparations for scaling production of the enhanced mattresses and obtaining required certifications/accreditations for deployment to surface ships. In partnership with the small business, the Navy will integrate the Phase II developed enhanced shipboard mattress as a production product into deployed Naval vessels and transition finalized product to Naval Surface Force. Aim at scaling the production mattresses to commercial markets (e.g., recreational vehicles, garrison bunks). Continued research and development to ensure wide scale acceptance for increased sleep metrics is to be conducted simultaneously.

The final product to be delivered to the Navy should meet all dimensional and safety requirements as specified (MIL-PRF-32568), while also demonstrating increased comfort and sleep quality/ quantity for the users over previous Navy bedding.

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KEYWORDS: Sleep; fatigue; performance; mattress; berthing; comfort

Version 2

N25A-T018 TITLE: High-Power Microwave Phase Shifter

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials;Directed Energy (DE);Microelectronics

OBJECTIVE: Develop a high-power phase shifter and feed network in X-band capable of withstanding peak powers of above 100 MW and pulse durations of threshold of at least 10 ns with an objective of 200ns. The phase shifter ideally would be capable of operating within a fraction of a frequency band aligned to available waveguide, have insertion loss of < 0.5 dB with 0 – 360° phase control. The device should be sealed sufficient to maintain high vacuum which would be maintained by pumps located elsewhere along the connecting waveguides. The device should have a tuning agility to move across 180 degrees in less than 100 ms and have a tuning resolution better than 1 degree.

DESCRIPTION: Microwave phase shifters are commonly used for phase steering arrayed antennas but do not operate at the typical peak powers seen in High Power Microwave (HPM) systems. HPM sources may produce output powers of several GW which can be divided and fed to an arrayed antenna network. These sources are typically operated at high vacuum (< 10-5 torr) along with the output antenna and waveguide networks for insulation purposes. At such vacuum levels, field stresses up to 80 MV/m are tolerable for the short durations produced by these source technologies.

HPM sources in this regime have bandwidths less than 1% of the center frequency for any output pulse and 3 dB pulse widths less than 50 ns. In some instances, multiple or tunable sources may be used which have varying output frequencies from shot to shot, each with the sub 1% bandwidth.
HPM systems in this regime typically operate with maximum repetition frequencies of a 10s to 100s of Hz for burst durations around a second. If the tuning agility is sufficiently fast, it may be possible to adjust the phase of the antenna elements from shot to shot within a burst.

Based on these considerations, the following design requirements should be considered for the phase shifter:

- Input Power: 100 MW or greater
- Pulse duration: 10 ns to 200 ns
- Operational frequency: X-Band
- Bandwidth: 1% or more of design frequency
- Vacuum: < 10-5 torr
- Tuning resolution: < 1 degree
- Tuning agility: 1.8 degree / ms
- Phase control range: 0-360 degree
- Insertion loss of < 0.5 dB
- Minimize electrical scale of the component
- Minimize cost per device

PHASE I: Develop and numerically simulate a design demonstrating the performance capabilities of the phase shifter that meet the requirements above. Build scaled proof-of-concept hardware to validate modeling. Provide a plan for the Phase II effort.

PHASE II: In consultation with ONR, proceed to fabrication of a phase shifter to be tested at full power using government provided source hardware. In addition to single element testing construct a sub array of as few as 16 elements for testing of steering capability. Document the performance and design of the developed hardware and sub array.

PHASE III DUAL USE APPLICATIONS: HPM weapons of both defensive and offensive designs could utilize a developed phase shifter to integrate on a number of candidate platforms. In addition, HPM technology is also usable in high-power radar and the Electronic Attack (EA) subsystems for Electronic Warfare (EW). In consultation and with ONR approval, proceed to building multiple phase shifters to integrate into a full phased steered HPM system utilizing an arrayed antenna provided by the government. Implement a control system to coordinate the phase shifters. Extensive commercial applications include phased array antennas for radar systems, satellite communications and 5G networks with additional applications in test/measurement equipment for RF systems and medical devices.

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KEYWORDS: High Power Microwave; HPM; Microwave; Phase shifter; Electromagnetic; Beam-steering; Antenna

N25A-T019 TITLE: Passive Automatic Dependent Surveillance-Broadcast Validation to Support Collision Avoidance During Missions Requiring Emissions Control

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Integrated Sensing and Cyber

OBJECTIVE: Develop a passive means to validate Automatic Dependent Surveillance-Broadcast (ADS-B) tracks that is suitable for aircraft operating with no radio frequency emissions.

DESCRIPTION: The Navy desires a system that validates ADS-B independently of any active sensors. The use case for this STTR topic is a large U.S. military Unmanned Aircraft System (UAS) engaged in operations not conducted under International Civil Aviation Organization (ICAO) flight procedures. Procedures exist for military aircraft operations in international airspace consistent with the Convention on International Civil Aviation [Ref 1]. When following ICAO flight procedures is not practical and compatible with the mission, U.S. military aircraft must operate with due regard for the safety of all other aircraft including civilian aircraft consistent with DoDI 4540.01 [Ref 2]. Such a case might arise, for example, when the mission necessitates that radio frequency emissions be minimized or eliminated altogether.

The UAS commander must utilize all available resources and information in assessing an acceptable level of risk before conducting such operations [Ref 2]. At present, there is no technology by which a UAS might passively validate ADS-B tracks broadcast by other aircraft. The goal of this STTR topic is to create that technology and make new information available.

ADS-B is an aviation surveillance technology in which an aircraft periodically broadcasts its position and other related data, enabling it to be tracked. It consists of two distinct functions. "ADS-B Out" and "ADS-B In". "ADS-B Out" is active; it periodically broadcasts track information like identity, position, and velocity. "ADS-B In" is passive; it receives and processes "ADS-B Out" information transmitted by other aircraft [Ref 3]. Civilian flights in oceanic airspace must be conducted under Instrument Flight Rule procedures when operating at or above Flight Level 055 within the New York, Oakland, and Anchorage Oceanic Flight Information Regions. ADS-B is required under these rules. Oakland Center covers 18.7 million square miles of the Pacific Ocean, roughly 9.5% of the Earth's total surface area, making this the largest Area Control Center in the world by controlled surface area [Ref 4].

ADS-B tracks received by the UAS “ADS-B In” receiver inform collision avoidance maneuvers. Independent validation of the received data is necessary to prevent the UAS from responding to spoofed or erroneous tracks. In a permissive environment, Traffic Alert and Collision Avoidance System (TCAS) and radar could be acceptable means for validating ADS-B. Both of those are active so neither work when UAS operates under emission control. UAS with SIGNIT could conceivably validate the track’s relative bearing against the direction of arrival of the “ADS-B Out” broadcast but not all UAS have such systems. Modifying the ADS-B receiver to measure angle of arrival is cost prohibitive.

Comparing observed changes in received signal strength over time to the expected changes corresponding to changing distances between aircraft seems a promising approach to passively validating “ADS-B In.” This is a variation of the approach in Reference 5 that estimates distance to the other aircraft but needs an initial calibration. The Navy desires to determine whether the change in received signal strength correlates to the propagation path change over time, so the initial calibration might not be needed. Reference 6 shows that the propagation path loss agrees with free-space propagation. That experimental effort included a method to remove the effect of the alternate transmission from the two transponder antennas on the aircraft of interest. Reference 7 demonstrates a means to remove co-channel interference that degrades the signal strength measurement accuracy.

PHASE I: Plan and execute a series of experiments to passively validate "ADS-B In" by correlating changes in reported separation with actual changes in received signal strength. Take received signal strength's natural variability into consideration. While these experiments can be conducted with the receiver located on the ground consideration should be given to how the results may differ when implemented on aircraft.

PHASE II: Design and build a system that implements the techniques of Phase I and complies with flight critical software safety requirements. Conduct airborne experiments that demonstrate performance of the approach sufficient to certify that the system meets the requirement for independent validation.

PHASE III DUAL USE APPLICATIONS: Mature the capability sufficiently to transition as a software upgrade to airborne ADS-B system processing hardware. The technology is directly applicable to use by civil aviation particularly low-SWaP UAS incapable of supporting an active surveillance capability.

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KEYWORDS: Airspace access; Automatic Dependent Surveillance-Broadcast; ADS-B; emissions control; collision avoidance, safe separation; unmanned aerial systems; UAS

N25A-T020 TITLE: Generation of Infrared Synthetic Data for System Training and Testing

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Sustainment

OBJECTIVE: Develop and demonstrate synthetic generation of infrared imagery in a realistic ocean background for objects and vessels on the ocean surface in order to improve operator training and testing of infrared camera systems.

DESCRIPTION: There are a wide variety of high quality camera systems for examining the ocean surface from platforms in the air, sea surface or elsewhere. Synthesis of electro-optic data in the visible spectrum for such systems is a well-developed field and represented in many government-owner and commercially available software tools, particularly in the game industry. Products such as Unreal Engine, Unity and Blender provide tools for 3D image creation in a virtual world that captures physically realistic details of optical effects. The Navy seeks analogous solutions that operate in infrared bands such that imagery from infrared camera systems can be generated for purposes of operator training, system testing, and related applications. Proposals should address the commonly available infrared camera resolutions and consider the effects of all physically relevant infrared phenomena. These include, but are not limited to scattering, emissions sources, and environmental conditions from wind and the ocean.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and ONR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Identify a modeling approach for infrared spectrum imagery generation whose implementation can be justified on the basis of physical/optical realism, environmental effects, and in a computational feasibility. Analyze key design considerations assuming a commercially available electro-optic/infrared camera system of at least 640x480 pixel resolution. Assess the strengths and weaknesses of the proposed approach and report results for the solution to be developed in Phase II.

PHASE II: Develop and test a prototype for the proposed approach. Complete preliminary performance testing using open source surrogate data examples. Apply the prototypic software to generation of data given specifications provided by the Navy for up to 10 example objects on the ocean surface. It is possible, but not certain that the Phase II work will involve classified information.
It is possible that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: Phase III work will extensively test the prototype fabricated in Phase II and examine results under day and night conditions and well as performance in suboptimal environments and conditions. Potential dual use applications for this technology reside with automation of thermal sensing and imaging systems that require data for model training and system validation. In particular, sensors for evaluating structural heat loss in buildings and vehicles would benefit as well as automatic airborne sensing for search and rescue missions.

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KEYWORDS: synthetic data, imagery, infrared, simulation

N25A-T021 TITLE: Non-Wearable and Off-Engine Jet Noise Reduction

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Human-Machine Interfaces;Sustainment

OBJECTIVE: Develop and demonstrate a non-wearable, off-engine, jet noise reduction (JNR) solution that is capable of significantly reducing afterburning turbofan/turbojet jet noise levels during testing, ground and test cell operations, and take-off and landing events, thereby reducing the negative health and environmental impacts associated with jet noise.

DESCRIPTION: The U.S. Navy seeks a non-wearable, off-engine JNR solution to address the significant noise generated by afterburning jet engines during ground and indoor testing, aircraft taxiing, takeoff, and landing operations. Current state-of-the-art noise reduction technologies consist of:

- Barriers and headsets that attenuate and reflect sound
- Active sound sources that reduce noise through destructive interference
- A combination of the previous two principles

Engine mounted noise reduction technologies have been explored in the past, but such state-of-the-art solutions for tactical aircraft tend to negatively impact thrust performance while providing little noise reduction benefit for low bypass turbofan and turbojets, which makes them undesirable in a military application. Passive and Active Noise Control (ANC) equipped headsets are known to significantly reduce perceived jet noise levels, but they are insufficient for mitigating the sound emanating from modern afterburning engines. Further headset noise reduction can be accomplished through more invasive earpieces, but they come with other challenges. To further reduce perceived jet noise levels, the Navy is interested in non-wearable, off-engine, jet noise reduction technologies that complement the benefits of modern headset. This STTR topic focuses on:

- Developing advanced noise reduction techniques that can be implemented off-engine and benefit a broad coverage area
- Delivering a flexible solution that addresses the noise signatures of various U.S. Navy jet engines
- Developing a system that delivers noise reduction at the necessary amplitudes and frequency ranges needed to be effective
- Ensuring compactness and compatibility with current operating environments, minimizing need for infrastructure modifications
- Creating a solution that is energy efficient to minimize the burden of powering the system during extended operations
- Utilizing robust, durable technologies that can survive the harsh thermal, acoustic, vibratory, corrosive, and EMF environment of an aircraft carrier

Solutions should aim to:

- Achieving significant noise reduction (10dB+ desired) in the audible frequency range, especially at frequencies below 5kHz
- Providing tonal and broadband noise reduction

The resulting technology should:

- Function effectively at relevant distances from the jet engine exhaust centerline
- Be compact, lightweight, and portable to minimize impact on fleet operations.
- Be easily deployable, requiring a reasonable amount of time for setup and activation in various environments
- Operate reliably for extended periods of time in a typical Naval operating environment, which may consist of extreme weather and highly corrosive conditions
- Have broad application across engine operating range at various power levels
- Be responsive to rapid changes in engine power or acoustic disturbances

- Be responsive to changing noise source positions and environmental factors that influence acoustic propagation
- Ensure that noise is not increased beyond acceptable levels in an undesirable location

Example candidate technologies may include, but are not limited to:

- Off-Aircraft, distributed sound generators that synchronize with jet noise to destructively interfere at desired locations
- On-Aircraft, but off-engine, distributed sound generators that destructively interfere with jet noise
- Temporary fluid-based barriers, such as water/bubbly water curtains, that significantly attenuate/reflect/scatter jet noise with minimal impact to personnel/aircraft mobility. These fluid-based barriers should be sufficiently far from the jet plume to avoid interaction.

Please note that the Navy is not interested in the following types of technologies for this topic:

- Passive or dynamic solid barriers
- Passive or active noise cancelling headsets/headphones
- On-aircraft water injection system that interact with jet plume
- Engine, or nozzle, mounted flow actuators, tabs, or chevrons

PHASE I: Determine the feasibility and develop a concept for a non-wearable, off-engine jet noise reduction prototype system. The awardee will be expected to perform modeling and simulation to identify and design an innovative approach that meets the specified noise reduction, operational range, portability, and durability requirements. Phase I efforts should include the following:

1) Define and develop a detailed concept for the noise reduction system, including the underlying hardware, sensors, software, methodologies, and their integration.
2) Perform modeling and simulation to demonstrate the feasibility of the proposed concept in achieving the desired noise reduction within the specified parameters.
3) Develop a preliminary design for the prototype system, outlining key components, system architecture, and expected performance.
4) Provide a detailed report summarizing the concept development, feasibility analysis, and preliminary design. This report should include:
    a) Technical approach and innovation,
    b) Simulation results and analysis,
    c) Design specifications and proposed materials/components, and
    d) Implementation plan for Phase 2, including a projected timeline, milestones, and budget estimates.

The Phase I effort should culminate in a proof-of-concept demonstration, through simulation or fundamental testing, that substantiates the proposed technology's potential and justifies further investment in Phase II.

PHASE II: Develop, demonstrate, and validate a fully functional prototype of the non-wearable, off-engine, jet noise reduction solution based on the concept and feasibility analysis conducted in Phase I.

Phase II efforts should include the following:

1) Use the results from Phase I to design and fabricate a prototype system that meets the specified noise reduction, operational range, portability, and durability requirements.
2) Conduct testing to validate the performance of the prototype in reducing jet engine noise. Testing should first be carried out in a controlled environment and later in a realistic environment. The demonstration should showcase the system's effectiveness, ease of use, and operational reliability.

3) Provide detailed documentation of the prototype development efforts including hardware/software design, performance models, and simulation/test results.
4) Deliver a final report summarizing the development, demonstration, and validation of the prototype.
Phase II efforts should result in a validated prototype that demonstrates the capability to significantly reduce jet noise in a broad defined area.

PHASE III DUAL USE APPLICATIONS: Focus on the commercialization and deployment of the proposed technology within the U.S. Navy and other potential defense and civilian applications. The technology will become fully operational, validated, and production ready in Phase III and will be capable of significantly reducing jet noise levels during ground and indoor testing, aircraft taxiing, takeoff, and landing operations. The primary use cases for this technology beyond Phase II will be at Navy jet engine test facilities, on naval aircraft carriers and at Naval bases where jet noise is a significant concern. This technology has significant commercialization potential because it is applicable to commercial test facilities and airports as well. This technology could also be beneficial for powerplants and other industrial applications that use gas turbine engines.

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KEYWORDS: Jet Noise, Jet Noise Reduction, Active Noise Cancellation, Destructive Interference, Barriers, Noise Curtain, Turbofan, Turbojet, Gas Turbine, US Navy Carrier, US Navy Bases, Jet Engine Test Facilities

N25A-T022 TITLE: Photonic Bump Bonds

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials;Microelectronics;Space Technology

OBJECTIVE: Develop an enabler of heterogeneous photonic integrated circuits (PICs), the photonic equivalent of an electrical bump bond.

DESCRIPTION: Most radio frequency (RF) photonic components are created in the plane of the thin film substrate in order, or in part, to allow structures to be large in at least 1 dimension to compensate for, e.g., low electron-photon coupling. For optimum performance of any photonic circuit, the constituent components are best made in the material that individually optimizes their performance. Thus, the PIC of greatest application relevance requires moving the optical beams carrying the signals of interest between different chiplets of different chemical composition, lattice structure/spacing, and sometimes different substrates. Various methodologies are at modest TRL levels to interconnect such edge (of chiplet) emitting species including butt free space coupling, microlens, photonic wire bonds, V-grove alignment, and tapered transitions. However, the optical modes are generally very compact within the launching and receiving waveguide structures, so sub-micron 3D alignment of the 2 sides is crucial to the optical insertion loss. The tendency for most epoxy glues holding actual fibers in place to change shape and size (hence torques on fibers) when cured or subjected to temperature excursions is a major issue. Thus, routinely getting the per facet optical loss down below 0.1 dB is so far illusive. However, in the case of VCSELs, which emit light perpendicular to the substrate, the commercial market for high bandwidth data flow through arrays of fibers has driven the development of an interposer technology which can today yield the desired 0.1 dB loss per facet metric. This STTR topic seeks the development of a technique for planar photonic devices to turn their output beams from the in plane to vertical direction so that they can be packaged together in 3D manner, ideally from bumps located within the body of the chiplets, not just along its edges. Such bumps would allow the highest possible performance component chiplets to be combined into the smallest possible finished die, minimizing the discrepancy between today's extremely dense digital circuits and photonics. The same sort of bumps should be used to provide the multi-chip wiring module using low loss SiN waveguides and normal CMOS interconnects and wiring structures.

PHASE I: During the base, the right angle turn method described in the proposal shall be developed, at least by simulation, to the point where the technical risk of failure is arguably low. A first experimental demonstration is preferred. The non-SiP component can be of any type of functionality, so little effort shall be expended on improving its internal operation. However, the test plan included in the original proposal must explain how the performance of the joint will be quantitively measured. Write the preliminary Phase II proposal. During the Phase I option, if exercised, complete the proof-of-concept demonstration and negotiate the Phase II contract using a technically detailed proposal and clear work plan.

PHASE II: Work to improve the performance and reliability of the demonstrated link method(s). That is, work to improve the yield of joints with insertion losses of less than 0.1 dB, with threshold performance of >85% and objective level of >97% yield. Invent and demonstrate a way to achieve self-alignment of the mating structures. Deliver test articles to government for verification of performance claims.

PHASE III DUAL USE APPLICATIONS: Demonstrate the utility of this packaging technology by integrating a heterogeneous PIC having more than 5 optical components made from more than 2 different materials and relevance to a military function.

Development of a packaging technology with substantially improved internal integration losses will allow planar RF photonic circuits to be sold in a wider array of settings than currently possible. Attention will

then be focused on what functional performance the rf photonic circuit can provide, not have to continuously fight with excessively large noise figures created by the floor of the end-to-end insertion losses.

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KEYWORDS: bump bonds; vertical packaging; heterogeneous integration; 3D packaging; self-alignment; body pads

N25A-T023 TITLE: Robotic Manipulation Learning for Repair of Marine Diesel Engines

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Human-Machine Interfaces;Trusted AI and Autonomy

OBJECTIVE: Develop autonomous robotic manipulation capabilities using artificial intelligence (AI) to repair and maintain marine diesel engines operating during long missions without human intervention. The overall objective of this topic is to provide the means to extend the duration of missions with uncrewed surface vessels and to provide greater assurance of mission completion.

DESCRIPTION: The Navy is developing a family of uncrewed surface vessels. Current designs utilize marine diesel engines for propulsion. These vessels are expected to conduct autonomous operations on long distance missions without sailors being placed aboard to maintain equipment. It would be desirable to have robotic systems onboard capable of inspection and basic maintenance tasks to enable these vessels to achieve the mission objective area. Due to bandwidth limitations, the ability to conduct teleoperated robotic manipulation is unlikely. There have been recent advances in AI and machine learning (ML) for robotic manipulation that exploit copious internet online video manipulation examples or learning from human demonstration, combined with foundation models and reinforcement learning to achieve autonomous manipulation. However, the manipulation tasks selected are often pick and place, or typical kitchen tasks. There is a need to transfer ML for manipulation to the more demanding tasks involved in equipment maintenance, which often involve tool use, significant torques, deformable (soft/squishy) object handling, difficult to access locations on equipment, or bimanual manipulations. This STTR topic focuses on manipulation, not on robotic locomotion. Maintaining marine diesel engines to sustain fully operational status involves a number of manipulation and inspection tasks. These manipulations include valve opening, rotation or depression of switches, rotation, removal or replacement of lids, adding fluids, adding or removing rubber gaskets and washers, catching liquids in containers, and tightening/untightening bolts, and reattaching loose components. Recent advances in computer vision have been made for object identification that will be needed for both manipulation and inspection. Inspection tasks include checking for deformations, cracks, leaks and loose components (brackets, bolts, hangers), recognition of warning lights, and reading gauges or status message text. This topic seeks the development of robotic manipulation technology capable of such tasks. This would include machine vision, robotic arms, and end effectors, with possible tool attachments. Operation at sea presents mechanical stability challenges, so design concepts are needed that would allow the robotic assembly to be stabilized, yet reach the essential engine components from the appropriate perspective. This could include sliding on a frame around the engine, or locomotion on a stable base.

PHASE I: Develop a design for robotic manipulation capable of the basic tasks involved in diesel engine maintenance, exploiting emerging research on ML of dexterous manipulation. This ML can be based on databases of video for engine repair manipulation tasks or on learning from human demonstration with reinforcement learning. The robotic manipulation concept includes designs or identification of robotic arms, end effectors, stable base and sensors for machine vision. Design a frame or base and means of shifting the position of the arm(s) to reach different areas of a diesel engine. Assess the feasibility of tool use, or end effector adaptations to accomplish basic elements of engine maintenance. Identify suitable cameras and machine vision processors capable of inspections that support diesel engine maintenance. Assess the feasibility of tactile or torque sensing to enable closed loop control of manipulators. The overall approach chosen should be mindful of the ultimate goal of autonomous manipulation, without teleoperation.

PHASE II: Develop prototype hardware and software to validate the concepts developed in Phase I. This can include commercial off-the-shelf (COTS) robotic components. Demonstrate the learning and execution of maintenance relevant robotic manipulation tasks on a diesel engine, or close surrogate.

Possible tasks include valve opening, rotation or depression of switches, rotation, removal or replacement of lids, adding fluids, adding or removing rubber gaskets and washers, catching liquids in containers, and tightening/untightening bolts, and reattaching loose components. While the training of this system can involve teleoperation, the final demonstration should show autonomous manipulation to execute the sample tasks. Develop a prototype machine vision system to work with the robotic manipulators for basic inspection tasks and for closed loop control of the manipulation task.

PHASE III DUAL USE APPLICATIONS: Demonstrate and evaluate the robotic manipulation technology at a Naval LBES facility with marine diesel engines and on board a Navy uncrewed surface vessel with a marine diesel powerplant in coordination with PMS 406.
This technology has strong commercial potential since many commercial vessels use marine diesel engines. This technology would also be applicable to fleets of trucks using diesel engines.

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KEYWORDS: Robotics; manipulation; maintenance; machine learning; diesel engines; machine vision

Version 2

N25A-T024 TITLE: In-situ Seabed Physical Properties Probe (ISPPP)

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Sustainment

OBJECTIVE: Develop a probe to rapidly gather in-situ data on physical properties of seabed and shallow sub-subsurface sediments in either a tethered (reusable) or expendable form factor.

DESCRIPTION: Characterization of sediments on the seabed and in the shallow-subsurface is a vital requirement for research, offshore infrastructure development, acoustic propagation measurements, and target detection. Traditional methods of characterizing these sediments usually require shipboard sample collection and laboratory testing or specialized in-situ testing equipment requiring complex handling and data processing steps. Free-fall penetrometers, acoustic probes, and environmental sampling sensors are routinely deployed individually to estimate the physical properties of seabed sediments for use in geologic and geophysical assessment of seabed character, geoacoustic inversions, and transmission/bottom loss calculation; however, the time required for sample collection or in-situ measurement to data availability prohibits the utilization of these data in dynamic seabed environments. These instruments are also rarely used together, from autonomous platforms, or in expendable form factors. This STTR topic seeks development of an in-situ seabed physical properties probe to provide rapid seabed and shallow sub-surface sediment characterization in a tethered or expendable form factor.

PHASE I: Develop a concept for a seabed sediment physical properties probe capable of measuring/estimating environmental parameters (e.g., temperature, salinity) in the water column and seabed/sub-seabed (pore water), sediment sound speed, and sediment strength properties (e.g., undrained shear strength, friction angle and cohesion), with reporting of estimated seabed/shallow sub-surface character (e.g., sound speed ratio, grain size, porosity). Describe the mechanistic underpinnings of the probe and support it with models and simulations. Simulations should demonstrate the measurement resolution and uncertainty in sediment classification as a function of penetrated sediment depth in a representative ocean environment. The conceptual design should operate at ocean depths up to or greater than 4000m.

PHASE II: Develop, build, and demonstrate a hardware version of the probe based on the Phase I concept. The demonstration probe does not have to be expendable but should demonstrate the capabilities for acquisition and transmission of the sediment characteristics/physical properties stated in the requirements. Demonstrate that the probe is capable of penetrating the seabed and providing measurements at accurate sub-surface depths. Develop a capability for operation across a range of ocean depths and seabed types, with accurate measurement geo-positioning. Develop a plan in consultation with the Navy for demonstrating the probe to include compliance with environmental regulations. After demonstrations, develop a design for a tethered (reusable) version deployable from autonomous platforms or an expendable version of the probe with compact form factors that can communicate with remote receivers.

PHASE III DUAL USE APPLICATIONS: The expected transition will be a tethered or expendable acoustic in-situ measurement probe that can be used for seabed characterization and geotechnical surveys for naval or civil applications (e.g., reconnaissance geotechnical surveys, post-disaster seabed stability assessment, seabed AUV survey groundtruth measurements). The tethered version can be deployed from autonomous vehicles. The expendable version can be deployed by hand or with use of a launcher from the deck of a ship for operational surveys or in support of basic and applied research.

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KEYWORDS: Underwater Acoustics; seabed characterization; physical properties; Geoacoustic Inversion; geotechnical survey; sediment

N25A-T025 TITLE: Expendable Air-sea Profiling Observations in Hazardous Weather Conditions

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Integrated Sensing and Cyber;Sustainment;Trusted AI and Autonomy

OBJECTIVE: Develop an improved lower atmospheric observing capability that is hardened for hazardous weather environments to better capture vertical, horizontal, and temporal variations in atmospheric profiles of dynamics and thermodynamics to the air-sea interface compared to standard radiosonde profiling.

DESCRIPTION: Accurate assessment of the environment is a critical prerequisite for forecasting future environmental states and decision making from those forecasts. Remote and hazardous operating environments are among the most challenging to observe and constrain in a forecast system. Understanding and analyzing hazardous weather conditions, such as convective precipitation and tropical cyclones, as well as complex air-sea interactions such as sea ice formation and break-up in marginal ice zones, remain particularly difficult. Although recent initiatives have aimed to enhance the quality and accuracy of environmental observations (see References), operational meteorological and oceanographic (METOC) agencies still rely heavily on legacy profiling and remote sensing techniques. There is often insufficient quantity of observations due to cost per unit as well as quality of data due to the operating environment difficulties on the sensors. This STTR topic aims to build on these recent efforts by developing advanced observing technologies that can provide higher-quality data, complementing existing METOC systems in challenging environments.

This STTR topic solicits novel research for the development of a robust technological candidate aimed at improving expendable profiling of air-sea phenomena in hazardous weather conditions. In particular, a compact, easily deployable, and competitively priced alternative to radiosondes is sought that will provide a higher quantity and quality of air-sea interface related observations in areas such as tropical cyclones and polar lows where more precise information of dynamic and thermodynamic properties such as surface drag and fluxes can be constrained. Proposed work should clearly indicate where the effort will expand beyond current radiosonde capability in one or more of the following: spatial observations, temporal observations, sampling rate, price per unit, durability in hazardous conditions, uncertainty characterization, and ability to target/collocate with other observational platforms. The performer should also aim for a production schedule that allows for the prototyping dozens or even hundreds of units during the validation phase, with the potential to scale up to larger operational thresholds for commercialization or transition.

PHASE I: Determine technical feasibility for improved expendable in-situ profiling observations in hazardous weather conditions of dynamics and thermodynamics influenced by air-sea interactions. Perform background review of strengths and limitations of current observing platforms with an emphasis on measurement uncertainties and logistical challenges. From related operational and research work in this area, identify and design an improved concept of measurement techniques that include novel approaches to addressing the limitations identified in the review. Where appropriate, perform a comparative analysis with current state-of-the-art platforms (such as radiosondes or well calibrated remote sensing) including uncertainty specifications and real-world results from validated field efforts. Required Phase I deliverables will include a report on the state of the science, an overview of previous/current concepts of operations and their accuracy, and the proposed engineering, hardware, software, and/or platform upgrades that will enhance precision, accuracy, and/or quantity of air-sea in-situ observations in challenging environmental conditions. In particular, reporting must include a discussion of how the proposed effort will uniquely enable new environmental observations as compared to radiosondes and must include a comparison of strengths/weaknesses in cost, longevity, sampling rate, and spatial/temporal coverage.

PHASE II: Using results from the Phase I, proposed effort should develop, fabricate, demonstrate, validate, and iterate on expendable profiling technology. Work in the Phase II should particularly focus on calibration and demonstration of improved observational accuracy in hazardous weather conditions. Multiple development spirals of improved engineering prototypes and real-world demonstration in conjunction with currently used sensing technologies are needed to understand sensor characteristics and robustness in desired conditions. To the extent possible, technology deployment should leverage existing systems for ground, ship, and/or aircraft-based delivery mechanisms, thereby supporting comparative calibration/validation metrics and facilitating transition from a prototype to operational system. Part of this effort must also include data ingest and assimilation into operational forecast models with a comparative analysis of added value and analysis/forecast improvement. Phase II work should use multiple prototype demonstration efforts to refine and focus innovative measurement aspects and improve data measurement quality, in addition to refining the concept of operations and solidifying the observational niche of the proposed technology.

PHASE III DUAL USE APPLICATIONS: Phase III work should build upon the previous prototyping and demonstrations to effectively prove the operational concept and mature the technology into a new expendable profiling platform that can be commercialized. The technology must be calibrated and validated in real world conditions and compared to observations that are currently used in an operational meteorology and oceanography environment, such as in-situ stationary platforms, other expendable data, and remote sensing. The expected state of the technology should be a system that can deploy and track observations in real-time of expendable sensors from ground, ship and/or via aircraft. Success rate must match that of currently operational radiosonde technology, with improvements in specific hazardous weather conditions (such as tropical cyclones, polar lows, heavy convective precipitation, large wave events) with a demonstrated decrease in failure rates and modes. Final transition/commercialization will entail regular production of well characterized sensors that can be deployed operationally and in field missions for targeted air-sea profiling observations to be transmitted and used in real-time for environmental forecast and analysis.

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KEYWORDS: Radiosonde; expendable; profile; meteorological and oceanographic; METOC; meteorology; oceanography; autonomous; temperature; humidity; wind; flux; marine atmospheric boundary layer; tropical cyclone; polar low

N25A-T026 TITLE: Automatic Readiness and Training Systems Enhanced with Artificial Intelligence, Machine Learning, and Optimization

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Computing and Software;Integrated Sensing and Cyber;Trusted AI and Autonomy

OBJECTIVE: Develop an automated, artificially intelligence-enhanced system for scalable effective training and monitoring of skilled physical tasks. Proposed solution should be demonstrated in a domain of interest to the Department of Defense such as maintenance, operation of military equipment, surgery, manufacturing, or medical tests.

DESCRIPTION: Ensuring the readiness of the military requires the execution of a large number of skilled physical tasks in a variety of areas such as aircraft maintenance, operation of detection systems, medical care, production of advanced materials or equipment, and medical tests. Thus, it is critical to train and certify personnel to perform these tasks and to provide monitoring that supports ongoing quality improvement, through both individual feedback on performance and data collection for optimizing how the task is performed. The training environments range from centralized facilities to infrastructure-poor and geographically isolated locations with limited network connectivity.

Historically, training and performance feedback has been provided via direct supervision of trainees by experts. This approach is limited by the number of qualified experts available and the geographical location of trainees. A single expert can train and supervise a limited number of personnel at a time and must travel to the trainees' location. This prevents rapidly scaling capabilities relying on skilled physical tasks. When the experts available are too few to provide adequate ongoing supervision, it can also result in low-quality work-product. Moreover, recording data on how the work is performed can be time consuming, requiring tradeoffs between data collection and the amount of work accomplished. Finally, when experts leave their roles, knowledge is lost regarding how to perform specific tasks. While some training can be provided via written documents and videos alone without direct supervision, the quality of this training is low and such methods do not provide effective supervision or data collection.

The Navy seeks an approach for scalable effective training and monitoring of physical tasks that are mission critical and require extended concentration. This approach should be able to rapidly train and monitor thousands of personnel across many geographical areas with only one expert. It should use video and audio inputs to automatically record and process information about the work done to support individual feedback, optimization, and integration with other efforts that would benefit from information about work status. User input should not be required. It is envisioned that artificial intelligence and machine learning will play a key role in this approach as recent advances in these areas are poised to support automating many of the tasks that have historically been performed by individual experts.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and ONR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Provide a viable concept with detailed methodology to address the general training problem in the Description. A specific area of training in a domain of interest to the Department of Defense should also be identified, perform testing, and provide preliminary results. The concept should be generalizable beyond one specific area of training.

PHASE II: Develop a prototype for testing, demonstrating, and validating that the method improves performance for novice individuals on a specific area of training. Depending on the specific area chosen, this could result in the some of the work being classified.

It is probable that the work under this effort will be classified under Phase II (see the Description section for details).

PHASE III DUAL USE APPLICATIONS: Transition the prototype to a mature capability that will be used for a broad range of different areas of training.

The ideal end-state should be usable on handheld devices to leverage the video and audio technology there to provide input to the automated system, which would allow for feedback. Industrial and government sectors are expected to procure varying levels of this technology depending on the level of sophistication required by their training requirements.

REFERENCES:

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2. Yazdi, M. "Acquiring and Sharing Tacit Knowledge in Failure Diagnosis Analysis." Journal of Failure Analysis and Prevention, Vol 19(2), 2019, pp, 369-386. doi.org/10.1007/s11668-024-01904-y
3. Yilmaz, R; Bakhaidar, M and Del Maestro, RF. "Real-Time multifaceted artificial intelligence vs In-Person instruction in teaching surgical technical skills: a randomized controlled trial." Scientific Reports, 14(1), 2024. doi.org/10.1038/s41598-024-65716-8
4. Brooks Babin, LisaRe and Garven, Alice J. (Sena). "Tacit Knowledge Cultivation as an Essential Component of Developing Experts: A Literature Review." Journal of Military Learning, April 2019. https://www.armyupress.army.mil/Journals/Journal-of-Military-Learning/Journal-of-Military-Learning-Archives/JML-Apr-2019/Babin-Garvin-Tacit-Knowledge/
5. "National Industrial Security Program Executive Agent and Operating Manual (NISP), 32 U.S.C. § 2004.20 et seq. (1993)." https://www.ecfr.gov/current/title-32/subtitle-B/chapter-XX/part-2004

KEYWORDS: Artificial intelligence; machine learning; applied optimization; training

Version 2

N25A-T027 TITLE: Dynamic Thermal Management Suit

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials;Human-Machine Interfaces

OBJECTIVE: Develop and demonstrate a suit to be worn to protect an individual from thermal exposures in water (28-99°F) diving, capable of maintaining individual at a core body temperature of 95-99°F and maintaining the temperature of diver's extremities above 53.6°F. Suit shall be designed using new approaches and materials, that can insulate an individual submerged in water from hypo and hyperthermia. The system should work in helmeted or unhelmeted and in tethered and untethered conditions. The system should be capable of functioning in a highly contaminated environment.

DESCRIPTION: Thermal management in an aqueous environment places a significant challenge for human physiology due to the high thermal conductance of water versus air. Most locations in which military divers must dive require thermal protection. There is not currently a rugged, reliable system that can maintain normal physiologic temperature for the diver.

There are several engineering challenges. First in an untethered system the diver must carry any power source. Second, there is a specific requirement to keep the extremities warm enough to maintain manual dexterity. Third a tube suit or other wet suit materials compress at depth resulting in reduced insulation. Prior attempts have resulted in bulky designs that are difficult to both put on, take off and impedes mobility. Fourth and finally, when using a tethered system there may be heat loss/gain due to traversing the length of the tether resulting in reduced cooling or heating for the diver.

Consideration should be given to an innovative dynamic approach (e.g., additive manufacturing, metamaterials, bio-inspired hydrophobic gels) that shields the body from direct contact and allows the diver to maintain a normal body temperature. Innovative solutions that are dynamic and can adjust to cold and hot environments are most desirable. Mobility, and don and doff times should be comparable or better to those of current wetsuits.

PHASE I: Define and develop a design for a thermal-protective diving suit that is capable of use in a great range of water temperatures (28-99°F) and maintains an internal temperature that maintains a healthy core body temperature (95-99°F) and a temperature of > 53.6°F in the extremities for a 10-hour threshold duration. Prepare designs that are sufficiently detailed to specify all materials needed, their availability, how they will be implemented, and the overall thermal suit thickness. Specify how the design buffers the external water temperature to the body and will adapt in changing conditions. The suit material, its seams (both intergarment and at wrists, neck, and feet), and any closing mechanisms must stand up to typical underwater diver activities and approximately 100 dives. The design created in Phase I should lead to plans to build a prototype unit in Phase II.

Human factor and human subject testing are critical in follow-on Phases of this topic. Please carefully review the requirements of approval for proposals that include testing of human subject and compliance with Institutional Review Board (IRB): https://www.nre.navy.mil/work-with-us/how-to-apply/compliance-and-protections/research-protections. IRB approval is needed prior to award of Phase II. Performers must submit a proposal for IRB approval during Phase I.

PHASE II: Develop, fabricate, lab-test (in variable conditions), and provide two suits/systems for form, fit, and function evaluation by operational Navy divers in maritime environments. Within the period of performance, revise the design and refabricate an additional 10 units based on feedback.

PHASE III DUAL USE APPLICATIONS: Assist the Navy in transitioning the technology to operational use, support the Navy for test, validation, and qualification of the system for use by military divers, and

develop commercial variants suitable for commercial, scientific, and recreational divers. Create a marketing plan for reaching civilian users and mass production, to bring the per unit cost down to under five hundred dollars.

REFERENCES:

1. Piantadosi, C. A.; Ball, D. J.; Nuckols, M. L. and Thalmann, E. D. "Manned Evaluation of the NCSC Diver Thermal Protection (DTP) Passive System Prototype." US Naval Experimental Diving Unit Technical Report (NEDU-13-79), 1979. http://archive.rubicon-foundation.org/xmlui/bitstream/handle/123456789/3356/NEDU_1979_13.pdf?sequence=1
2. Bardi, Jason S. "How Does Fur Keep Animals Warm in Cold Water?" American Physical Society, College Park, MD, November 23, 2015. https://www.aps.org/units/dfd/pressroom/news/2015/upload/7pr-animals2015.pdf
3. Nuckols, M.L. “Analytical modeling of a diver dry suit enhanced with micro-encapsulated phase change materials.” Ocean Engineering, Volume 26, Issue 6, 1999, pp. 547-564. ISSN 0029-8018, https://doi.org/10.1016/S0029-8018(98)00001-8 https://www.sciencedirect.com/science/article/pii/S0029801898000018

KEYWORDS: Diving, Wetsuit, Drysuit, Hypothermia, Hyperthermia, Personal Protective Equipment, PPE

Version 2

N25A-T028 TITLE: Rapid Cryogenic Cooling of Superconducting Systems

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Advanced Materials;Directed Energy (DE);Renewable Energy Generation and Storage

OBJECTIVE: Develop a novel process, technique, or material for rapidly cooling a large thermal mass or distributed thermal load from room temperatures down to cryogenic temperatures to enable operation of a superconducting system within roughly 24 hours as compared to several days.

DESCRIPTION: High-temperature superconducting (HTS) magnet systems are being developed by the Navy for the minesweeping mission. These large-diameter magnets (on the order of 2 m) utilize standard heat exchanger technology with circulating cryogenic helium to distribute cooling throughout the magnet and may take several days to achieve operational temperatures. The Navy is also pursuing HTS degaussing systems which have a more distributed load and use circulating cryogenic helium in a cryostat. It is desired to reduce the time to cool down the system to operational temperatures so that the Navy's mission can be completed with reduced operational timelines. Future potential applications of HTS technologies include motors, generators, cables, etc., and the issue of cool down can be an important consideration in the viability of these concepts.

Heat pipes, often used for cooling electronics, utilize the phase change of a working fluid to dramatically boost the heat transfer across the length of the pipe. Unfortunately, this working fluid must be "tuned" to a particular temperature range as once the fluid freezes, the heat exchange benefit is lost. For cryogenic temperatures, Nitrogen has been explored; however, this limits the cold side temperature to 63 K. Many HTS Systems utilize colder temperatures to increase the ampacity of the wire or compensate for lost ampacity due to magnetic interaction with the material. In these situations, 55 K or below is a relatively common target temperature.

This STTR topic seeks to develop a cryogenic cooling method, and/or heat exchanger technology, capable of dramatically reducing the cooling time to improve operational readiness of Naval HTS systems. The proposed solution should be applicable to rapidly cool down a superconducting magnet that has a large thermal mass on the order of 2,000 lbs from room temperature to around 40-50 K by either direct conduction, or convection with a heat exchanger and circulating helium cryogenic gas. Additionally, there will be further consideration given to proposed solutions that can provide rapid cooling to a more distributed heat load as in a long length cryostat on the order of 600 ft containing HTS wires (HTS cable) with helium gas flowing within the cryostat at roughly 10-15 g/s.

The desire is to apply the rapid cooling method to either application to reduce the temperature from 300 K to below 50 K in as short of a time as possible. Ideally the time to cool will be within a 24 hr period. During this time, it is expected that there will be power available. The use of alternate cryogens such as liquid nitrogen may be considered but it should be noted that liquid nitrogen alone will not achieve the desired temperatures below 50 K. The use of toxic or flammable cryogens or cryogenic materials will most likely not be considered due to safety issues with using or handling these materials. Finally, the proposed solution should also include any auxiliary hardware that would be necessary such as helium circulators, heat exchangers, and cryorefrigerators.

PHASE I: Define and develop concepts for rapid cryogenic cooling. Determine the technical feasibility of the concepts meeting the desired performance specifications. Perform an analysis via modeling, experimentation, etc. Identify characteristics of the technology and nominal performance. Upon identification of a feasible solution, perform a cost estimate, for both prototype development and full-scale production. The Phase I Option, if exercised, includes a detailed design and specifications to build a prototype during a Phase II effort.

PHASE II: Develop, demonstrate, and validate a functional prototype of a rapid cryogenic cooling method and/or heat exchanger with characterization of key performance metrics at the proposer's facility or other suitable test center identified by the awardee. Establish cool down curves for various cryogenic temperature gradients with a representative thermal mass. Deliver the Phase II prototype to the Navy for further testing, or integration into a larger system.

PHASE III DUAL USE APPLICATIONS: Integrate the Phase II developed cryogenic cooling method and/or heat exchanger into a military or commercial superconducting system to achieve lower operational employment timelines and demonstrate ability to transition the technology. If successful demonstration of the technology is achieved, the transition of the development will lead to lower operational employment timelines for superconducting systems. This will enhance Fleet readiness when deploying superconducting systems in the field. There are several superconducting systems that are currently being transitioned to the Fleet and this technology may be implemented in future upgrades.
Applications in the commercial sector may include superconducting systems being developed for the wind power generation market, resilient power grid, superconducting propulsions for aviation, or existing medical devices such as MRIs.

REFERENCES:

1. Green, Michael. "Methods of speeding up the cool-down of superconducting magnets that are cooled using Small coolers at temperatures below 30 K." OP Conference Series: Materials Science and Engineering, Volume 1240, Advances in Cryogenic Engineering: Proceedings of the Cryogenic Engineering Conference (CEC) 2021, 19-23 July 2021, Virtual Conference, USA. 10.1088/1757-899X/1240/1/012139
2. Kwon, Daniel and Sedwick, Raymond. "Cryogenic Heat Pipe for Cooling High-Temperature Superconducting Coils." Cryogenics, 23, 2009, pp. 514-523. 10.1016/j.cryogenics.2009.07.005
3. Bai, Lizhan; Zhang, Lianpei; Lin, Guiping; He, Jiang and Wen, Dong-sheng. "Development of cryogenic loop heat pipes: A review and comparative analysis." Applied Thermal Engineering, 89, 2015, pp.180-191. 10.1016/j.applthermaleng.2015.06.010
4."Magnetic and Acoustic Generation Next Unmanned Superconducting Sweep (MAGNUSS)." ONR Special Notice N00014-21-S-SN12, 2021. https://sam.gov/opp/ddd803b6d03e42e9b6dbcde611800317/view
5. "High Temperature Superconducting Degaussing System." US Patent Number US 7,451,719 B1, November 2008.

KEYWORDS: Cryogenic Temperature; Heat Exchanger; Rapid Cool Down; Cryogenic Cooling; Superconducting Systems; Cryogenic Thermal Mass; Heat Removal

DEPARTMENT OF THE AIR FORCE
25.A SMALL BUSINESS TECHNOLOGY TRANSFER (STTR)
PROPOSAL SUBMISSION INSTRUCTIONS

The Air Force intends these Phase I proposal submission instructions to clarify the Department of Defense (DoD) Broad Agency Announcement (BAA) as it applies to the topics solicited herein. **Offerors must ensure proposals meet all requirements of the STTR 25.A BAA posted on the Defense SBIR/STTR Innovation Portal (DSIP) at the proposal submission deadline date/time.**

Applicants are encouraged to thoroughly review the DoD Program BAA and register for the DSIP Listserv to remain apprised of important programmatic and contractual changes.

- Full component-specific instructions and topic descriptions are available on DSIP at https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations. Be sure to select the tab for the appropriate BAA cycle.

Please ensure all e-mail addresses listed in the proposal are current and accurate. The DAF is not responsible for ensuring notifications are received by firms changing mailing address/e-mail address/company points of contact after proposal submission without proper notification to the AF. **If changes occur to the company mail or email addresses or points of contact after proposal submission, the information must be provided to the AF SBIR/STTR One Help Desk.** The message shall include the subject line, "25.A Address Change".

Points of Contact:

- General information related to the AF SBIR/STTR program and proposal preparation instructions, contact the AF SBIR/STTR One Help Desk at usaf.team@afsbirsttr.us. All applicants have ample opportunity to request clarifying information. **The DAF encourages applicants to request clarifying information as early as possible, as delays in such requests constrain the DAF's ability to provide satisfactory resolution to applicant concerns.**
- Questions regarding the DSIP electronic submission system, contact the DoD SBIR/STTR Help Desk at dodsbirsupport@reisystems.com.
- For technical questions about the topics during the pre-announcement and open period, please reference the DoD SBIR 25.A BAA.
- Air Force SBIR/STTR Contracting Officer (CO):
    - Mr. Daniel J. Brewer, Daniel.Brewer.13@us.af.mil

General information related to the AF Small Business Program can be found at the AF Small Business website, http://www.airforcesmallbiz.af.mil/. The site contains information related to contracting opportunities within the AF, as well as business information and upcoming outreach events. Other informative sites include those for the Small Business Administration (SBA), www.sba.gov, and the Procurement Technical Assistance Centers (PTACs), http://www.aptacus.us.org. These centers provide Government contracting assistance and guidance to small businesses, generally at no cost.

PHASE I PROPOSAL SUBMISSION: The DoD STTR 25.A Broad Agency Announcement, https://www.dodsbirsttr.mil/submissions/login, includes all program requirements. Phase I efforts should address the feasibility of a solution to the selected topic's requirements.

PHASE I PROPOSAL FORMAT

Complete proposals must include all of the following:

Volume 1: DoD Proposal Cover Sheet

Note: If selected for funding, the proposal's technical abstract and discussion of anticipated benefits will be publicly released. Therefore, do not include proprietary information in this section.

Volume 2: Technical Volume
Volume 3: Cost Volume
Volume 4: Company Commercialization Report
Volume 5: Supporting Documents
Volume 6: Fraud, Waste, and Abuse Training
Volume 7: Disclosures of Foreign Affiliations or Relationships to Foreign Countries

DoD PROPOSAL COVER SHEET (VOLUME 1)

Complete the proposal Cover Sheet in accordance with the instructions provided via DSIP. The technical abstract should include a brief description of the program objective(s), a description of the effort, anticipated benefits and commercial applications of the proposed research, and a list of keywords/terms. The technical abstract of each successful proposal will be submitted to the Office of the Secretary of Defense (OSD) for publication and, therefore, must not contain proprietary or classified information.

TECHNICAL VOLUME (VOLUME 2):

The Technical Volume should include all graphics and attachments but should not include the Cover Sheet, which is completed separately as Volume 1. The Phase I technical volume (uploaded in Volume 2) shall contain the required elements found below. Ensure that all graphics are distinguishable in black and white.

The Phase I Technical Volume page/slide limits identified for the topics do not include the Cover Sheet, Cost Volume, Cost Volume Itemized Listing (a-h). The Technical Volume must be no smaller than 10-point on standard 8-1/2" x 11" paper with one-inch margins. Only the Technical Volume and any enclosures or attachments count toward the page limit. In the interest of equity, pages/slides in excess of the stated limits will not be reviewed. The documents required for upload into Volume 5, "Other", do not count toward the specified limits.

These instructions supplement the 25.A STTR BAA. In addition to the requirements found in the 25.A STTR BAA, applicants are required to provide the following information in Volume 2:

Key Personnel: Identify in the Technical Volume all key personnel who will be involved in this project; include information on directly related education, experience, and citizenship.

- A technical resume of the principal investigator, including a list of publications, if any, must be included. Only one principal investigator/project manager can be designated to a proposal at any given time.
- Concise technical resumes for subcontractors and consultants, if any, are also useful.
- Identify all U.S. permanent residents to be involved in the project as direct employees, subcontractors, or consultants.
- Identify all non-U.S. citizens expected to be involved in the project as direct employees, subcontractors, or consultants. For all non-U.S. citizens, in addition to technical resumes, please provide countries of origin, the type of visa or work permit under which they are performing and an explanation of their anticipated level of involvement on this project, as appropriate. Additional information may be requested during negotiations in order to verify the foreign citizen's eligibility to participate on a contract issued as a result of this announcement. **Note:** Do not upload information such as Permanent Resident Cards (Green Cards), birth certificates, Social Security Numbers, or other PII to the DSIP system.

Phase I Statement of Work Outline
NOTE: The DAF uses the work plan outline as the initial draft of the Phase I Statement of Work (SOW). Therefore, **do not include proprietary information in the work plan outline**. To do so will necessitate a request for revision, if selected, and may delay award.

Include a work plan outline in the following format:
Scope: List the effort's major requirements and specifications.
Task Outline: Provide a brief outline of the work to be accomplished during the Phase I effort.
Milestone Schedule
Deliverables
Progress reports
Final report with SF 298

COST VOLUME (VOLUME 3)
Cost information should be provided by completing the Cost Volume in DSIP and including the Cost Volume Itemized Listing specified below. The Cost Volume detail must be adequate to enable Air Force personnel to determine the purpose, necessity and reasonability of each cost element. Provide sufficient information (a.-g. below) regarding funds use. The DSIP Cost Volume and Itemized Cost Volume Information will not count against the specified page limit. The itemized listing also may be submitted in Volume 5 under the "Other" dropdown option.

a. **Direct Cost Materials**: Justify costs for materials, parts, and supplies with an itemized list containing types, quantities, prices and where appropriate, purpose. Material costs may include the costs of such items as raw materials, parts, subassemblies, components, and manufacturing supplies.

b. **Other Direct Costs**: This category includes, but is not limited to, specialized services such as machining, milling, special testing or analysis, and costs incurred in temporarily using specialized equipment. Proposals including leased hardware must include an adequate lease v. purchase justification.

c. **Direct Labor**: Identify key personnel by name, if possible, or by labor category, if not. Direct labor hours, labor overhead and/or fringe benefits, and actual hourly rates for each individual are also necessary for the CO to determine whether these hours, fringe rates, and hourly rates are fair and reasonable.

d. **Travel**: Travel costs must relate to project needs. Break out travel costs by trip, number of travelers, airfare, per diem, lodging, etc. The number of trips required, as well as the destination and purpose of each, should be reflected. Recommend budgeting at least one trip to the Air Force location managing the contract.

e. **Subcontracts**: Involvement of consultant in the project's planning and/or research stages may be appropriate. If so, describe in detail and include information in the Cost Volume. A minimum of 40% of each STTR project must be conducted by the SBC and a minimum of 30% of the effort performed by the single partnering Research Institution. Deviations from these performance of work requirements are not permitted. The STTR funded work percentage calculation considers both direct and indirect costs after removal of the SBC's proposed profit. Support subcontract costs with copies of executed agreements. The documents must adequately describe the work to be performed. At a minimum, include a Statement of Work (SOW) with a corresponding detailed Cost Volume for each planned subcontract. Additionally, see DoD SBIR 23.3 BAA for more information regarding the required Allocation of Rights Agreement.

f. **Special Tooling, Special Test Equipment, and Material**: The inclusion of equipment and materials will be carefully reviewed relative to need and appropriateness to the work proposed. Special tooling and

special test equipment purchases must, in the CO's opinion, be advantageous to the Government and relate directly to the effort. These toolings or equipment should not be of a type that an offeror would otherwise possess in the normal course of business. These may include items such as innovative instrumentation and/or automatic test equipment.

g. **Consultants**: Provide a separate agreement letter for each consultant. The letter should briefly state what service or assistance will be provided, the number of hours required, and the hourly rate.

NOTE: If no exceptions are taken to an offeror's proposal, the Government may award a contract without negotiations. . Therefore, the offeror's initial proposal should contain the offeror's best terms from a cost or price and technical standpoint. If there are questions regarding the award document, contact the Phase I CO identified on the cover page. The Government reserves the right to reopen negotiations later if the CO determines doing so to be necessary.

COMPANY COMMERCIALIZATION REPORT (VOLUME 4)

Completion of the CCR as Volume 4 of the proposal submission in DSIP is required. Please refer to the DoD STTR 25.A BAA for full details on this requirement. Information contained in the CCR will not be considered by the Air Force during proposal evaluations.

SUPPORTING DOCUMENTS VOLUME (VOLUME 5)

The following documents may be required if applicable to your proposal:

1. DD Form 2345: For proposals submitted under export-controlled topics, either International Traffic in Arms or Export Administration Regulations (ITAR/EAR), a copy of the certified DD Form 2345, Militarily Critical Technical Data Agreement, or evidence of application submission must be included. The form, instructions, and FAQs may be found at the United States/Canada Joint Certification Program website, http://www.dla.mil/HQ/InformationOperations/Offers/Products/LogisticsApplications/JCP/DD2345Ins tructions.aspx. DD Form 2345 approval will be required if proposal if selected for award.
2. Verification of Eligibility of Small Business Joint Ventures (Attachment 3 to the DOD STTR 25.A BAA)
3. Technical Data Rights Assertions (if asserting data rights restrictions)

FRAUD, WASTE, AND ABUSE TRAINING (VOLUME 6)

Note that the FWA Training must be completed prior to proposal submission. When training is complete and certified, DSIP will indicate completion of the Volume 6 requirement. The proposal cannot be submitted until the training is complete.

DISCLOSURES OF FOREIGN AFFILIATIONS OR RELATIONSHIPS TO FOREIGN COUNTRIES (VOLUME 7)

Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please be aware that the Disclosures of Foreign Affiliations or Relationships to Foreign Countries WILL NOT be accepted as a PDF Supporting Document in Volume 5 of the DSIP proposal submission. Do not upload any previous versions of this form to Volume 5. For additional details, please refer to the DoD SBIR Program BAA.

DISCRETIONARY TECHNICAL AND BUSINESS ASSISTANCE (TABA)

The Air Force does not participate in the Discretionary Technical and Business Assistance (TABA) Program. Proposals submitted in response to DAF topics shall not include TABA.

AIR FORCE PROPOSAL EVALUATIONS

Proposals will be evaluated for overall merit in accordance with the criteria discussed in the 25.A BAA. DAF is seeking varying technical/scientific approaches and/or varying and new technologies that would be responsive to the problem statement(s) and area(s) of interest in the topic. Multiple procurements are planned and anticipated to be awarded as a result of the topic, each proposal is considered a separate procurement and will be evaluated on its own merit, and that the Government may award all, some, or none of the proposals. Any per-award or per-topic funding caps are budgetary estimates only, and more or less funding may become available. Funding decisions are made with complete disregard to the other awards under the same topic.

In accordance with 15 USC 638(vv) (Section 4 of the SBIR and STTR Extension Act of 2022), and the Deputy Secretary of Defense Memorandum; Subject: Defense Small Business Innovation Research and Small Business Technology Transfer Due Diligence Program dated May 13, 2024, the DAF will review all proposals submitted in response to this BAA to assess security risks presented by small business concerns seeking a Federally funded award. The DAF will use information provided by the small business concern in response to the Disclosure of Foreign Affiliations or Relationships to Foreign Countries and the proposal to conduct a risk-based due diligence review on the cybersecurity practices, patent analysis, employee analysis, and foreign ownership of a small business concern, including the small business concern and employees of the small business concern to a foreign country, foreign person, foreign affiliation, or foreign entity. The DAF will also assess proposals utilizing open-source analysis and analytical tools, for the nondisclosures of the information set forth in 15 U.S.C. 638(g)(13). If DAF assesses that a small business concern has security risk(s), DAF will review the proposal, the evaluation, and the security risks and may decide not to select the proposal for award based upon a totality of the review.

MAJORITY OWNERSHIP IN PART BY MULTIPLE VENTURE CAPITAL, HEDGE FUND, AND PRIVATE EQUITY FIRMS

Small business concerns that are owned in majority part by multiple venture capital operating companies (VCOCs), hedge funds, or private equity funds are not eligible to submit applications or receive awards for Department of Air Force Topics.

PERFORMANCE OF WORK REQUIREMENTS AND LOCATION OF WORK

For both Phase I and Phase II, a minimum of 40% of each STTR award must be conducted by the awardee and a minimum of 30% of the effort must be performed by the single partnering Research Institution. Note, applicants and awardees may partner with multiple entities that separately meet the definition of a "Research Institution" as indicated in the SBIR BAA. Applicants may use only one partnering Research Institution to meet STTR eligibility requirements. The DAF will not consider requests for deviations to these performance of work requirements.

All R/R&D work must be performed in the United States. Based on a rare and unique circumstance, the DAF may approve a particular portion of the R/R&D work to be performed or obtained in a country outside of the United States. The awarding Funding Agreement officer must approve each specific condition in writing. Applicants seeking this approval must make such a request with their initial proposal submission. The DAF will not consider these requests prior to proposal submission.

DAF USE OF SUPPORT CONTRACTORS

Restrictive notices notwithstanding, proposals may be handled for administrative purposes only, by support contractors. These support contractors may include, but are not limited to TEC Solutions, Inc., APEX, Oasis Systems, Riverside Research, Peerless Technologies, HPC-COM, Mile Two, Montech, Wright Brothers Institute, and MacB (an Alion Company). In addition, only Government employees and technical personnel from Federally Funded Research and Development Centers (FFRDCs) MITRE and

Aerospace Corporations working under contract to provide technical support to AF Life Cycle Management Center and Space and Missiles Centers may evaluate proposals. All support contractors are bound by appropriate non-disclosure agreements. Contact the AF SBIR/STTR CO Daniel J. Brewer (Daniel.Brewer.13@us.af.mil) with concerns regarding the use of support contractors.

PROPOSAL STATUS AND FEEDBACK

The Principal Investigator (PI) and Corporate Official (CO) indicated on the Proposal Cover Sheet will be notified by e-mail regarding proposal selection or non-selection. Small Businesses will receive a notification for each proposal submitted. Please read each notification carefully and note the Proposal Number and Topic Number referenced.

Automated feedback will be provided for Phase I proposals designated Not Selected. Additional feedback may be provided at the sole discretion of the DAF.

IMPORTANT: Proposals submitted to the DAF are received and evaluated by different organizations, handled by topic. Each organization operates within its own schedule for proposal evaluation and selection. Updates and notification timeframes will vary. If contacted regarding a proposal submission, it is not necessary to request information regarding additional submissions. Separate notifications are provided for each proposal.

The Air Force anticipates that all proposals will be evaluated and selections finalized within approximately 90 calendar days of solicitation close. Refrain from contacting the BAA CO for proposal status before that time.

Refer to the DoD STTR 25.A BAA for procedures to protest the Announcement.
As further prescribed in FAR 33.106(b), FAR 52.233-3, Protests after Award should be submitted to: Air Force SBIR/STTR Contracting Officer Daniel J. Brewer, Daniel.Brewer.13@us.af.mil.

AIR FORCE SUBMISSION OF FINAL REPORTS

All Final Reports will be submitted to the awarding DAF organization in accordance with Contract instructions. Companies will not submit Final Reports directly to the Defense Technical Information Center (DTIC).

PHASE II PROPOSAL SUBMISSIONS

DAF organizations may request Phase II proposals while Phase I technical performance is ongoing or at any time after the conclusion of the period of performance. This decision will be based on the awardee's technical progress, as determined by an DAF Technical Point of Contact review using the Phase II review criteria outlined above.

Phase II is the demonstration of the technology found feasible in Phase I. Only Phase I awardees are eligible to submit a Phase II proposal. All Phase I awardees will be sent a notification with the Phase II proposal submittal date and detailed Phase II proposal preparation instructions. If the physical or email addresses or firm points of contact have changed since submission of the Phase I proposal, correct information shall be sent to the AF SBIR/STTR One Help Desk. Phase II dollar values, performance periods, and proposal content will be specified in the Phase II request for proposal.

NOTE: The DAF primarily makes STTR Phase I and II awards as Firm-Fixed-Price contracts. However, awardees are strongly urged to work toward a Defense Contract Audit Agency (DCAA)-approved accounting system. If the company intends to continue work with the DoD, an approved accounting system will allow for competition in a broader array of acquisition opportunities, including award of

Cost-Reimbursement types of contracts. Please address questions to the Phase II CO, if selected for award.

All proposals must be submitted electronically via DSIP by the date indicated in the Phase II proposal instructions. Note: Only ONE Phase II proposal may be submitted for each Phase I award.

AIR FORCE SBIR/STTR PROGRAM MANAGEMENT IMPROVEMENTS

The DAF reserves the right to modify the Phase II submission requirements. Should the requirements change, all Phase I awardees will be notified. The DAF also reserves the right to change any administrative procedures that will improve management of the DAF SBIR/STTR Program at any time.

DAF STTR Phase I Topic Index

Topic Number	Topic Title	Proposal Maximum Value*	Proposal Maximum PoP**	Technical Volume (Volume 2) Page/Slide Limit***
AF25A-T001	Low Cost Technologies for Manufacturing of Thin Film Lithium Niobate	$200,000.00	6 Months	20 Pages/Slides
AF25A-T002	MOVE_IT - Moving Operations for Vertical lift Integration Technology -- STTR	$200,000.00	6 Months	20 Pages/Slides
AF25A-T003	Use of Laser Welding for Fabrication of Nickel Based Superalloy Aerostructures and Propulsion Components	$200,000.00	6 Months	20 Pages/Slides
AF25A-T004	Supply Chain Analytics: AI to Distill Data into Insight	$200,000.00	6 Months	20 Pages/Slides

*Proposals in excess of this amount will not be considered for award.
**Proposals in excess of this duration will not be considered for award.
***Pages/slides in excess of this number will not be considered during proposal evaluations.

TOPIC NUMBER: AF25A-T001

TITLE: Low Cost Technologies for Manufacturing of Thin Film Lithium Niobate

TECHNOLOGY AREAS: Sensors; Space Platforms; Information Systems; Air Platform; Battlespace

OBJECTIVE: This topic seeks to develop alternative low-cost manufacturing processes for producing high quality thin film lithium niobate (TFLN) that do not rely on the prevailing "Smart-Cut" ion implantation technique. These processes should be scalable and versatile, able to produce a variety of film thicknesses, crystallographic orientations, and on multiple substrates at low cost with minimal NRE. Proposed technologies should have clear compatibility with existing TFLN use cases and show a path towards widespread commercial use, ideally by providing homogenous 3-4 inch wafers of TFLN on relevant substrates such as quartz, but other approaches suitable for volume device manufacturing will also be considered. In either case a demonstration RF or photonic device must be fabricated using the TFLN produced under this effort. Films on the order of 300nm to several microns thick should be producible with desired total thickness variation of less than 3 percent and a surface roughness of less than 1 nanometer. The overall goal is a manufacturing process capable of several hundred wafers or equivalent devices per year at a cost of less than $2000 per wafer or a greater than 25 percent cost reduction. Ideally these technologies would also be transferrable to other thin film electro-optic materials such as Barium titanate.

DESCRIPTION: Thin Film Lithium Niobate (TFLN), which consists of a thin film (300nm+) of lithium niobate, traditionally on insulator, is a key electro-optic material for next generation sensor and communication systems needed by the U.S. Space Force, Air Force, and DoD at large. TFLN offers significant improvements over bulk lithium niobate in processing, size, weight, and performance as well as enabling new capabilities in photonics, ultra-wideband Radio Frequency, surface acoustic wave filters (SAW) and quantum sensing and computing.

Currently the leading technology of choice for TFLN manufacturing is ion slicing, otherwise known as "Smart-Cut", which uses ion implantation to induce a cleavage plane at some depth determined by ion dose and energy in the bulk lithium niobate wafer. This wafer is then bonded to the handle substrate wafer, annealed, and exfoliated. This leaves the damaged ion implant layer as a thin film on the handle wafer, which can then be further annealed and polished using chemical mechanical polishing (CMP) to remove defects and improve uniformity. CMP can also be used exclusively to grind away most of the bulk wafer, but this process has not yet achieved the uniformity of Smart-Cut at the wafer scale. While the Smart-Cut process is capable of producing highly uniform films, it has high capital investment costs (or high service cost using 3rd parties) for the ion implantation step. It also suffers from high non-recurring engineering costs for different TFLN wafer compositions, as well as increased cost for thicker films due to the increased ion dose required.

This effort should mature, adapt, or improve upon alternative existing or demonstrated thin film manufacturing technologies to produce high quality TFLN for use by researchers and in DoD and commercial systems with reduced cost and capacity to scale. Offerors should plan to demonstrate the technology at the chip or wafer scale and with enough iterations to extract meaningful statistical data on scalability, yield, and cost. Work plans are expected to build towards the objectives stated above through process development and use this data to map out the parameter space of the manufacturing technology. Aside from establishing viability, yield, and cost, a key component is to demonstrate the ability to quickly and cheaply adapt the technology to produce TFLN of varying thickness, crystal geometry, and/or substrate. Offerors are expected to work with researchers, device designers or manufacturers to ensure

compatibility of their process and to produce a demonstration device using their material. For any proposals requiring heterogenous integration, that should be demonstrated under this effort. Preference will be given towards those technologies that can produce homogenous lithium niobate on insulator at the 3-4 inch wafer scale. Desired characterization includes surface roughness, total thickness variation, crystallinity, flatness, and film thickness. Bond strength should be tested at relevant operating temperatures. The final report should include a qualitative and quantitative cost and scaling analysis comparing the developed technology to the current technologies described above.

PHASE I: Under a Phase I award the awardee shall provide a qualitative and quantitative study of the proposed solution along with experimental proof of their process. This includes justifying, insofar as possible, the scientific and technical merit and feasibility of ideas appearing to have commercial potential. At a minimum, the offeror must: (1) Demonstrate capability and competence with the proposed manufacturing process by providing initial results producing thin film lithium niobate or past work with a comparable material, in which case they shall also provide technical justification for applicability to lithium niobate and the topic objectives. (2) Provide a comparison with the existing processes described in the topic description in the context of the stated objectives. This must specifically include substantiated cost estimates for obtaining or producing the proposed wafer compositions using established techniques and estimated cost savings. (3) Define a complete, clear, realistic, and actionable plan with the proposed solution and the AF customer. Describe if/how the demonstration can be used by other DoD or Governmental customers.

PHASE II: Under the Phase II effort, the awardee shall sufficiently develop a manufacturing process capable of producing thin film lithium niobate and characterize parameters including surface roughness, total thickness variation, crystallinity, flatness, and film thickness. Through sufficient statistical data the manufacturing process will be evaluated by establishing a baseline cost per wafer or cost savings per fabricated device, a cost-scaling estimate for volume manufacturing (to 1000s of wafers or 10000s of devices a year), and yield estimates. The offeror must demonstrate through fabrication the ability to modify the manufacturing process to produce multiple compositions of varying film thickness and/or substrate and document the non-recurring engineering (NRE) cost and process to do so. Finally, suitability for RF or photonics applications must be demonstrated through device fabrication and/or heterogenous integration. Temperature and mechanical stability of bonded material should be verified to relevant device operating conditions. At the end of the effort material shall be provided to Department of the Air Force for independent verification. Identification of manufacturing/production issues and or business model modifications required to further improve product or process relevance to improved sustainment costs and material availability should be documented. These Phase II awards are intended to provide a path to commercialization, not the final step for the proposed solution.

PHASE III DUAL USE APPLICATIONS: The awardee may pursue commercialization of the thin film lithium niobate manufacturing process developed in Phase II for transitioning expanded mission capability to a broad range of potential government and civilian users and alternate mission applications. Direct access with end users and government customers will be provided with opportunities to receive Phase III awards for providing the government additional research & development, or direct procurement of products and services developed in coordination with the program. This work should meet Technology Readiness Level 5 and Manufacturing Readiness Level 4 before entry into a Phase III. For technologies requiring heterogenous integration, special tooling or skillsets, or other large capital investment, suitable partnerships should be identified (e.g. a foundry such as AIM Photonics). Phase III candidates should have a viable plan to scale to high volume (1000s of wafers or 10000s of devices) and plan to supply both the DoD and commercial markets.

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2. Chia-Cheng Wu, Ray-Hua Horng, Dong-Sing Wuu, Tsai-Ning Che, Shih-Shian Ho, Chia-Jen Ting and Hung-Yin Tsai; "Thinning Technology for Lithium Niobate Wafer by Surface Activated Bonding and Chemical Mechanical Polishing." Japanese Journal of Applied Physics, Vol 45, Page 3822. (2006);

3. Ryo Takigawa, Eiji Higurashi, Tadatomo Suga, Tetsuya Kawanishi; "Room-temperature transfer bonding of lithium niobate thin film on micromachined silicon substrate with Au microbumps."Sensors and Actuators A: Physical, Vol 264, Pages 274-281. (2017);

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5. Adcock, J.C., Ding, Y. Quantum prospects for hybrid thin-film lithium niobate on silicon photonics. Front. Optoelectron. 15, 7 (2022);

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KEYWORDS: thin-film; lithium niobate; lithium niobate on insulator; low cost; chemical mechanical polishing; wafer manufacturing; heterogeneous integration; wafer bonding; microwave photonics

TOPIC NUMBER: AF25A-T002

TITLE: MOVE_IT - Moving Operations for Vertical lift Integration Technology -- STTR

TECHNOLOGY AREAS: Battlespace

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with section 3.5 of the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: This topic seeks to perform system engineering, concept exploration, analysis, and modeling and simulation to help USSF determine what infrastructure needs are for Initial Operating Capability (IOC).

DESCRIPTION: The Department of the Air Force and Space Force are exploring the rocket transportation capabilities for DoD logistics and the Air Force Research Laboratory (AFRL) is currently assessing emerging rocket capabilities across the commercial vendor base, and its potential use for quickly transporting DoD materiel across the globe. The Air and Space Forces need to be prepared to ship DoD cargo via rocket once this mode of transport is operational. The recent advances in the U.S. commercial launch market are presenting the need for assessment and maturation of system-of-systems concepts in rocket transportation for DoD logistics by the United States Air Force and Space Force (USAF/USSF).

The goal of this effort is to assist USSF in understanding all the steps needed from the time a Rocket Cargo request is generated all the way to delivery. This includes payload storage, new mobility focused payload processing & validation, CONOPS, and required infrastructure therein, etc. Help identifying estimated short-falls hindering them from rapidly achieving IOC for the emerging rocket transportation mode. The focus is on infrastructure and payload processing. An infrastructure example might be to use existing modeling tools to show how the tool can allow the USSF to set up a fuel supply network, with production, fuel storage points, and launch sites as the nodes, all connected by pipelines, roads, or rails. Capacity for throughput and storage is tracked to answer questions like, “if this pipeline is cut, how many days can this space port operate using only truck-delivered fuel?” Or “if there is a mishap on a launch pad, what is the minimum safe distant and how can traffic be re-routed?” This effort should NOT be looking to build new modeling software but take advantage of existing modeling tools identify steps needed from call-up to delivery that can optimize logistics flow for both support at the Space Port and the overall system. Payload processing examples could be new ideas for payload storage and the CONOPS to get this cargo loaded onto the rocket or developing new methods and cutting-edge tools to rapidly validate existing payloads and/or materiel against the rigors of space flight (such as re-entry) and survivability of landing. Traditionally, the strenuous qualification and certification process for space hardware has been a lengthy affair, often spanning months or even years. This extended timeline poses a significant stumbling block for cargo and materiel transported by rocket.

AFRL is supporting the Operational Imperatives and Great Power Competition by seeking to help identify, develop, and deliver processes, workflows, models, simulations, and documentation for rocket

transportation to be effective, competitive, and affordable. This analysis is critical to optimally design efficient processes, CONOPS, and a space port that can provide the required launch readiness in the presence of possible adversary interdiction attempts. As new processes and workflows are generated, they will need to be integrated into existing space port systems.

The main deliverables will be analysis, simulations, process definitions, workflows, documentation, sub-scale experiments, prototypes, tests, and demonstrations that advance the operational imperatives.

PHASE I: This topic is intended to develop methodology and feasibility for Initial Operating Capability (IOC) conditions in order to help USSF determine current infrastructure gaps. Methodology should be model-based and employ system engineering fundamental concepts. Design parameters and timelines should be specified that show how to obtain the USSF's IOC requirements.

PHASE II: The awardee shall sufficiently develop concepts and/or processes and perform analysis in order to conduct multiple modeling and simulation experiments. These M&S efforts will include cost benefit analysis, system-of-systems studies, experimentation, and evaluation of operational imperatives to enable future concepts. M&S and experimentation should explore a wide range of integrating commercial capabilities to support the operational imperatives. These capabilities should consider areas that are unique to military operations, logistics, mission planning, mission execution, and base sustainment.

PHASE III DUAL USE APPLICATIONS: Phase III shall include upgrades to the analysis, M&S, T&E results and provide mature simulations of system concepts. Phase III shall provide a business plan and address the ability to transition technology and system concepts to commercial applications. The adapted non-Defense commercial solutions shall provide expanded mission capability for a broad range of potential Governmental and civilian users and alternate mission applications. Integration and other technical support to operational users may be required.

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IISE 2024 Conference

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KEYWORDS: Agent-based Simulation, GIS, Logistics, Rockets, Discrete-Event Simulation,

TOPIC NUMBER: AF25A-T003

TITLE: Use of Laser Welding for Fabrication of Nickel Based Superalloy Aerostructures and Propulsion Components

TECHNOLOGY AREAS: Weapons; Space Platforms; Air Platform

OBJECTIVE: This topic seeks to develop process parameters for laser welding of high performance Nickel based superalloy components for air and space structures and propulsion

DESCRIPTION: This topic is intended to develop laser welding parameters for thicknesses greater than 0.25 inches of high temperature superalloys with high precision and repeatability when compared to electron beam welding. The offeror shall sufficiently develop the laser welding process for high performance Nickel based superalloys to conduct demonstrations to meet Air Force needs. Identification of manufacturing/production issues required to further improve cost, rates, availability, and safety, should be documented.

PHASE I: The Phase I effort should demonstrate the following: laser welded samples of high performance Nickel based superalloys, the scientific and technical merit of laser welding compared to Electron Beam welding. Develop laser welding parameters for thicknesses greater than 0.25 inches of high temperature superalloys with high precision and repeatability. The offeror shall sufficiently investigate the laser welding process for high performance Nickel based superalloys to meet AF needs. Some of the tasks include but not limited to are : (1) Perform industry survey for equipment suppliers for laser welders, (2) Down select a robotic laser welding equipment supplier, (3) Develop laser welding process with and w/o filler metal wire and evaluate high performance Nickel based superalloy parts from casting, wrought, additive manufacturing, etc. and evaluate weldments.

PHASE II: Under the Phase II effort, the awardee shall sufficiently develop the technical approach, product, or process in order to conduct a small number of relevant demonstrations. Identification of manufacturing/production issues and or business model modifications required to further improve product or process relevance to improved sustainment costs, availability, or safety, should be documented. These Phase II awards are intended to provide a path to commercialization, not the final step for the proposed solution. Some of the tasks include but not limited to are : (1) Perform industry survey for equipment suppliers for laser welders, (2) Down select a robotic laser welding equipment supplier, (3) Develop laser welding process with filler metal wire and evaluate materials (high temperature Nickle based super alloy casting, wrought, Additive Manufacturing, etc.) and evaluate weldments.

PHASE III DUAL USE APPLICATIONS: The awardee will pursue commercialization of the various technologies developed in Phase II for transitioning expanded mission capability to a broad range of potential government and civilian users and alternate mission applications. The expected scope but not limited to is : (1) Seamless integration with automation solution for workpiece handling, loading and unloading of components and created sub-assemblies and (2) Implement digital engineering for real-time parameter and data streaming for statistical process control (SPC) process data reporting and storage capability. Direct access with end users and government customers will be provided with opportunities to receive Phase III awards for providing the government additional research & development, or direct procurement of products and services developed in coordination with the program.

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2. Grünenwald, Stefan. "High Power Fiber Laser Welding of Thick Section Materials–Process Performance and Weld Properties." (2019).

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KEYWORDS: Multi-axis robotic laser welding; laser welding of thick sections; high power laser welding; laser welding of Inconel alloys; laser welding of Haynes alloys; laser welding of airframes, laser welding of spaceframes

TOPIC NUMBER: AF25A-T004

TITLE: Supply Chain Analytics: AI to Distill Data into Insight

TECHNOLOGY AREAS: Air Platform; Information Systems; Nuclear; Electronics; Materials; Weapons

OBJECTIVE: The end state we aim to achieve with this Phase I STTR project centers on developing a foundational AI-driven toolchain specifically tailored for enhancing the operational resilience of the Air Force and Space Force through superior supply chain analytics. This toolchain will incorporate cutting-edge artificial intelligence to seamlessly integrate with existing defense supply chain frameworks, significantly enhancing real-time decision-making capabilities.

Our objective is to provide a robust proof of concept that demonstrates the feasibility of translating complex supply chain data into actionable insights through an intuitive user interface designed for logistics subject matter experts (SMEs). This will be achieved by leveraging advanced machine learning techniques, including Retrieval-Augmented Generation (RAG), to ensure the reliability and accuracy of data outputs.

Moreover, the project will establish a secure, scalable architecture that integrates with the Air Force Research Laboratory's (AFRL) Earth 616 program's existing graph databases and blockchain infrastructure. This integration aims to enhance the transparency and security of data processes within the Defense Industrial Base (DIB), setting the stage for Phase II development and eventual deployment.

Ultimately, the success of this project will be measured by its ability to reduce the reliance on large data analytics teams, thereby speeding up the response times to critical supply chain challenges and allowing for a more agile adaptation to dynamic operational environments. By the project's conclusion, we intend to have laid the groundwork for a transformative tool that not only meets the immediate needs of the Air Force and Space Force but also has significant potential for broader commercial application within complex supply chain systems across various sectors.

DESCRIPTION: To meet the objectives of this Phase I STTR project, which involves initial research into developing an AI-driven toolchain for improving supply chain analytics within the Defense Industrial Base (DIB), the following work plan is suggested.

Preliminary Data Architecture Development:

• Objective: Establish a basic framework for data integration that supports AI capabilities, aligning with the Earth 616 program's existing technologies.

• Approach: Conduct a feasibility study to identify the optimal data structures and integration points between AI models and existing DIB databases, focusing on scalability and security for future phases.

AI Model Conceptualization:

• Objective: Design preliminary AI models that utilize large language models (LLMs) and Retrieval-Augmented Generation (RAG) to interpret and respond to logistics queries.

• Approach: Develop initial AI prototypes that demonstrate basic understanding and processing of supply chain data. This includes setting up the groundwork for advanced AI development in Phase II by exploring different machine learning frameworks and algorithms.

User Interface (UI) Prototyping:

• Objective: Design a simple, intuitive user interface prototype that enables direct interaction of logistics SMEs with the AI toolchain.

• Approach: Sketch the initial design of the UI that facilitates easy input of queries and displays AI-generated insights in a clear, understandable manner. This task will involve basic mock-ups and flow diagrams to outline the user interaction without fully implementing the interface.

Initial Security Framework Setup:

• Objective: Outline a security strategy to protect data integrity and comply with military standards, focusing on blockchain integrations.
• Approach: Identify key security requirements and potential technologies that can be incorporated into the toolchain. Develop a basic security plan that includes data encryption and access controls suitable for a research prototype.
Exploratory Testing and Validation:
• Objective: Conduct initial testing of the AI prototypes and data architecture to validate their potential for Phase II expansion.
• Approach: Implement rudimentary tests to evaluate the conceptual models and frameworks for functionality and feasibility. This early-stage testing will focus on identifying major flaws or limitations that could impede further development.
Documentation and Reporting:
• Objective: Produce detailed documentation of the research findings, methodologies, and initial prototypes developed during Phase I.
• Approach: Compile a comprehensive report that includes technical descriptions, test results, and conceptual plans for the next phase of development. This document will serve as a foundational resource for scaling the project.
This Phase I effort is designed to establish the viability and strategic direction for a more detailed and resource-intensive development in Phase II, ensuring that the groundwork is laid for a transformative tool that meets the future needs of the Air Force and Space Force.

PHASE I: The objective of this STTR Phase I project is to create an AI-driven toolchain to enhance decision-making within the Defense Industrial Base (DIB) by simplifying complex data analysis for the Air Force and Space Force. The project will develop a proof of concept demonstrating the feasibility of using AI to operationalize data analytics and improve operational readiness.
Background: Building upon the successes of the Earth 616 program, this project seeks to further integrate and leverage AI and machine learning technologies to address the complexities of modern military supply chains. The Earth 616 program demonstrated the effectiveness of using advanced data architectures like graph databases and blockchain for enhancing data visibility and integrity. This project will expand on those technologies to create an intuitive, accessible interface for non-technical personnel.
Scope of Work:
• Data Architecture Development: Design and develop a preliminary data architecture that integrates seamlessly with existing DIB systems, focusing on scalability and security.
• AI Model Development: Prototype AI models that leverage natural language processing and machine learning to transform complex logistics data into understandable insights.
• Interface Design and Usability Testing: Design a user-friendly interface that allows straightforward interaction via natural language queries and conduct usability testing with target end-users to ensure the interface meets their needs.
• Feasibility Testing and Validation: Execute initial tests to validate the AI toolchain across predefined use cases, which include scenarios of logistical disruptions and supply chain anomalies.
Expected Outcomes: By the end of Phase I, we expect to have a validated proof of concept that:
• Demonstrates the AI toolchain's ability to effectively interpret and analyze complex supply chain data.
• Showcases a secure, user-friendly interface that simplifies user interaction and accelerates decision-making processes.
• Provides initial performance metrics and feedback from potential end-users within the military logistics community.
Future Work: Phase II will expand on the initial models and framework developed in Phase I, enhancing

the toolchain's capabilities and integrating additional data sources for greater predictive accuracy. It will also involve broader testing with end-users to refine the toolchain and ensure it meets operational needs across different military environments. The ultimate goal is to deploy a fully operational system that can be used across the Air Force to bolster supply chain resilience and decision support.

Period of Performance Objectives and Expectations: During the Phase I period, the project will focus on establishing technical feasibility through:

• Development of a Basic AI Model: Establishing baseline functionality that can handle routine supply chain queries.
• Prototype Interface Development: Ensuring the interface is intuitive and meets the basic needs of logistics personnel.
• Initial Use Case Validation: Using specific scenarios such as sudden demand spikes or logistics route disruptions to test the model's responsiveness and accuracy.
• Operational Parameters Setup: Defining and documenting the operational parameters under which the AI toolchain will be developed and tested, ensuring alignment with DIB operational standards and requirements.

PHASE II: The objective of the Phase II effort will be to develop and validate an advanced Generative Artificial Intelligence (GenAI) toolchain designed to integrate seamlessly with the Earth 616 program. By leveraging the unique datasets associated with the KC-46 program, alongside Air Force and Defense Logistics Agency (DLA) Enterprise data sources, this effort aims to enhance decision-making within the Defense Industrial Base (DIB). The goal is to provide sophisticated, real-time analytics that directly support logistics and supply chain subject matter experts (SMEs), ultimately improving operational readiness and resilience.

Prototyping Expectations: Prototyping in Phase II will focus on developing a GenAI toolchain that operates in concert with existing Earth 616 infrastructure. Key aspects of the toolchain will include:

• Seamless Integration with Earth 616: The GenAI toolchain will be designed to integrate with Earth 616's graph databases and blockchain-based datastores. This integration is intended to ensure secure and context-aware data management, enabling SMEs to make informed decisions based on the most current and relevant data.
• Customization for Specific Data Sources: The toolchain will be tailored to effectively utilize data from the KC-46 program and other relevant Air Force and DLA Enterprise datasets, demonstrating its applicability and value in real-world logistics environments.

Operating Parameters: The toolchain will be developed to operate within defined parameters to ensure it meets the requirements of existing Air Force systems, while also being adaptable to future needs:

• Interoperability with Earth 616: The toolchain must operate seamlessly within Earth 616's distributed ledger technology (DLT) and existing database architectures, ensuring smooth integration into current operations.
• Data Handling: It will be capable of processing and analyzing specific data types and formats used within KC-46 and DLA logistics environments, with an emphasis on ensuring data integrity and security.
• Compliance: The toolchain will adhere to all applicable military cybersecurity and data handling standards, ensuring it is fully compliant with DoD requirements.

Testing Requirements: To validate the effectiveness of the toolchain, a comprehensive testing strategy

will be implemented:

• Functional Testing: This will ensure the toolchain integrates correctly with Earth 616 systems and manages data flows efficiently and securely.
• Performance Testing: The toolchain's ability to process and analyze logistics data from the KC-46 program will be evaluated, with a focus on efficiency, accuracy, and relevance.
• User Acceptance Testing: SMEs from logistics and supply chain sectors will participate in user acceptance testing to refine the toolchain's usability and functionality, ensuring it meets operational needs.

Success Criteria: Success in Phase II will be measured by the toolchain's ability to:

• Enhance Decision-Making: The toolchain should provide measurable improvements in the accuracy, relevance, and timeliness of logistics decisions, helping SMEs make more informed and effective decisions.
• Demonstrate Operational Value: The toolchain will be evaluated on its ability to improve overall operational efficiency and resilience, particularly in scenarios requiring rapid decision-making and response.
• Achieve User Satisfaction: The toolchain's usability and functionality will be assessed through user feedback, with an emphasis on ensuring it meets the needs of SMEs and other key stakeholders.

Future Work: Upon successful validation in Phase II, the toolchain may be expanded in Phase III to support broader applications across additional Air Force and DLA programs. The focus will be on full operational deployment, with further development aimed at enhancing the toolchain's predictive and prescriptive analytics capabilities, ensuring it continues to provide valuable support for military logistics and supply chain management.

PHASE III DUAL USE APPLICATIONS: Objective and Scope: Phase III of this project will focus on transitioning the Generative Artificial Intelligence (GenAI) toolchain developed in Phase II into full operational use within both the Department of Defense (DoD) environments, particularly the Air Force and Space Force, and exploring potential commercial applications. The primary goal is to ensure that the toolchain can be seamlessly integrated into existing military operations and civilian sectors that benefit from advanced supply chain analytics.

Expected Technology Readiness Level (TRL) at Phase III Entry: The GenAI toolchain is expected to enter Phase III with a Technology Readiness Level (TRL) of 6. This indicates that the technology will have demonstrated a high fidelity prototype in a relevant environment (i.e., within the Earth 616 framework using real Air Force data). The Phase II activities are designed to validate the toolchain's functionalities, ensuring it meets the operational needs and integration requirements for broader deployment.

Transition Planning: The transition plan for Phase III includes several key components:

1. Operational Deployment: Collaborate with Air Force Program Executive Offices (PEOs) and Systems Program Offices (SPOs) to facilitate the deployment of the toolchain across various Air Force supply chain operations. This includes integration with existing logistics and data analytics platforms used by the Air Force and Space Force.

2. Training and Support: Develop comprehensive training programs for end-users to ensure they can effectively utilize the GenAI toolchain. This will include both online and in-person training modules, focusing on how to use the toolchain for enhancing decision-making in supply chain management.
3. Continuous Improvement and Updates: Establish a feedback loop with end-users to continuously improve the toolchain functionalities based on real-world use and evolving needs. This will involve regular updates and patches to the software to address any emerging challenges and incorporate new data sources as they become available.
4. Commercial Sector Transition: Identify and engage with commercial partners in industries such as manufacturing and logistics that could benefit from enhanced predictive analytics and data visualization capabilities. The toolchain will be adapted to meet the specific needs of these industries, ensuring it provides value in both military and civilian contexts.
5. Compliance and Certification: Ensure all necessary compliance with DoD cybersecurity standards and commercial data protection regulations. This will include obtaining any necessary certifications to deploy the technology in sensitive or regulated environments.

Expected Outcomes: By the end of Phase III, the GenAI toolchain is expected to be fully operational within the DoD and beginning to gain traction in commercial sectors. It should demonstrate significant improvements in supply chain decision-making accuracy, efficiency, and responsiveness, contributing to enhanced operational readiness and reduced costs.

Future Work: Looking beyond Phase III, future work will focus on expanding the toolchain's capabilities to incorporate emerging AI technologies and data analytics methodologies. This will include exploring the use of machine learning models for predictive maintenance and the integration of Internet of Things (IoT) data to further enhance supply chain visibility and resilience.

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KEYWORDS: Artificial Intelligence; Machine Learning; Supply Chain Logistics; Knowledge Graphs; Scenario Generation; Data Analytics; Distributed Ledger Technology; Blockchain; Operational Readiness; Military Logistics; Data Integration; AI-Enabled Decision Support; Resilient Supply Chains; Data Security; User Interface Design

Defense Health Agency (DHA)
2025.A Small Business Technology Transfer (STTR)
Proposal Submission Instructions

INTRODUCTION

The Defense Health Agency (DHA) Small Business Technology Transfer (STTR) Program seeks small businesses with strong research and development capabilities to pursue and commercialize medical technologies. DHA STTR encourages participation in innovation and entrepreneurship by women and socially or economically disadvantaged persons.

Proposers responding to a topic in this BAA must follow all general instructions provided in the Department of Defense (DoD) STTR Program BAA. The instructions below are important to review, as they include agency-specific requirements that add to or deviate from the DoD Program BAA. Proposals that do not address agency-specific requirements may be considered non-responsive. Only Government personnel will evaluate proposals submitted under this solicitation cycle.

Proposers are encouraged to thoroughly review the DoD Program BAA and register for the DSIP Listserv to remain apprised of important programmatic and contractual changes.

- The DoD Program BAA is located at: Active Solicitations (dodsbirsttr.mil). Be sure to select the tab for the appropriate BAA cycle.
- Register for the DSIP Listserv at: Sign Up form (campaigner.com).

Specific questions about the DHA STTR Program and agency-specific instructions included below should be directed to: DHA SBIR/STTR Program Management Office (PMO) Email: usarmy.detrick.medcom-usamrmc.mbx.dhpsbir@health.mil

PHASE I PROPOSAL GUIDELINES

The Defense SBIR/STTR Innovation Portal (DSIP) is the official portal for DoD SBIR/STTR proposal submission. Proposers are required to submit proposals via DSIP. Proposals submitted by any other means will be disregarded. Detailed instructions regarding registration and proposal submission via DSIP are provided within the DoD STTR Program BAA.

Technical Volume (Volume 2)

The technical volume is not to exceed **20 pages** and must follow the format and content requirements provided in the DoD STTR Program BAA. Do not duplicate the electronically-generated Cover Sheet or insert information associated with the Technical Volume into other sections of the proposal, as these will count toward the 20-page limit.

Only the electronically-generated Cover Sheet and Cost Volume are excluded from the 20-page limit. Technical Volumes that exceed the 20-page limit will be deemed non-compliant and will not be evaluated.

Cost Volume (Volume 3)

The Phase I Base amount must not exceed $250,000 over a 6-month period of performance. Costs must be clearly identified on the Proposal Cover Sheet (Volume 1) and in Volume 3.

Please review the updated Percentage of Work (POW) calculation details included in the DoD Program BAA. Deviations from the POW requirements are not permitted.

Travel must be justified and should relate to direct Research Development Test & Evaluation (RDT&E) Technology Readiness Level (TRL)-increasing task costs. Travel costs must include the purpose of the trip(s), number of trips, origin and destination, length of trip(s), and number of personnel.

Company Commercialization Report (CCR) (Volume 4)
Completion of the CCR as Volume 4 of the proposal submission in DSIP is required. Please refer to the DoD STTR Program BAA for full details on this requirement. Information contained in the CCR will be considered by DHA during proposal evaluations.

Supporting Documents (Volume 5)
Volume 5 is provided for proposing SBCs to submit additional documentation to support the Coversheet (Volume 1), Technical Volume (Volume 2), and the Cost Volume (Volume 3). Please refer to the DoD Program BAA for more information.

Fraud, Waste and Abuse Training (Volume 6)
Fraud, Waste and Abuse training material can be found in the Volume 6 section of the proposal submission module in DSIP and must be thoroughly reviewed once per year to proceed with proposal submission.

Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)
Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please be aware that the Disclosures of Foreign Affiliations or Relationships to Foreign Countries WILL NOT be accepted as a PDF Supporting Document in Volume 5 of the DSIP proposal submission. Do not upload any previous versions of this form to Volume 5. For additional details, please refer to the DoD Program BAA.

PHASE II PROPOSAL GUIDELINES
Phase II proposals may only be submitted by Phase I awardees unless a topic is otherwise marked. Phase II is the demonstration of the technology found feasible in Phase I. Specific due date, content, and submission requirements of the Phase II proposal will be provided by the DHA STTR PMO; typically during month five of the Phase I contract.

Due to limited funding, the DHA STTR Program reserves the right to limit awards under any topic. Only proposals considered to be of superior quality will be funded. Small businesses submitting a proposal are required to develop and submit a Commercialization Strategy describing feasible approaches for transitioning and/or commercializing the developed technology in their Phase II proposal, including plans to pursue appropriate seed-stage investment. This plan shall be included in the Technical Volume.

The Cost Volume must contain a budget for the entire 24-month Phase II period in an amount not to exceed the maximum dollar amount of $1,300,000. Budget costs must be submitted using the Cost Volume format (accessible electronically on the DoD submission site) and shall be presented side-by-side on a single Cost Volume Sheet.

DHA STTR Phase II proposals have seven volumes: Proposal Cover Sheets; Technical Volume; Cost Volume; Company Commercialization Report; Supporting Documents; Fraud, Waste, and Abuse; Disclosures of Foreign Affiliations or Relationships to Foreign Countries.

The Technical Volume has a 40-page limit including: table of contents, pages intentionally left blank, references, letters of support, appendices, technical portions of subcontract documents (e.g., statements of

work and resumes) and any attachments. Technical Volumes that exceed the 40-page limit will be deemed non-compliant and will not be evaluated.

DISCRETIONARY TECHNICAL AND BUSINESS ASSISTANCE (TABA)
The DHA STTR Program does not participate in the Technical and Business Assistance (formerly the Discretionary Technical Assistance Program). Contractors shall not submit proposals that include Technical and Business Assistance.

The DHA STTR Program offers a Commercialization Readiness Program providing assistance to small businesses that receive Phase I and Phase II awards.

EVALUATION AND SELECTION
All proposals will be evaluated in accordance with the evaluation criteria listed in the DoD STTR Program BAA.

Proposing firms will be notified of selection or non-selection status for a Phase I award within 90 days of the closing date of the BAA. Non-selected companies may request feedback within 15 calendar days of the non-select notification. The Corporate Official identified in the firm's proposal shall submit the feedback request to the STTR Office at usarmy.detrick.medcom-usamrmc.mbx.dhpsbir@health.mil. Feedback is provided in an official PDF via email to the Corporate Official identified in the firm proposal within 60 days of receipt of the request. Requests for oral feedback will not be accommodated. If contact information for the Corporate Official has changed since proposal submission, a notice of the change on company letterhead signed by the Corporate Official must accompany the feedback request.

NOTE: Feedback is not the same as a FAR Part 15 debriefing. Acquisitions under this solicitation are awarded via "other competitive procedures". Therefore, offerors are neither entitled to nor will they be provided FAR Part 15 debriefs.

Refer to the DoD STTR Program BAA for procedures to protest the Announcement. Protests after Award, as further prescribed in FAR 33.106(b), FAR 52.233-3, should be submitted to:

Ms. Samantha L. Connors SBIR/STTR Chief, Contracts Branch 8
Contracting Officer
U.S. Army Medical Research Acquisition Activity
Email: samantha.l.connors.civ@health.mil

AWARD AND CONTRACT INFORMATION
Phase I awards will total up to $250,000 for a 6-month effort and will be awarded as Firm-Fixed-Price Purchase Orders.

Phase II awards will total up to $1,300,000 for a 24-month effort and will typically be awarded as Firm-Fixed-Price contracts. If a different contracting type is preferred, such as Cost-Plus, the rationale as to why must be included within the proposal.

Phase I/Phase II awardees will be informed of contracting specialist, contracting officer, and Technical Point of Contact/Contract Officer Representative upon award.

ADDITIONAL INFORMATION

RESEARCH INVOLVING HUMAN SUBJECTS, HUMAN SPECIMENS/DATA, OR ANIMAL RESEARCH

The DHA STTR Program highly discourages offerors from proposing animal or human use research during Phase I due to the significant lead time required to prepare documentation and secure approval, which could substantially delay the funding and performance of the Phase I award.

Prior to contract award when an IRB is indicated, proposers must demonstrate compliance with relevant regulatory approval requirements that pertain to proposals involving human subjects, human specimens, or research with animals. If necessary, approvals are not obtained within two months of notification of selection, the decision to award may be terminated.

Offerors are expressly forbidden to use, or subcontract for the use of, laboratory animals in any manner without the express written approval of the U.S. Army Medical Research and Development Command (USAMRDC) Animal Care and Use Review Office (ACURO). Written authorization to begin research under the applicable protocol(s) proposed for this award will be issued in the form of an approval letter from the USAMRDC ACURO to the recipient. Modifications to previously approved protocols require re-approval by ACURO prior to implementation.

Research under this award involving the use of human subjects, to include the use of human anatomical substances or human data, shall not begin until the USAMRDC's Office of Human Research Oversight (OHRO) provides formal authorization. Written approval to begin a research protocol will be issued from the USAMRDC OHRO, under separate notification to the recipient. Written approval from the USAMRDC OHRO is required for any sub-recipient using funds from this award to conduct research involving human subjects. If the Offeror intends to submit research funded by this award to the U.S. Food and Drug Administration, Offerors shall propose a regulatory strategy for review.

*NOTE: Exempt animal or human research use shall also reflect 'yes' on the proposal coversheet for USAMRDC ACURO and OHARO records.

Non-compliance with any provision may result in withholding of funds and or termination of the award.

For more information please visit the MRDC Office of Human Research Oversight website here:
USAMRDC: USAMRDC: Office of Human Research Oversight (OHRO) (health.mil)

FEDERAL FACILITY USE

The DHA STTR Program highly discourages small business concerns (SBCs) from subcontracting to a federal facility and/or utilizing for testing due to the significant lead time required to secure approval, which could substantially delay the performance of the award.
Use of federal facilities is prohibited without an approved waiver from the DHA SBIR/STTR Office.

An SBC whose proposed work includes federal facility use is required to provide a written justification, uploaded to the Supporting Documents (Volume 5), that includes the following information:

1. An explanation of why the SBIR/STTR research project requires the use of the federal facility, including data that verifies the absence of non-federal U.S. facilities, in support of the overall mission and research area.
2. Evidence that there is no applicable U.S. facility that has the ability or expertise to perform the specified work.
3. Why the Federal Agency will not and cannot fund the use of the Federal facility or personnel for the SBIR/STTR project with non-SBIR/STTR funds.

The DHA STTR Program has the right of refusal. Companies that fail to meet requirements specified above will be at risk of delay to award or funding.

If the proposal is selected, the U.S. Army Medical Research Acquisition Activity (USAMRAA) will assist in establishing the waiver for DHA SBIR/STTR Office approval. If approved, the proposer will subcontract directly with the federal facility and not a third-party representative.

Transfer of funds between a company and a Military Lab must meet the following APAN 15-01 requirements (the full text of this notice can be found at https://usamraa.health.mil/SiteAssets/APAN%2015-01%20Revised%20Feb%202018.pdf):

(1) The DoD Intramural Researcher must obtain a letter from his/her commanding officer or Military Facility director authorizing his/her participation in the Extramural Research project. This letter must be provided to the Extramural Organization for inclusion in the proposal or application.

(2) The DoD Intramural Researcher must also coordinate with his/her local RM office (or equivalent) to prepare a sound budget and justification for the estimated costs. Where there are no DoD-established reimbursement rates [e.g., institution review board (IRB) fees, indirect cost rates, etc.], the Military Facility's RM office (or equivalent) must provide details of how the proposed rates were determined. The DoD Intramural Researcher must use the budget and justification form enclosed in APAN 15-01 when developing the estimated costs and provide it to the Extramural Organization for inclusion in the proposal or application.

(3) The Extramural Research proposal or application must include a proposed financial plan for how the Military Facility's Intramural Research costs will be supported [i.e., directly funded by DoD, resources (other than award funds) provided by the Awardee to the Military Facility, or award funds provided by the Awardee to the Military Facility (in accordance with the requirements below)].

(4) The DoD Intramural Researcher should also coordinate with his/her technology transfer office.

INTERNATIONAL TRAFFIC IN ARMS REGULATION (ITAR)
For topics indicating ITAR restrictions or the potential for classified work, limitations are generally placed on disclosure of information involving topics of a classified nature or those involving export control restrictions, which may curtail or preclude the involvement of universities and certain nonprofit institutions beyond the basic research level. Small businesses must structure their proposals to clearly identify the work that will be performed that is of a basic research nature and how it can be segregated from work that falls under the classification and export control restrictions. As a result, information must also be provided on how efforts can be performed in later phases, such as Phase III, if the university/research institution is the source of critical knowledge, effort, or infrastructure (facilities and equipment).

System for Award Management (SAM)
As stated in FAR Provision 52.204-7 System for Award Management, proposing Small Business Concerns must be registered in SAM, https://sam.gov, at the time of submission and shall continue to be registered until time of award, during performance, and through final payment of any contract, basic agreement, basic ordering agreement, or blanket purchasing agreement resulting from this solicitation.

DHA STTR 25.A Topic Index

DHA25A-001	Development of Organoid Screening Tool for High-Throughput Compound Screening to be Used in Military-Relevant Exposures

DHA25A-001 TITLE: Development of Organoid Screening Tool for High-Throughput Compound Screening to Be Used in Military-Relevant Exposures

OUSD (R&E) CRITICAL TECHNOLOGY AREA(S): Combat Casualty Care

OBJECTIVE: This STTR project aims to develop an advanced, high-throughput platform using human-derived organoids to predict compound exposure, toxicity, and efficacy. The platform will integrate cutting-edge organoid technologies with automated systems for novel compounds testing to create a comprehensive, physiologically relevant model for safety and effectiveness assessment.

DESCRIPTION: Military personnel frequently face various types of trauma in the line of duty (1), including but not limited to blast injuries, burns, fractures, and traumatic brain injuries (TBI). These injuries often result in complex physiological responses that are challenging to study and treat effectively. Inflammation is associated with these disorders and may underlie the risk for health declines (2). This innovative proposal aims to revolutionize the novel drug development by creating a cutting-edge high-through put screening platform that would combine organoid technology (3), microfluidics, automation, and artificial intelligence. Many start-ups are currently working on domain of organoid technology; however, their focus is on tumor related organoids or patient derived organoids. There is a need to develop methods that swiftly and inexpensively produce thousands of organoids for screening purpose. Here, we seek to address critical challenge by requesting an efficient method that has a potential to accelerate the pace of new drug discovery innovation. Traditional 2D cell cultures and animal models have limitations in replicating the complexity of human tissue responses to trauma and potential therapeutic compounds.

There is a critical need for a more efficient and accurate screening method to evaluate potential therapeutic compounds for trauma-related injuries such traumatic brain injuries. This will benefit the drug discovery field in facilitating pre-clinical testing of more personalized treatments and will refine studies on the effects of environmental exposures risk to humans. These organoids can not only used for testing new therapeutic targets but can also allow us to expand our knowledge of the molecular mechanisms of environmental exposures. Such a method should be able to mimic the 3D structure and functionality of human tissues while allowing for high-throughput screening of multiple compounds simultaneously (4). This STTR topic will fulfill multiple portfolio areas of DHA (environmental exposures/ traumatic brain injury) and aligns well with the FDA Modernization Act 2.0 (5) by proposing an innovative approach to drug testing that could potentially reduce or eliminate the need for animal testing in certain stages of new drug development. This model will provide a more physiologically relevant model for predicting human responses to novel drugs (such as anti-inflammatory compounds) compared to traditional 2D cell cultures or animal models. Thus, this will help with accelerating the drug development timeline by enabling rapid, high-throughput screening of drug candidates. Current barriers its widespread use, e.g. in routine safety testing are the cost and complexity of organoid culture.

PHASE I: The plan for this topic is to develop and optimize protocols for generating and maintaining a diverse array of organoids derived from human and animal (e.g. mouse) pluripotent or adult stem cells. These protocols should be translated to normal and diseased organoids to accurately model military-relevant trauma responses. This would require optimized organoid culture conditions to ensure long-term viability and functionality. Once the organoid conditions are established, this would require designing and fabricating a microfluidic chip capable of housing multiple organoid types. Thus, allowing for the assessment of novel compound effects on multiple organ systems. These two components need to be integrated into automated systems for precise drug administration, real-time monitoring of organoid responses, and data collection. It is important to demonstrate the integration of biosensors to organoid set-up followed by demonstrating an automated data collection and storage systems. Prepare final report and outline plans for Phase II development.

PHASE II: The Phase II should consider combined prototype (biosensor and organoid setup) that will require machine learning algorithms to analyze multi-organ response data and generate predictive models for drug toxicity and efficacy. The goal here is to create a user-friendly interface mimicking military relevant trauma such a TBI for researchers to test drug candidate data. This would require an established model showing trauma outcome such as oxidative stress, inflammation etc. This trauma derived organoid will be programmed as fully functional mini-organs, hence will preserve the genetic and epigenetic makeup of diseased state. The organoids will be screened using established therapeutic compounds. Once the tests are completed, analyze, and interpret data from validation studies and then scaling up the system for higher throughput capabilities. It is recommended to optimize the system for long-term studies (up to 30 days) and implement quality control measures throughout the automated processes. Finally developing a validated organoid-driven prototype capable of testing multiple organ systems. In this phase, plan for regulatory documentation and potential submissions along with comprehensive business plan for potential commercialization. Compounds thoroughly tested using this technology have a potential to proceed to clinical trials, upon FDA approval.

PHASE III DUAL USE APPLICATIONS: Phase III work is typically oriented towards technology transition to Acquisition Programs of Record and/or commercialization of STTR research or technology. In Phase III, the performer is expected to seek funding from non-STTR government sources and/or the private sector to develop or transition the prototype into a viable product or service for sale in the military or private sector markets. The Phase III description must include the "vision" or "end-state" of the research. It must describe one or more specific Phase III military applications and/or supported S&T or acquisition program as well as the most likely path for transition of the STTR from research to operational capability. Additionally, the Phase III section must include (a) one or more potential commercial applications OR (b) one or more commercial technology that could be potentially inserted into defense systems because of this project. For this topic, both military and non-military medical markets should be considered.

Realization of a dual-use technology applicable to both the military and civilian use could be achieved via making commercial partners with cell culture-based companies or organoid based start-ups. Military relevant exposures may include compounds/conditions relevant to military environments (chemical warfare agents, extreme stress). Thus the target applications, including drug discovery, as well as fields like organoid technology, toxicology, personalized medicine, and organ development, must be identified at this stage. Similar to the Government sector, organoid platform in the civilian sector could utilize by screening FDA approved drugs for repurposing, toxicity screening of environmental chemicals etc.

REFERENCES:

1. Straud CL, Siev J, Messer S, Zalta AK. Examining military population and trauma type as moderators of treatment outcome for first-line psychotherapies for PTSD: A meta-analysis. J Anxiety Disord. 2019 Oct;67:102133. doi: 10.1016/j.janxdis.2019.102133. Epub 2019 Aug 18. PMID: 31472332; PMCID: PMC6739153. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6739153/

2. Jessica Gill, Hyunhwa Lee, Taura Barr, Tristin Baxter, Morgan Heinzelmann, Hannah Rak, Vincent Mysliwiec,
Lower health related quality of life in U.S. military personnel is associated with service-related disorders and inflammation, Psychiatry Research, Volume 216, Issue 1,2014, Pages 116-122, ISSN 0165-1781, https://doi.org/10.1016/j.psychres.2014.01.046.

3. Lampart Franziska L., Iber Dagmar , Doumpas Nikolaos Organoids in high-throughput and high-content screenings. Frontiers in Chemical Engineering, 5, 2023

https://www.frontiersin.org/journals/chemical-engineering/articles/10.3389/fceng.2023.1120348

4. Zhou JQ, Zeng LH, Li CT, He DH, Zhao HD, Xu YN, Jin ZT, Gao C. Brain organoids are new tool for drug screening of neurological diseases. Neural Regen Res. 2023 Sep;18(9):1884-1889. doi: 10.4103/1673-5374.367983. PMID: 36926704; PMCID: PMC10233755. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10233755/

5. Zushin PH, Mukherjee S, Wu JC. FDA Modernization Act 2.0: transitioning beyond animal models with human cells, organoids, and AI/ML-based approaches. J Clin Invest. 2023 Nov 1;133(21):e175824. doi: 10.1172/JCI175824. PMID: 37909337; PMCID: PMC10617761. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10617761/

KEYWORDS: Organoids, high-throughput screening, tissue, inflammation, trauma, Biofluidic, algorithms

Appendix A
TECHNICAL PROPOSAL TEMPLATE (VOLUME 2)

INSTRUCTIONS

These instructions and template apply to DoD SBIR/STTR Phase I topics and provide general guidelines for completing the Phase I Technical Volume. Information provided in the Service/Component-specific instructions for the topic of interest take precedence over any instructions listed below.

The template (beginning on the following page) is the format model that may be used to prepare the Phase I Technical Volume. Do not include the instructions provided on this page or any bracketed [ ] guidance in the template.

Disclosure

Offerors that include in their proposals data which they do not want disclosed to the public for any purpose, or used by the U.S. Government except for evaluation purposes, must:

(1) Mark the first page of each Volume of the Submission with the following legend:
"This proposal includes data that must not be disclosed outside the Government and must not be duplicated, used, or disclosed-in whole or in part-for any purpose other than to evaluate this proposal. If, however, a contract is awarded to this offeror as a result of-or in connection with-the submission of this data, the Government has the right to duplicate, use, or disclose the data to the extent provided in the resulting contract. This restriction does not limit the Government's right to use information contained in this data if it is obtained from another source without restriction. The data subject to this restriction are contained in pages [insert numbers or other identification of sheets]";

(2) Mark each sheet of data it wishes to restrict with the following legend:
"Use or disclosure of data contained on this page is subject to the restriction on the first page of this volume."

Format

The Technical Volume shall meet the following requirements:

- Please refer to Service/Component-specific topic instructions for the page limit and how a technical volume is handled if the stated page count is exceeded. It is the proposing firm's responsibility to verify that the Technical Volume does not exceed the page limit after upload to DSIP. Unless otherwise noted, all content in the Technical Volume will count toward the limit.
- Single column format, single-spaced typed lines.
- Standard 8 ½" x 11" paper format.
- Page margins one inch on all sides. A header and footer may be included in the one-inch margin.
- **The header on each page of the Technical Volume should contain your company name, topic number, and DSIP-assigned proposal number.**
- No font smaller than 10-point. For headers, footers, imbedded tables, figures, images, or graphics that include text, a font size of smaller than 10-point is allowable, though proposers are cautioned that the text may be unreadable by evaluators.

Do not lock or encrypt the uploaded file. Do not include or embed active graphics such as videos, moving pictures, or other similar media in the document.

Delete this instruction page and begin the Technical Volume starting with the following page.

[Title]

Volume 2: Technical Volume

[Note: Remove the disclosure statement below if not applicable to your proposal. Refer to Instructions.]

This proposal includes data that must not be disclosed outside the Government and must not be duplicated, used, or disclosed – in whole or in part – for any purpose other than to evaluate this proposal. If, however, a contract is awarded to this offeror as a result of – or in connection with – the submission of this data, the Government has the right to duplicate, use, or disclose the data to the extent provided in the resulting contract. This restriction does not limit the Government's right to use information contained in this data if it is obtained from another source without restriction. The data subject to this restriction are contained in pages <insert numbers or other identification of sheets>.

1. Identification and Significance of the Problem or Opportunity.

[Define the specific technical problem or opportunity addressed and its importance.]

2. Phase I Technical Objectives.

[Enumerate the specific objectives of the Phase I work, including the questions the research and development effort will try to answer to determine the feasibility of the proposed approach.]

3. Phase I Statement of Work (include Subcontractors and/or Research Institutions).

(a) [Provide an explicit, detailed description of the Phase I approach. If a Phase I option is required or allowed by the Component (refer to Component-specific instructions for topic of interest), describe appropriate research activities which would commence at the end of Phase I base period should the Component elect to exercise the option. The Statement of Work should indicate what tasks are planned, how and where the work will be conducted, a schedule of major events, and the final product(s) to be delivered. The Phase I effort should attempt to determine the technical feasibility of the proposed concept. The methods planned to achieve each objective or task should be discussed explicitly and in detail. This section should be a substantial portion of the Technical Volume.

(b) The BAA may contain topics that have been identified by the Program Manager as research or activities involving Human/Animal Subjects and/or Recombinant DNA. If Phase I performance includes performance of these kinds of research or activities, please identify the applicable protocols and how those protocols will be followed during Phase I. Please note that funds cannot be released or used on any portion of the project involving human/animal subjects or recombinant DNA research or activities until all of the proper approvals have been obtained. **SBCs proposing research involving human and/or animal use are encouraged to separate these tasks in the technical proposal and cost proposal in order to avoid potential delay of contract award.**]

4. Related Work.

[Describe significant activities directly related to the proposed effort, including any conducted by the principal investigator, the proposing firm, consultants, or others. Describe how these activities interface with the proposed project and discuss any planned coordination with outside sources. The Technical Volume must persuade evaluators of the proposer's awareness of the state of the art in the topic. Describe any previous work not directly related but similar to the proposed effort. Provide the following: (1) a short description, (2) the client for which work was performed (including the Government Point of Contact to be contacted including e-mail address and phone number), and (3) date of performance including project completion.]

5. <u>Relationship with Future Research or Research and Development</u>.

(a) [State the anticipated results of the proposed approach if the project is successful.

(b) Discuss the significance of the Phase I effort in providing a foundation for a Phase II research or research and development effort.

(c) Identify the applicable clearances, certifications and approvals required to conduct Phase II testing. Outline the plan for ensuring timely completion of stated authorizations in support of a Phase II research or research and development effort.]

6. <u>Commercialization Strategy.</u>

[Describe in approximately one page the SBC's strategy for commercializing this technology in DoD, other Federal Agencies, and/or private sector markets. Provide specific information on the market need the technology will address and the size of the market. Also include a schedule showing the quantitative commercialization results from the project that your company expects to achieve.]

7. <u>Key Personnel</u>.

[Identify key personnel who will be involved in the Phase I effort including information on directly related education and experience. A concise technical resume of the principal investigator, including a list of relevant publications (if any), must be included (Please do not include Privacy Act Information). All resumes will count toward the page limit for Volume 2, as specified in the Component-specific instructions.]

[Principal Investigator Name]
[School, Degree, Year]

Relevant Experience
[A concise description of the principal investigator's relevant technical experience and its application to this topic.]

Relevant Awards or Patents
[List any awards received or patents granted or applications submitted for work related to this topic.]

Relevant Publications
[List any publications relevant to this topic.]

[Repeat this format as necessary to address the qualifications of all key personnel.]

8. <u>Foreign Citizens</u>.

[Identify any foreign citizens or individuals holding dual citizenship expected to be involved on this project as a direct employee, subcontractor, or consultant. For these individuals, please specify their country of origin, the type of visa or work permit under which they are performing and an explanation of their anticipated level of involvement on this project. The proposal may be deemed nonresponsive if the requested information is not provided. You may be asked to provide additional information during negotiations in order to verify the foreign citizen's eligibility to participate on an SBIR/STTR contract. Supplemental information provided in response to this paragraph will be protected in accordance with the Privacy Act (5 U.S.C. 552a), if applicable, and the Freedom of Information Act (5 U.S.C. 552(b)(6)). Note: If no foreign nationals will be involved in proposed work, the word "None" can be substituted for the table.]

Name [include direct employees, subcontractors, and consultants]	Foreign National (Yes/No)	Country of Origin	Type of Visa or Work Permit	Level of Involvement (Role)

9. Facilities/Equipment.

[Describe available instrumentation and physical facilities necessary to carry out the Phase I effort. Justify equipment purchases in this section and include detailed pricing information in the Cost Volume. State whether or not the facilities where the proposed work will be performed meet environmental laws and regulations of federal, state (name), and local Governments for, but not limited to, the following groupings: airborne emissions, waterborne effluents, external radiation levels, outdoor noise, solid and bulk waste disposal practices, and handling and storage of toxic and hazardous materials.]

10. Subcontractors/Consultants.

[Propose efforts as applicable to either SBIR or STTR as follows:

SBIR. Involvement of a university or other subcontractors or consultants in the project may be appropriate. A minimum of two-thirds of the research and/or analytical work in Phase I, as measured by direct and indirect costs, must be carried out by the proposing small business firm, unless otherwise approved in writing by the Contracting Officer. SBIR efforts may include subcontracts with Federal Laboratories and Federally Funded Research and Development Centers (FFRDCs). A waiver is not required for the use of Federal Laboratories and FFRDCs; however, proposers must certify their use of such facilities on the proposal cover sheet. Subcontracts with other Federal organizations are not permitted. Note that universities cannot publicly release information related to Export Controlled/ITAR restricted topics. (Refer to the DoD SBIR/STTR Broad Agency Announcement for detailed eligibility requirements as it pertains to the use of subcontractors/consultants.)

STTR. Involvement of a Research Institution in the project is required. A minimum of 40 percent of the research and/or analytical work in Phase I, as measured by direct and indirect costs, must be conducted by the proposing small business firm, and a minimum of 30 percent of the research and/or tasks in Phase I, as measured by direct and indirect costs, must be conducted by a single Research Institution. STTR efforts may include subcontracts with Federally Funded Research and Development Centers (FFRDCs). A waiver is not required for the use of Federal Laboratories, but they do not qualify as a Research Partner; proposers may only subcontract to Federal Laboratories within the remaining 30 percent and must certify their use of such facilities on the proposal cover sheet. Subcontracts with other Federal organizations are not permitted. Note that universities cannot publicly release information related to Export Controlled/ITAR restricted topics. (Refer to the DoD SBIR/STTR Broad Agency Announcement for detailed eligibility requirements as it pertains to the use of subcontractors/consultants.]

11. Prior, Current or Pending Support of Similar Proposals or Awards.

[If a proposal submitted in response to this BAA is substantially the same as another proposal that was funded, is now being funded, or is pending with another Federal Agency, another or the same DoD Service/Component, you must disclose this on the proposal cover sheet and provide the following information:

a) Name and address of the Federal Agency(s) or DoD Component to which a proposal was submitted, will be submitted, or from which an award is expected or has been received.

b) Date of proposal submission or date of award.

c) Title of proposal.

d) Name and title of principal investigator for each proposal submitted or award received.

e) Title, number, and date of BAA(s) or solicitation(s) under which the proposal was submitted, will be submitted, or under which award is expected or has been received.

f) If award was received, provide contract number.

g) Specify the applicable topics for each proposal submitted or award received.

Note: If this does not apply, state in the proposal "No prior, current, or pending support has been provided for proposed work."]

12. <u>Identification and Assertion of Restrictions on the Government's Use, Release, or Disclosure of Technical Data or Computer Software.</u>

The Offeror asserts for itself, or the persons identified below, that the Government's rights to use, release, or disclose the following technical data or computer software should be restricted:

Technical Data or Computer Software to be Furnished with Restrictions	Basis for Assertion	Asserted Rights Category	Name of Person or Organization Asserting Restrictions
[(LIST)]	[(LIST)]	[(LIST)]	[(LIST)]

[Completion of this table and submission of the proposal constitutes signature for the information listed in the table above.]

[**ADDITIONAL INFORMATION/INSTRUCTION:** Rights in technical data, including software, developed under the terms of any contract resulting from proposals submitted in response to this BAA generally remain with the contractor, except that the Government obtains a royalty-free license to use such technical data only for Government purposes during the period commencing with contract award and ending twenty years after completion of the project under which the data were generated. This data <u>must</u> be marked with the restrictive legend specified in DFARS 252.227-7018 Class Deviation 2020-O0007. Upon expiration of the twenty-year restrictive license, the Government has unlimited rights in the SBIR/STTR data. During the license period, the Government may not release or disclose SBIR/STTR data to any person other than its support services contractors except: (1) For evaluation purposes; (2) As expressly permitted by the contractor; or (3) A use, release, or disclosure that is necessary for emergency repair or overhaul of items operated by the Government. See DFARS clause 252.227-7018 Class Deviation 2020-O0007 "Rights in Noncommercial Technical Data and Computer Software – Small Business Innovation Research (SBIR) Program."

If a proposer plans to submit assertions in accordance with DFARS 252.227-7017 Class Deviation 2020-O0007, those assertions must be identified and assertion of use, release, or disclosure restriction MUST be included with your proposal submission. The contract cannot be awarded until assertions have been approved. Please note that only the table is included in the page limitation; any supporting data

concerning the contract/grant number and awarding agency, as well as planned use or need of the data asserted, can be provided in Volume 5, Supporting Documents.

The following instructions apply to the fields in the table above (Identification and Assertion of Restrictions on the Government's Use, Release, or Disclosure of Technical Data or Computer Software).

1) For technical data (other than computer software documentation) pertaining to items, components, or processes developed at private expense, identify both the deliverable technical data and each such item, component, or process. For computer software or computer software documentation identify the software or documentation.

2) Generally, development at private expense, either exclusively or partially, is the only basis for asserting restrictions. For technical data, other than computer software documentation, development refers to development of the item, component, or process to which the data pertain. The Government's rights in computer software documentation generally may not be restricted. For computer software, development refers to the software. Indicate whether development was accomplished exclusively or partially at private expense. If development was not accomplished at private expense, or for computer software documentation, enter the specific basis for asserting restrictions.

3) Enter asserted rights category (e.g., Government purpose license rights from a prior contract, rights in SBIR/STTR data generated under another contract, limited, restricted, or government purpose rights under this or a prior contract, or specially negotiated licenses).

4) Corporation, individual, or other person, as appropriate.

Enter “none” when all data or software will be submitted without restrictions.]

Appendix B
DEFINITIONS

The following definitions from the SBA SBIR/STTR Policy Directive, the Federal Acquisition Regulation (FAR) and other cited regulations apply to this BAA.

Commercialization

The process of developing products, processes, technologies, or services, and the production and delivery (whether by the originating party or others) of the products, processes, technologies, or services for Federal Government or commercial markets purchase or use.

Cooperative Research and Development

An SBC and a research institution jointly conduct R&D. For purposes of the STTR Program, the SBC performs 40 percent of the work, and the single research institution performs not less than 30 percent of the work. For purposes of the SBIR Program, this refers to work a research institution conducts as the SBC's subcontractor. The proposing SBC must conduct at least two-thirds of the research and/or analytical work in Phase I.

Covered Individual

An individual who contributes in a substantive, meaningful way to the scientific development or execution of a R&D project proposed to be carried out with a DoD-funded award. DoD has further designated covered individuals as including all proposed key personnel.

Essentially Equivalent Work

Work that is substantially the same research, which is proposed for funding in more than one contract proposal or grant application submitted to the same federal agency or submitted to two or more different federal agencies for review and funding consideration; or work where a specific research objective and the research design for accomplishing the objective are the same or closely related to another proposal or award, regardless of the funding source.

Export Control

The International Traffic in Arms Regulations (ITAR), 22 CFR Parts 120 through 130, and the Export Administration Regulations (EAR), 15 CFR Parts 730 through 799, apply to all projects with military or dual-use applications that develop beyond fundamental research, which is basic and applied research ordinarily published and shared broadly within the scientific community. Details are available here.

NOTE: Export control compliance statements found in the individual Service/Component-specific proposal instructions are not meant to be all inclusive. They do not remove any liability from the submitter to comply with applicable ITAR or EAR export control restrictions or from informing the U.S. Government of any potential export restriction as fundamental R&D efforts proceed.

Federal Laboratory

In 15 U.S.C. §3703, it means any laboratory, any federally funded R&D center (FFRDC), or any center established under 15 U.S.C. §§ 3705 & 3707 that a federal agency owns, leases, or otherwise uses and the Federal Government funds, whether the U.S. Government or the contractor operates.

Federally Funded Award

A Phase I, Phase II (including Direct to Phase II, sequential Phase II/subsequent Phase II and cross-agency Phase II), or Phase III SBIR or STTR award made using a funding agreement.

Foreign Affiliation

Under 15 U.S.C. § 638(e)(16), foreign affiliation means a funded or unfunded academic, professional, or institutional appointment or position with a foreign government or government-owned entity, whether full-time, part-time, or voluntary (including adjunct, visiting, or honorary). This includes appointments or positions deemed adjunct, visiting, or honorary with research institutions located in a foreign country of concern.

Foreign Country of Concern

In 15 U.S.C. § 638(e)(17), foreign country of concern means the People's Republic of China, the Democratic People's Republic of Korea, the Russian Federation, the Islamic Republic of Iran, or any other country the Secretary of State determines as a country of concern.

Foreign Entity

Foreign entity means any branch, partnership, group or sub-group, association, estate, trust, corporation or division of a corporation, non-profit, academic institution, research center, or organization that foreign owners, foreign investors, foreign management, or a foreign government establish, direct, or control.

Foreign Government

Foreign government means any government or governmental body, organization, or instrumentality, including government owned-corporations, other than the U.S. Government or U.S. state, territorial, tribal, or jurisdictional governments or governmental bodies. The term includes, but is not limited to, non-U.S. national and subnational governments, including their respective departments, agencies, and instrumentalities.

Foreign National

Foreign National (also known as Foreign Person) under 22 CFR 120.16 mean any natural person who is not a lawful permanent resident as defined by 8 U.S.C. § 1101(a)(20) or who is not a protected individual as defined by 8 U.S.C. § 1324b(a)(3). It also means any foreign corporation, business association, partnership, trust, society, or any other entity or group that is not incorporated or organized to do business in the United States, as well as international organizations, foreign governments and any agency or subdivision of foreign governments (e.g., diplomatic missions).

"Lawfully admitted for permanent residence" means the status of having been lawfully accorded the privilege of residing permanently in the United States as an immigrant in accordance with the immigration laws, such status not having changed.

"Protected individual'' means an individual who (A) is a citizen or national of the United States, or (B) is an alien who is lawfully admitted for permanent residence, is granted the status of an alien lawfully admitted for temporary residence under 8 U.S.C. § 1160(a) or 8 U.S.C. § 1255a(a)(1), is admitted as a

refugee under 8 U.S.C. § 1157, or is granted asylum under Section 8 U.S.C. § 1158; but does not include (i) an alien who fails to apply for naturalization within six months of the date the alien first becomes eligible (by virtue of period of lawful permanent residence) to apply for naturalization or, if later, within six months after November 6, 1986, and (ii) an alien who has applied on a timely basis, but has not been naturalized as a citizen within two years after the date of the application, unless the alien can establish that the alien is actively pursuing naturalization, except that time consumed in the Service's processing the application shall not be counted toward the two-year period.

Fraud, Waste and Abuse

a. Fraud includes any false
b. about a material fact or any intentional deception designed to deprive the United States unlawfully of something of value or to secure from the United States a benefit, privilege, allowance, or consideration to which an individual or business is not entitled.
c. Waste includes extravagant, careless or needless expenditure of government funds, or the consumption of government property, that results from deficient practices, systems, controls, or decisions.
d. Abuse includes any intentional or improper use of government resources, such as misuse of rank, position, or authority or resources.
e. The SBIR/STTR Program training related to Fraud, Waste and Abuse is available here. See Section 1.12 for reporting fraud, waste, and abuse.

Funding Agreement

Any contract, grant, or cooperative agreement entered between any Federal Agency and any SBC for the performance of experimental, developmental, or research work, including products or services, Federal Government-funded in whole or in part. DoD Services/Components will only use contracts and other transaction authority (OTA) agreements for all SBIR/STTR awards.

Historically Black Colleges and Universities, and Minority-Serving Institutions

Department of Education list for historically Black colleges and universities and minority-serving institutions.

HUBZone Certified Small Business Concern

An SBC with SBA certification under the Historically Underutilized Business Zones (HUBZone) Program (13 C.F.R. § 126) as a HUBZone firm listed in the dynamic small business search (DSBS).

Malign Foreign Talent Recruitment Program

As defined in 42 U.S.C § 19237, the term “malign foreign talent recruitment program” means-

(A) any program, position, or activity that includes compensation in the form of cash, in-kind compensation, including research funding, promised future compensation, complimentary foreign travel, things of non de minimis value, honorific titles, career advancement opportunities, or other types of remuneration or consideration directly provided by a foreign country at any level (national, provincial, or local) or their designee, or an entity based in, funded by, or affiliated with a foreign country, whether or not directly sponsored by the foreign country, to the targeted individual, whether directly or indirectly stated in the arrangement, contract, or other documentation at issue, in exchange for the individual-

(i) engaging in the unauthorized transfer of intellectual property, materials, data products, or

other nonpublic information owned by a United States entity or developed with a Federal R&D award to the government of a foreign country or an entity based in, funded by, or affiliated with a foreign country regardless of whether that government or entity provided support for the development of the IP, materials, or data products;

(ii) being required to recruit trainees or researchers to enroll in such program, position, or activity;

(iii) establishing a laboratory or company, accepting a faculty position, or undertaking any other employment or appointment in a foreign country or with an entity based in, funded by, or affiliated with a foreign country if such activities are in violation of the standard terms and conditions of a Federal R&D award;

(iv) being unable to terminate the foreign talent recruitment program contract or agreement except in extraordinary circumstances;

(v) through funding or effort related to the foreign talent recruitment program, being limited in the capacity to carry out a R&D award or required to engage in work that would result in substantial overlap or duplication with a Federal R&D award;

(vi) being required to apply for and successfully receive funding from the sponsoring foreign government's funding agencies with the sponsoring foreign organization as the recipient;

(vii) being required to omit acknowledgment of the recipient institution with which the individual is affiliated, or the Federal research agency sponsoring the R&D award, contrary to the institutional policies or standard terms and conditions of the Federal R&D award;

(viii) being required to not disclose to the Federal research agency or employing institution the participation of such individual in such program, position, or activity; or

(ix) having a conflict of interest or conflict of commitment contrary to the standard terms and conditions of the Federal R&D award; and

(B) a program that is sponsored by-

(i) a foreign country of concern or an entity based in a foreign country of concern, whether or not directly sponsored by the foreign country of concern;

(ii) an academic institution on the list developed under section 1286(c)(8) of the John S. McCain National Defense Authorization Act for Fiscal Year 2019 (10 U.S.C. 2358 note; 1 Public Law 115–232) ; or

(iii) a foreign talent recruitment program on the list developed under section 1286(c)(9) of the John S. McCain National Defense Authorization Act for Fiscal Year 2019 (10 U.S.C. 2358 note; 1 Public Law 115–232).

Performance Benchmark Requirements

SBCs with multiple SBIR/STTR awards must meet minimum performance requirements to be eligible to apply for a new Phase I or Direct-to-Phase II award. The purpose of these requirements is to ensure that Phase I SBCs with multiple prior SBIR/STTR awards are making progress towards commercializing the work done under those awards. The Phase I to Phase II transition rate addresses the extent to which an awardee progresses a project from Phase I to Phase II. The commercialization benchmark addresses the extent to which an awardee has moved past Phase II work towards commercialization.

The SBIR and STTR Extension Act of 2022 (Pub. L. 117-183) amended the benchmarks' applications for more experienced firms. Detailed information on benchmark calculations and increased performance standards for more experienced firms can be found here.

Personal Conflict of Interest

A situation in which an individual has a financial interest, personal activity, or relationship that could impair the employee's ability to act impartially and in the best interest of the government when performing under the contract. (A de minimis interest that would not "impair the employee's ability to act impartially and in the best interest of the government" is not covered under this definition.)

Among the sources of personal conflicts of interest are:

a. The covered employee's, their close family members' or other members of their household's financial interests;
b. Other employment or financial relationships (including seeking or negotiating for prospective employment or business); and
c. Gifts, including travel.

Financial interests referred to in this definition's first paragraph may arise from:

a. Compensation, including wages, salaries, commissions, professional fees, or fees for business referrals;
b. Consulting relationships (including commercial and professional consulting and service arrangements, scientific and technical advisory board memberships, or serving as an expert witness in litigation);
c. Services provided in exchange for honorariums or travel expense reimbursements;
d. Research funding or other forms of research support;
e. Investment in the form of stock or bond ownership or partnership interest (excluding diversified mutual fund investments);
f. Real estate investments;
g. Patents, copyrights, and other IP interests; or
h. Business ownership and investment interests.

Principal Investigator/Program Manager

The principal investigator/project manager is the proposing SBC-designated individual who provides the scientific and technical direction to a funding agreement-supported project.

Proprietary Information

Proprietary information is any information that a SBC considers to be non-public information the SBC owns and is marked accordingly.

Research Institution

Any organization located in the United States that is:

a. A university.
b. A nonprofit institution as defined in Section 4(5) of the Stevenson-Wydler Technology Innovation Act of 1980.
c. A contractor-operated federally funded R&D center, as identified by the National Science Foundation in accordance with the government-wide FAR issued in accordance with the Office of Federal Procurement Policy Act Section 35(c)(1). A list of eligible FFRDCs is here.

Research or Research and Development

Any activity that is:

a. A systematic, intensive study directed toward greater knowledge or understanding of the subject studied.
b. A systematic study directed specifically toward applying new knowledge to meet a recognized need; or
c. A systematic knowledge application toward the production of useful materials, devices, systems, or methods, including design, development, and improvement of prototypes and new processes to meet specific requirements.

Research Involving Animal Subjects

All activities involving animal subjects shall be conducted in accordance with DoDI 3216.01 "Use of Animals in DoD Programs," 9 C.F.R. parts 1-4 "Animal Welfare Regulations," National Academy of Sciences Publication "Guide for the Care & Use of Laboratory Animals," as amended, and the Department of Agriculture rules implementing the Animal Welfare Act (7 U.S.C. §§ 2131-2159), as well as other applicable federal and state law and regulation and DoD instructions.

"Animal use" protocols apply to all activities that meet any of the following criteria:

a. Any research, development, test, evaluation or training, (including experimentation) involving an animal or animals.
b. An animal is defined as any living or dead, vertebrate organism (non-human) that is being used or is intended for use in research, development, test, evaluation or training.
c. A vertebrate is a member of the subphylum Vertebrata (within the phylum Chordata), including birds and cold-blooded animals.

See DoDI 3216.01 for definitions of these terms and more information about the applicability of DoDI 3216.01 to work involving animals.

Research Involving Human Subjects

All research involving human subjects shall be conducted in accordance with 32 C.F.R. § 219 "The Common Rule," 10 U.S.C. § 980 "Limitation on Use of Humans as Experimental Subjects," and DoDI 3216.02 "Protection of Human Subjects and Adherence to Ethical Standards in DoD-Supported Research," as well as other applicable federal and state law and regulations, and DoD Services/Component guidance. Proposing SBCs must be cognizant of and abide by the additional restrictions and limitations imposed on the DoD regarding research involving human subjects, specifically as they regard vulnerable populations (DoDI 3216.02), recruitment of military research subjects (DoDI 3216.02), and informed consent and surrogate consent (10 U.S.C. § 980) and chemical and biological agent research (DoDI 3216.02). Food and Drug Administration regulation and policies may also apply.

"Human use" protocols apply to all research that meets any of the following criteria:

a. Any research involving an intervention or an interaction with a living person that would not be occurring or would be occurring in some other fashion but for this research.
b. Any research involving identifiable private information. This may include data/information/specimens collected originally from living individuals (broadcast video, web-use logs, tissue, blood, medical or personnel records, health data repositories, etc.) in which the identity of the subject is known, or the identity may be readily ascertained by the investigator or associated with the data/information/specimens.

See DoDI 3216.02 for definitions of these terms and more information about the applicability of DoDI 3216.02 to research involving human subjects.

Research Involving Recombinant DNA Molecules

Any recipient performing research involving recombinant DNA molecules and/or organisms and viruses containing recombinant DNA molecules shall comply with the National Institutes of Health Guidelines for Research Involving Recombinant DNA Molecules, dated January 2011, as amended. The guidelines can be found at: https://osp.od.nih.gov/wp-content/uploads/2016/05/NIH_Guidelines.pdf. Recombinant DNA is defined as (i) molecules that are constructed outside living cells by joining natural or synthetic DNA segments to DNA molecules that can replicate in living cells or (ii) molecules that result from the replication of those described in (i) above.

Service-Disabled Veteran-Owned Small Business (SDVOSB)

A service-disabled veteran or service-disabled veterans-owned and controlled SBC defined in Small Business Act 15 USC § 632(q)(2) and SBA's implementing SDVOSB regulations (13 CFR 125).

Small Business Concern (SBC)

A concern that meets the requirements set forth in 13 C.F.R. § 121.702 (available here).

An SBC must satisfy the following conditions on the date of award:

a. Is organized for profit, with a place of business located in the United States, which operates primarily within the United States, or which makes a significant contribution to the United States economy through payment of taxes or use of American products, materials or labor;
b. Is in the legal form of an individual proprietorship, partnership, limited liability company, corporation, joint venture, association, trust or cooperative, except that if the concern is a joint venture, each entity to the venture must meet the requirements set forth in paragraph (c) below;
c. Is more than 50 percent directly owned and controlled by one or more individuals (who are citizens or permanent resident aliens of the United States), other SBCs (each of which is more than 50 percent directly owned and controlled by individuals who are citizens or permanent resident aliens of the United States), or any combination of these; and
d. Has, including its affiliates, not more than 500 employees. (See here for definition of an affiliate.)

Subcontract

A subcontract is any agreement, other than one involving an employer-employee relationship, including consultants, the funding agreement awardee enters calling for supplies or services for the performance of the original funding agreement.

Subcontractor

Subcontractor means any supplier, distributor, vendor, firm, academic institution, research center, or other person or entity that furnishes supplies or services pursuant to a subcontract, at any tier.

United States

United States means the fifty states, the territories and possessions of the Federal Government, the Commonwealth of Puerto Rico, the Republic of the Marshall Islands, the Federated States of Micronesia, the Republic of Palau, and the District of Columbia.

Women-Owned Small Business Concern

An SBC where one or more women own at least 51 percent, or in the case of any publicly owned business, women own at least 51 percent of the stock, and women control the management and daily business operations.

Appendix C
POTENTIAL APPLICABLE FEDERAL ACQUISITION REGULATION, DEFENSE FEDERAL ACQUISITION REGULATION SUPPLEMENT CLAUSES

Note: Green cells are potential required Federal Acquisition Regulation (FAR) and Defense Federal Acquisition Regulation Supplement (DFARS) clauses. Blue cells are potential required FAR and DFARS clauses, when applicable.

Clause Number	Title	Date	When Applicable
52.203-17	Contractor Employee Whistleblower Rights	Nov-23	
52.203-19	Prohibition on Requiring Certain Internal Confidentiality Agreements or Statements	Jan-17	
52.204-10	Reporting Executive Compensation and First-Tier Subcontract Awards	Jun-20	
52.204-13	System for Award Management Maintenance	Oct-18	
52.204-18	Commercial and Government Entity Code Maintenance	Aug-20	
52.204-19	Incorporation by Reference of Representations and Certifications	Dec-14	
52.204-21	Basic Safeguarding of Covered Contractor Information Systems	Nov-21	
52.204-23	Prohibition on Contracting for Hardware, Software, and Services Developed or Provided by Kaspersky Lab and Other Covered Entities	Dec-23	
52.204-25	Prohibition on Contracting for Certain Telecommunications and Video Surveillance Services or Equipment	Nov-21	
52.204-27	Prohibition on a ByteDance Covered Application	Jun-23	
52.204-30	Federal Acquisition Supply Chain Security Act Orders—Prohibition	Dec-23	
52.209-06	Protecting the Government's Interest When Subcontracting with Contractors Debarred, Suspended, or Proposed for Debarment	Nov-21	
52.209-10	Prohibition on Contracting with Inverted Domestic Corporations	Nov-15	
52.219-06	Notice of Total Small Business Set-Aside	Nov-20	
52.219-08	Utilization of Small Business Concerns	Feb-24	
52.219-28	Post-Award Small Business Program Representation	Feb-24	
52.222-03	Convict Labor	Jun-03	
52.222-19	Child Labor-Cooperation with Authorities and Remedies	Feb-24	
52.222-21	Prohibition of Segregated Facilities	Apr-15	
52.222-25	Affirmative Action Compliance	Apr-84	
52.222-26	Equal Opportunity	Sep-16	
52.222-36	Equal Opportunity for Workers with Disabilities	Jun-20	
52.222-50	Combating Trafficking in Persons	Nov-21	

Clause Number	Title	Date	When Applicable
52.225-01	Buy American-Supplies	Oct-22	
52.225-13	Restrictions on Certain Foreign Purchases	Feb-21	
52.226-07	Drug-Free Workplace	May-24	
52.226-08	Encouraging Contractor Policies to Ban Text Messaging While Driving	May-24	
52.227-01 Alt I	Authorization and Consent - Alternate I (Apr-84)	Jun-20	
52.227-02	Notice and Assistance Regarding Patent and Copyright Infringement	Jun-20	
52.227-11	Patent Rights-Ownership by the Contractor	May-14	
52.227-20	Rights in Data-SBIR Program	May-14	
52.232-11	Extras	Apr-84	
52.232-23	Assignment of Claims	May-14	
52.232-25	Prompt Payment	Jan-17	
52.232-33	Payment by Electronic Funds Transfer - System for Award Management	Oct-18	
52.232-39	Unenforceability of Unauthorized Obligations	Jun-13	
52.232-40	Providing Accelerated Payments to Small Business Subcontractors	Mar-23	
52.232-01	Disputes	May-14	
52.233-04	Applicable Law for Breach of Contract Claim	Oct-04	
52.242-15	Stop-Work Order	Aug-89	
52.243-01 Alt V	Changes-Fixed-Price Alternate V (Apr-84)	Aug-87	
52.244-06	Subcontracts for Commercial Products and Commercial Services	Feb-24	
52.246-09	Inspection of Research and Development (Short Form)	Apr-84	
52.252-02	Clauses Incorporated by Reference	Feb-98	
52.252-06	Authorized Deviations in Clauses	Nov-20	
52.253-01	Computer Generated Forms	Jan-91	
252.203-7000	Requirements Relating to Compensation of Former DoD Officials	Sep-11	
252.203-7002	Requirement to Inform Employees of Whistleblower Rights	Dec-22	
252.204-7000	Disclosure of Information	Oct-16	
252.204-7003	Control of Government Personnel Work Product	Apr-92	
252.204-7008	Compliance with Safeguarding Covered Defense Information Controls	Oct-24	
252.204-7009	Limitations on the Use or Disclosure of Third-Party Contractor Reported Cyber Incident Information	Jan-23	
252.204-7012	Safeguarding Covered Defense Information and Cyber Incident Reporting (DEVIATION 2024-O0013)	May-24	
252.204-7016	Covered Defense Telecommunications Equipment or Services—Representation	Dec-19	

Clause Number	Title	Date	When Applicable
252.204-7017	Prohibition on the Acquisition of Covered Defense Telecommunications Equipment or Services—Representation	May-21	
252.204-7018	Prohibition on the Acquisition of Covered Defense Telecommunications Equipment or Services	Jan-23	
252.204-7019	Notice of NISTSP 800-171 DoD Assessment Requirements	Nov-23	
252.204-7020	NIST SP 800-171 DoD Assessment Requirements	Nov-23	
252.204-7022	Expediting Contract Closeout	May-21	
252.204-7024	Notice on the use of the Supplier Performance Risk System	Mar-23	
252.227-7016	Rights in Bid or Proposal Information	Jan-23	
252.227-7018	Rights in Noncommercial Technical Data and Computer Software--Small Business Innovation Research (SBIR) Program (DEVIATION 2020-O0007) (Jul 23)	Nov-23	
252.227-7019	Validation of Asserted Restrictions--Computer Software	Jan-23	
252.227-7025	Limitations on the Use or Disclosure of Government-Furnished Information Marked with Restrictive Legends	Jan-23	
252.227-7030	Technical Data--Withholding of Payment	Mar-00	
252.227-7037	Validation of Restrictive Markings on Technical Data	Jan-23	
252.227-7039	Patents--Reporting of Subject Inventions	Apr-90	
252.232-7003	Electronic Submission of Payment Requests and Receiving Reports	Dec-18	
252.232-7006	Wide Area WorkFlow Payment Instructions	Jan-23	
252.232-7010	Levies on Contract Payments	Dec-06	
252.235-7010	Acknowledgment of Support and Disclaimer	May-95	
252.235-7011	Final Scientific or Technical Report	Dec-19	
252.243-7001	Pricing of Contract Modifications	Dec-91	
252.244-7000	Subcontracts for Commercial Items	Nov-23	
52.203-03	Gratuities	Apr-84	Exceeding the simplified acquisition threshold.
52.203-05	Covenant Against Contingent Fees	May-14	Exceeding the simplified acquisition threshold.
52.203-06 or Alt I	Restrictions on Subcontractor Sales to the Government or ALT I	Nov-21	Exceeding the simplified acquisition threshold, Alt I commercial products or commercial services.
52.203-07	Anti-Kickback Procedures	Jun-20	Contracts exceeding $150,000.
52.203-08	Cancellation, Rescission, and Recovery of Funds for Illegal or Improper Activity	May-14	Exceeding the simplified acquisition threshold.
52.203-10	Price or Fee Adjustment for Illegal or Improper Activity	May-14	Exceeding the simplified acquisition threshold.
52.203-12	Limitation on Payments to Influence Certain Federal Transactions	Jun-20	Contracts exceeding $150,000.
52.204-02	Security Requirements	Mar-21	May require access to classified information; cost contract (see 16.302) for research and development with an educational institution is contemplated.

Clause Number	Title	Date	When Applicable
52.212-04	Contract Terms and Conditions—Commercial Products and Commercial Services	Nov-23	Commercial products or commercial services.
52.212-05	Contract Terms and Conditions Required to Implement Statutes or Executive Orders-Commercial Items	May-24	Commercial products or commercial services.
52.219-14	Limitations on Subcontracting	Oct-22	Set aside for small business and the contract amount is expected to exceed the simplified acquisition threshold.
52.222-35	Equal Opportunity for Veterans	Jun-20	Contracts exceed $150,000.
52.222-37	Employment Reports on Veterans	Jun-20	If contract contains 52.222-35.
52.222-40	Notification of Employee Rights Under the National Labor Relations Act	Dec-10	Exceeding the simplified acquisition threshold.
52.222-54	Employment Eligibility Verification	May-22	Contracts exceed $150,000.
52.223-03	Hazardous Material Identification and Material Safety Data	Feb-21	Requires the delivery of hazardous materials.
52.229-03	Federal, State, and Local Taxes	Feb-13	A fixed-price contract is contemplated; and the contract is expected to exceed the simplified acquisition threshold.
52.232-02	Payment under Fixed-Price Research and Development Contracts	Apr-84	Fixed Price R&D.
52.233-03	Protest After Award	Aug-96	Exceed the simplified acquisition threshold.
52.242-13	Bankruptcy	Jul-95	Exceed the simplified acquisition threshold.
52.242-17	Government Delay of Work	Apr-84	Supplies other than commercial or modified-commercial products.
52.245-01	Government Property	Sep-21	When property is expected to be furnished.
52.245-09	Use and Charges	Apr-12	When the clause at 52.245-1 is included.
52.246-04	Inspection of Services-Fixed Price	Aug-96	Services, or supplies that involve the furnishing of services, when a fixed-price contract is contemplated, and the contract amount is expected to exceed the simplified acquisition threshold.
52.246-16	Responsibility for Supplies	Apr-84	Supplies, services involving the furnishing of supplies, or research and development, when a fixed-price contract is contemplated, and the contract amount is expected to exceed the simplified acquisition threshold.
52.246-23	Limitation of Liability	Feb-97	Exceed the simplified acquisition threshold.
52.247-34	F.o.b. Destination	Nov-91	When the delivery term is f.o.b. destination.
252.204-7015	Notice of Authorized Disclosure of Information for Litigation Support	Jan-23	Commercial products and commercial services.
252.209-7004	Subcontracting with Firms that are Owned or Controlled by the Government of a Country that is a State Sponsor of Terrorism	May-19	Contracts value of $150,000 or more.
252.211-7003	Item Identification and Valuation	Jan-23	For supplies, and for services involving the furnishing of supplies.
252.223-7001	Hazard Warning Labels	Dec-91	Requires submission of hazardous material data sheets.
252.223-7008	Prohibition of Hexavalent Chromium	Jan-23	For supplies, maintenance and repair services, or construction.
252.225-7001	Buy American and Balance of Payments Program	Feb-24	Acquisition of commercial products and commercial services.

Clause Number	Title	Date	When Applicable
252.225-7002	Qualifying Country Sources as Subcontractors	Mar-22	(i) 252.225-7001, Buy American and Balance of Payments Program. Use if one or more is included: (ii) 252.225-7021, Trade Agreements. (iii) 252.225-7036, Buy American - Free Trade Agreements - Balance of Payments Program.
252.225-7012	Preference for Certain Domestic Commodities	Apr-22	Commercial products and commercial services.
252.225-7052	Restriction on Acquisition of Certain Magnets, Tantalum, and Tungsten	May-24	Products and commercial services, that exceed the simplified acquisition threshold.
252.225-7056	Prohibition Regarding Business Operations with the Maduro Regime	Jan-23	Commercial products and commercial services.
252.225-7060	Prohibition on Certain Procurements from the Xinjiang Uyghur Autonomous Region	Jun-23	Products utilizing funds appropriated or otherwise made available for any fiscal year.
252.225-7972	Prohibition on the Procurement of Foreign-Made Unmanned Aircraft Systems (DEVIATION 2020-O0015)	May-20	
252.225-7967	Prohibition Regarding Russian Fossil Fuel Business Operations (DEVIATION 2024-O0006, Revision 1)	Feb-24	Exceeding the simplified acquisition threshold.
252.228-7001	Ground and Flight Risk	Mar-23	Acquisition, development, production, modification, maintenance, repair, flight, or overhaul of aircraft owned by or to be delivered to the Government.
252.228-7005	Mishap Reporting and Investigation Involving Aircraft, Missiles, and Space Launch Vehicles	Nov-19	Acquisition, development, production, modification, maintenance, repair, flight, or overhaul of aircraft owned by or to be delivered to the Government.
252.235-7002	Animal Welfare	Dec-14	Uses live vertebrate animals.
252.235-7004	Protection of Human Subjects	Jul-09	Involving human subjects.
252.243-7002	Requests for Equitable Adjustment	Dec-22	Exceeding the simplified acquisition threshold.
252.245-7003	Contractor Property Management System Administration	Apr-12	Containing the clause at FAR 52.245-1, Government Property.
252.245-7005	Management and Reporting of Government Property	Jan-24	Containing the clause at FAR 52.245-1, Government Property.
252.247-7023	Transportation of Supplies by Sea	Jan-23	Except - those with an anticipated value at or below the simplified acquisition threshold.